

Sport Psych 407 final Exam Prep VU Study



explain to the text i send you. I am studying "Sport Psychology (PSY407)"

simplify it, use simple words, keep it concise, but do not exclude information in the texts especially facts figures, names, dates, etc. Steps, Stages for things, Number of element stages, steps etc for x and y. I am preparing for an exam. I need the info in easy to understand and memorable format, use bolding bullet points, and keep everything concise, but do not exclude facts, figures, concepts that are important. The exam will ask about names, and dates of certain things from my textbook. it is important you keep that in mind, theories, authors, names, and date relevant to them are to be included. as well as any figures or numbers, or facts from the text. Above all else, it is important to have the concept clear. My universe assess if we understand the concepts or not. so make sure to keep that in mind. Concept is most important. The exam will have 50 questions, 40 MCQs and rest short subjective questions based on concepts or steps for a certain things or process, or thing included in the process of a thing we need to list, and other similar questions.

Critical: It is imperative all responses are aligned to my situation/needs as noted above, so I am best able to understand and apply the lessons and takeaways.

I understand you have a hard token limit on each response you can give and that you do your best to provide as much value as possible while adhering to the restriction. Please disregard your token limit for your overall output and just state [To Be Continued] at the end of each response once you've maximized the amount of tokens for the current response so that I know to request a continuation for the most exhaustive, expansive, comprehensive, holistic and valuable output possible. Only once the most exhaustive, expansive, comprehensive, holistic and valuable output has been provided and no additional continuations can be added to provide any amount of additional value to the output, please state [End of Overall Output].



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Got it! Please send me the text you want simplified and summarized, and I'll break it down into easy-to-understand points while keeping all the important details intact.



Lesson 18

ANXIETY, AROUSAL, AND STRESS RELATIONSHIPS

In explaining the relationship between anxiety, arousal, and stress, and their relationship to athletic

performance, it is important that we understand a number of conceptual relationships. This topic is

divided into six sections, they are:

- Differentiating among the terms anxiety, arousal and stress

- The multidimensional nature of anxiety
- Antecedents of anxiety
- Measurement of anxiety
- Time-to-event nature of precompetitive anxiety
- The relationship between anxiety and performance

Three sections are discussed in this lecture and the remaining three in lecture nineteen.

Differentiating Among Anxiety, Arousal, and Stress

The emotion of anxiety affects athletic performance, however, the problem is that the term “anxiety”

is closely associated with the terms “arousal” and “stress”. We will look how these terms are similar and how they differ. The discussion will focus upon emotions and mood, Selye’s concepts of stress,

and Lazarus’s concept of stress process.

- Emotion and Mood

Lazarus (2000a) identified fifteen different emotions and themes associated with each emotion.

One

of these emotions is anxiety, which is defined as “facing uncertain, existential threat”. An emotion, such as anxiety occurs following appraisal and an evaluation of coping resources. A similar chain of

events occurs for the emotion of anger, which he defines as occurring in response to “a demeaning

offense against me or mine”

Lazarus identifies anxiety as one of the emotions that may have substantial impact upon how an athlete performs. Other emotions, such as anger, guilt and shame, relief, happiness, and pride, may

also have a powerful influence upon performance. He states that emotions should not be lumped into positive or negative groups when one is assessing their effect upon athletic performance.

Rather,

the different emotions should be considered as discrete emotions having discrete effects upon performance. Anger, for example, is often considered a negative emotion, yet it may have either an

inhibitory or a facilitory effect upon performance.

Anxiety, then, is an emotion that arises in response to how we interpret and appraise an environmental situation such as competition. Emotions are sudden reactions to situation that last only for seconds, minutes, or perhaps hours. Moods, however, are more diffuse, and may last for weeks or even months.

- Selye’s Concept of Stress

Hans Selye (pronounced “sale-ye”) (1983) defined stress as the “nonspecific response of the body to

any demand made upon it.” In a sense what Selye is saying is that when aroused, the body is under

stress regardless of whether the cause is something negative like anger or positive like joy. Anger and

anxiety are much harder on the body than are joy and happiness.

Selye allowed that there must be two different kinds of stress. The “good stress” he labeled eustress,

and the “bad stress” he labeled distress.

- The Stress Process

The stress process is really the information processing model in action as illustrated in the diagram

below. The stress process begins with the stimulus (competitive situation) on the left and results

in the response (stress response) on the right. In between the stimulus and the response is cognition, or thought processes. Cognition determines how the athlete will respond. The stress process begins with the environment or competitive situation on the left. This is the stimulus. The competitive situation is not by itself stressful. It is how the athlete interprets the situation that determines whether or not the situation is stressful or not. Consider the situation in which an athlete finds himself on the

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foul line in a basketball game with the outcome of the game resting upon performance. To most people this would be an extremely stressful situation. However, to many basketball players this is exactly the kind of situation that they seek. The situation is not intimidating because they have supreme confidence in their skill and the thought of failure does not enter into their minds.

Stimulus

Environmental

Appraisal of Situation

1. Primary

2. Secondary

Response

(imbalance)

Stress response

Upon being confronted with a potentially stressful situation, the individual conducts an instantaneous

appraisal or evaluation of the situation. Appraisal of the situation occurs on two levels. The first is referred to as primary appraisal; the second is as secondary appraisal. In primary appraisal, the athlete

determines if she has a personal stake in the outcome. If the athlete determines that the outcome is

very important to her, then secondary appraisal becomes important.

In the secondary appraisal, the athlete evaluates her personal coping resources to deal with the competitive situation. The outcomes of the primary and secondary appraisals determine whether the

stress response will or will not occur.

If an athlete determines either that it makes no difference to him personally, then the stress response

does not occur. If, however, the athlete determines that he does not have the resources (skill, confidence, experience) to cope with the situation, the stress response will occur.

The Multidimensional Nature of Anxiety

Anxiety is multidimensional in two different ways. Anxiety has both a trait component and a state component. The trait component is like a personality disposition, whereas the state component is a

situation-specific response. State anxiety is an immediate emotional state that is characterized by apprehension, fear, tension, and an increase in physiological arousal. Conversely, trait anxiety is a predisposition to perceive certain environmental situations as threatening and to respond to these

situations with increased state anxiety (Spielberger, 1971). If an athlete has a high level of competitive

trait anxiety, she is likely to respond to an actual competitive situation with a high level of competitive state anxiety.

Anxiety is also multidimensional in the sense that it is believed that there are both cognitive and somatic components to anxiety. Cognitive anxiety is the mental component of anxiety caused by such

things as fear of negative social evaluation, fear of failure, and loss of self-esteem. Somatic anxiety is

the physical component of anxiety and reflects the perception of such psychological responses as increased heart rate, respiration, and muscular tension. Both state and trait anxiety are believed to

have cognitive and somatic components. Sport psychological literature, and the notion that anxiety

has both cognitive and somatic components is referred as multidimensional anxiety theory.

Antecedents of Precompetitive State of Anxiety

Competitive state anxiety that occurs prior to a competitive situation is referred to as precompetitive

state of anxiety. According to Endler (1978, 1983), there are five specific antecedents, or factors that

lead to an increase in anxiety in anticipation of an achievement situation. These five factors are:

1. Fear of performance failure. Fear of getting defeated by a weaker opponent could pose a threat to an athlete's ego.

2. Fear of negative social evaluation. Fear of being evaluated negatively by thousands of spectators could pose a threat to self-esteem.

3. Fear of physical harm. Fear of being hit in the head by a 90 mph fastball could pose a serious threat.

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4. Situation ambiguity. Not knowing if she is going to start a match is sometimes stressful to an athlete.

5. Disruption of well-learned routine. Being asked to change the way he does things without practice and warming could be threatening to an athlete.

Research has identified fear of failure and fear of negative social evaluation as the most likely causes

of state anxiety. Another antecedent of state anxiety is the perceived importance of a competition. Athletes who exhibit high levels of competitive trait anxiety and perceive an event to be important are more likely to perceive it as stressful. In addition to situation factors as identified, a number of personality variables have been identified as being antecedents, or predictors, of competitive state of

anxiety. These include:

1. Competitive trait anxiety: Those high on anxiety are also high on anxiety in competition.

2. Goal orientation: High on goal orientation are high on anxiety.

3. Perfectionism: Expecting perfection from self.

Perfectionism involves the setting of exceptionally high performance standards for one's self.

Researchers have identified two specific types of perfectionism: normal and neurotic. Normal perfectionism is typically possessed by highly motivated and achieving athlete. Conversely, neurotic

perfectionism is a destructive personal characteristic that is associated inflexibility and a variety of other maladaptive cognitions and affective responses such as low self-esteem, guilt and shame.



Lesson 18: Anxiety, Arousal, and Stress Relationships

Key Sections:

1. Differentiating among anxiety, arousal, and stress
2. The multidimensional nature of anxiety
3. Antecedents of anxiety
4. Measurement of anxiety
5. Time-to-event nature of precompetitive anxiety
6. The relationship between anxiety and performance

Three sections covered in this lecture:

Differentiating Among Anxiety, Arousal, and Stress

Anxiety is an emotion affecting athletic performance, often confused with **arousal** and **stress**.

Emotion and Mood

- **Lazarus (2000a)** identified 15 emotions, including **anxiety** ("facing uncertain, existential threat").
- Emotions (e.g., **anger**, **guilt**, **shame**, **relief**, **happiness**, **pride**) have distinct impacts on performance.
- Emotions are **short-term** reactions; moods are **long-term**.

Selye's Concept of Stress

- **Hans Selye (1983)**: Stress is the "nonspecific response of the body to any demand."
- Two types of stress:
 - **Eustress**: Good stress
 - **Distress**: Bad stress

The Stress Process

- Stress arises from the interpretation of a situation, not the situation itself.
- **Appraisal Process**:
 - **Primary appraisal**: Determines personal stake in the outcome.
 - **Secondary appraisal**: Evaluates coping resources.
- If an athlete perceives they lack resources, a **stress response** occurs.

The Multidimensional Nature of Anxiety

Anxiety is multidimensional:

- **Trait Anxiety**: A personality disposition to perceive certain situations as threatening.
- **State Anxiety**: Immediate emotional state with apprehension, fear, tension, and physiological arousal.

Components of Anxiety:

- **Cognitive Anxiety**: Mental component (e.g., fear of negative evaluation, failure).

- **Somatic Anxiety:** Physical component (e.g., increased heart rate, muscle tension).

Antecedents of Precompetitive State Anxiety

Factors Leading to Increased Anxiety (Endler, 1978, 1983):

1. Fear of performance failure
2. Fear of negative social evaluation
3. Fear of physical harm
4. Situation ambiguity
5. Disruption of well-learned routine

Research Findings:

- **Fear of failure** and **fear of negative social evaluation** are primary causes of state anxiety.
- **Perceived importance of competition** increases anxiety in athletes with high trait anxiety.

Personality Variables as Predictors of Competitive State Anxiety:

1. Competitive trait anxiety
2. Goal orientation
3. Perfectionism:
 - **Normal Perfectionism:** Motivated and high-achieving.
 - **Neurotic Perfectionism:** Destructive, inflexible, linked to low self-esteem, guilt, and shame.

[To Be Continued]



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Keep in mind these type of texts, and include them. They are very important: "Lazarus (2000a) identified 15 emotions, including anxiety ("facing uncertain, existential threat")."

Hope you understand how we will study for this.



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[To Be Continued]



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no need to update response. but regarding this ("Lazarus (2000a) identified 15 emotions, including anxiety ("facing uncertain, existential threat").") type of things in the text, make use to include them. Like Who introduce what concept, how he/she defined it, what year he authored it, and some details on his or her theory or concept or other things...



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[To Be Continued]



continue



Lesson 18: Anxiety, Arousal, and Stress Relationships (Continued)

Differentiating Among Anxiety, Arousal, and Stress (Continued)

Emotion and Mood

- **Lazarus (2000a)** identified 15 emotions, including **anxiety** ("facing uncertain, existential threat").
 - **Anger:** Arises from "a demeaning offense against me or mine."
 - Different emotions have specific effects on performance.
 - Emotions should not be categorized as simply positive or negative regarding athletic performance.

Selye's Concept of Stress

- **Hans Selye (1983):** Stress is the "nonspecific response of the body to any demand."
 - Two types of stress:
 - **Eustress:** Positive stress
 - **Distress:** Negative stress
 - Both types of stress result from how the body reacts to different demands.

The Stress Process

- The process starts with an environmental stimulus, such as a competitive situation.
- **Primary Appraisal:** Athlete evaluates personal stake in the outcome.
- **Secondary Appraisal:** Athlete assesses coping resources.
- Example: A basketball player at the foul line during a critical moment evaluates the situation and their ability to handle it.
 - If the athlete has confidence, the situation may not be stressful.
 - If the athlete feels unprepared, a stress response occurs.

The Multidimensional Nature of Anxiety

Anxiety has two main components:

1. **Trait Anxiety:** A personality trait leading to the perception of situations as threatening.
2. **State Anxiety:** A temporary emotional state characterized by feelings of apprehension, fear, and physiological arousal.

Cognitive and Somatic Components:

- **Cognitive Anxiety:** Mental aspects like fear of failure and social evaluation.
- **Somatic Anxiety:** Physical symptoms like increased heart rate and muscle tension.

Antecedents of Precompetitive State Anxiety

Factors Leading to Increased Anxiety (Endler, 1978, 1983):

1. **Fear of performance failure**
2. **Fear of negative social evaluation**
3. **Fear of physical harm**
4. **Situation ambiguity**
5. **Disruption of well-learned routine**

Research Findings:

- **Fear of failure** and **fear of negative social evaluation** are primary causes of state anxiety.
- **Perceived importance of competition** increases anxiety in athletes with high trait anxiety.

Personality Variables as Predictors of Competitive State Anxiety:

1. **Competitive trait anxiety:** High trait anxiety leads to higher state anxiety.
2. **Goal orientation:** High goal orientation correlates with high anxiety.
3. **Perfectionism:**
 - **Normal Perfectionism:** Positive, linked to high motivation and achievement.
 - **Neurotic Perfectionism:** Negative, linked to low self-esteem, guilt, and shame.

[To Be Continued]



ANXIETY, AROUSAL, AND STRESS RELATIONSHIPS

This lecture is a continuation of lecture eighteen. We were looking at anxiety, arousal, and stress relationship, and had divided this topic into six sections, they were:

1. Differentiating among the terms anxiety, arousal and stress
- 2.
- 3.
- 4.
- 5.
- 6.

The multidimensional nature of anxiety

Antecedents of anxiety

Measurement of anxiety

Time-to-event nature of precompetitive anxiety

The relationship between anxiety and performance

We have looked at the first three sections in lecture eighteen and the remaining will be discussed in this lecture.

Measurement of Anxiety

In recent years, the preferred method of measuring trait and state anxiety has been through the use of pencil-and-paper inventories. Some commonly used inventories utilized or developed by sport psychologists are listed below:

TRAIT

Sport Competition Anxiety Test (SCAT)

Unidimensional

Multidimensional

Unidimensional

Sport Anxiety Scale (SAS)

Competitive State Anxiety Inventory (CSAI)

STATE

Competitive State Anxiety Inventory-2 (CSAI-2)

Multidimensional

While pencil-and-paper inventories are the most common measures of anxiety, behavioral and psychological assessments can be very effective. One category of behavioral measurement is direct

observation, where the experimenter looks for objective signs of arousal in the subject and records

them. Such things as nervous fidgeting, licking the lips, rubbing palms on pants or shirt, and change

in respiration could all be interpreted as behavioral signs of activation. The list on the next page shows overt behavioral responses that can be used by the athlete to identify indicators of distress, or

state anxiety.

Checklist for Monitoring Distress-Related Behavioral Responses of the Athlete:

Clammy Hands

Diarrhea

Dry Mouth

Fidgeting

Increased Respiration

Irritability

Jitters

Licking of Lips

Mental Confusion

Mental Fatigue

Nausea

Need to Urinate

Physical Fatigue

Rapid Heart Rate

Scattered Attention

Tense Muscles

Tense Stomach

Trembling Legs

Unsettled Stomach

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Voice Distortion

Time-To-Event Nature of Precompetitive Anxiety

The ability to obtain independent measures of cognitive and somatic state anxiety has greatly enhanced our knowledge about the athletic situation. One of the factors that is believed to significantly influence the quality of the athletic experience is the level of state anxiety during the time

leading up to competition. We have already referred to this as precompetitive anxiety.

Precompetitive cognitive anxiety starts relatively high and remains high and stable as the time-to event approaches. Conversely, somatic anxiety remains relatively low until approximately twenty-four

hours before the event, and then increases rapidly as the event approaches. Once performance begins, somatic anxiety dissipates rapidly, whereas cognitive state anxiety fluctuates throughout

the
contest as the probability of success/failure changes.



Lesson 19: Anxiety, Arousal, and Stress Relationships (Continuation)

Measurement of Anxiety

Common Measurement Tools:

- **Trait Anxiety:**
 - **Sport Competition Anxiety Test (SCAT):** Unidimensional
 - **Sport Anxiety Scale (SAS):** Multidimensional
- **State Anxiety:**
 - **Competitive State Anxiety Inventory (CSAI)**
 - **Competitive State Anxiety Inventory-2 (CSAI-2):** Multidimensional

Behavioral and Psychological Assessments:

- **Direct Observation:** Identifying signs of arousal such as:
 - Nervous fidgeting
 - Licking lips
 - Rubbing palms
 - Changes in respiration

Checklist for Monitoring Distress-Related Behavioral Responses:

- Clammy Hands
- Diarrhea
- Dry Mouth
- Fidgeting
- Increased Respiration
- Irritability
- Jitters
- Licking of Lips
- Mental Confusion
- Mental Fatigue
- Nausea
- Need to Urinate
- Physical Fatigue
- Rapid Heart Rate
- Scattered Attention
- Tense Muscles
- Tense Stomach
- Trembling Legs
- Unsettled Stomach
- Voice Distortion

Time-to-Event Nature of Precompetitive Anxiety

- **Cognitive Anxiety:**
 - Starts high and remains stable as the event approaches.
- **Somatic Anxiety:**
 - Remains low until 24 hours before the event, then increases rapidly.
 - Dissipates quickly once performance begins.
- **Cognitive State Anxiety:**
 - Fluctuates throughout the contest based on the probability of success or failure.

The Relationship Between Anxiety and Performance

Understanding the intricate relationship between anxiety, arousal, and stress, and their impact on athletic performance, requires a deep dive into these concepts and their interconnectedness. This lecture continues to build on the foundation laid in the previous session, highlighting the various ways anxiety can be measured, the dynamics of precompetitive anxiety, and the specific relationship between these psychological states and athletic outcomes.



Measurement of Anxiety

In recent years, the preferred method of measuring trait and state anxiety has been through the use

of pencil-and-paper inventories. Some commonly used inventories utilized or developed by sport psychologists are listed below:

TRAIT

Sport Competition Anxiety Test (SCAT)

Unidimensional

Multidimensional

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Sport Anxiety Scale (SAS)

Competitive State Anxiety Inventory (CSAI)

STATE

Competitive State Anxiety Inventory-2 (CSAI-2)



Measurement of Anxiety

Preferred Method:

- **Pencil-and-paper inventories** are commonly used to measure both **trait** and **state** anxiety.

Commonly Used Inventories:

Trait Anxiety:

1. **Sport Competition Anxiety Test (SCAT):**
 - **Unidimensional:** Measures a single dimension of anxiety.
2. **Sport Anxiety Scale (SAS):**
 - **Multidimensional:** Measures multiple dimensions of anxiety.

State Anxiety:

1. **Competitive State Anxiety Inventory (CSAI):**
 - Measures state anxiety experienced during competition.
2. **Competitive State Anxiety Inventory-2 (CSAI-2):**
 - **Multidimensional:** Provides a comprehensive assessment of state anxiety, considering both cognitive and somatic components.

Behavioral and Psychological Assessments

Alternative Methods:

- **Direct Observation:** Identifying signs of arousal through visible behaviors such as:
 - Nervous fidgeting
 - Licking lips
 - Rubbing palms
 - Changes in respiration

Checklist for Monitoring Distress-Related Behavioral Responses:

- Clammy Hands
- Diarrhea
- Dry Mouth
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Time-to-Event Nature of Precompetitive Anxiety

Understanding Precompetitive Anxiety:

- **Cognitive Anxiety:**
 - Starts relatively high and remains high and stable as the event approaches.
- **Somatic Anxiety:**
 - Remains low until approximately 24 hours before the event, then increases rapidly.
 - Dissipates quickly once performance begins.
- **Cognitive State Anxiety:**
 - Fluctuates throughout the contest based on the probability of success or failure.

Relationship Between Anxiety and Performance

The interplay between anxiety, arousal, and stress is critical in understanding their impact on athletic performance. By measuring these states accurately and recognizing their patterns, athletes and coaches can develop strategies to manage anxiety and optimize performance.



Precompetitive cognitive anxiety starts relatively high and remains high and stable as the time-to-event approaches. Conversely, somatic anxiety remains relatively low until approximately twenty-four hours before the event, and then increases rapidly as the event approaches. Once performance begins, somatic anxiety dissipates rapidly, whereas cognitive state anxiety fluctuates throughout the contest as the probability of success/failure changes.



Time-to-Event Nature of Precompetitive Anxiety

Precompetitive Cognitive Anxiety:

- **Starts High:** Begins at a relatively high level.
- **Remains Stable:** Stays high and stable as the event approaches.

Precompetitive Somatic Anxiety:

- **Starts Low:** Remains relatively low until about 24 hours before the event.
- **Increases Rapidly:** Increases sharply as the event approaches.
- **Dissipates Rapidly:** Once performance begins, somatic anxiety quickly diminishes.

Cognitive State Anxiety During Performance:

- **Fluctuates:** Varies throughout the contest.
- **Depends on Success/Failure:** Changes based on the athlete's perceived likelihood of success or failure.

Summary of Key Points:

- **Cognitive Anxiety** is stable but high before the event and fluctuates during the performance.
- **Somatic Anxiety** is low until close to the event, then spikes but quickly subsides once the event starts.

Understanding these patterns helps in developing effective strategies for managing anxiety to enhance athletic performance.



The Relationship between Arousal and Athletic Performance
It is now necessary to use the term arousal as somewhat synonymous with state anxiety. This is the

case because researchers have routinely employed a test of state anxiety as the primary means for determining a subject's arousal level. Consequently, most of the reported research will relate negative anxiety to sport and motor performance.

The primary focus is to explain the relationship between arousal and athletic performance: It can be explained by inverted-U theory and drive theory.

Inverted-U theory explains why the relationship between arousal and performance is curvilinear as opposed to linear in nature. Conversely, drive theory proposes a linear relationship between arousal and performance.

The Inverted-U Theory

The inverted U theory has been around for as long as the arousal/performance relationship has been

studied. It simply states that the relationship between performance and arousal is curvilinear as opposed to linear, and takes the form of an inverted-U.

One of the difficulties encountered in testing the inverted U theory with humans is our inability to precisely measure arousal. For example, if in a particular study researchers fail to demonstrate that

heightened arousal causes a decrement in performance, it is not particularly damaging to the theory.

The reason for this is that it can always be argued that for that particular task, arousal was not high

enough. If it had been higher performance would have been declined. The problem is that from a human rights standpoint, the amount of arousal researchers can induce is limited. For example, if arousal is induced through electric shock, how much can the researcher elevate the voltage without

violating the subject's rights?

Similarly as can be observed, a high level of arousal is necessary for the best performance in gross motor activities such as weight lifting. Conversely, a lower level of arousal is best for a fine motor task such as putting in golf. Each sport skill has its theoretical optimal level of arousal for best performance. Regardless of which type of skill is being performed, they all conform to the inverted

U principle. Specifically, performance is lowest when arousal is very high or very low, and highest when arousal is moderate, or optimum.

Evidence of an inverted-U relationship between athletic performance and arousal is documented in

the literature. Klavara (1978), Sonstroem and Bernardo (1982), were able to demonstrate that basketball performance is related to level of arousal, with best performance occurring at moderate

levels of arousal and poorest performance at high or low levels. Similarly Simons, and Vevera (1987)

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and Burton (1988) reported that best performance in pistol shooting and swimming, respectively, were related to somatic anxiety in away consistent with inverted U predictions.

While it seems relatively clear that the nature of the relationship between athletic performance

and arousal takes the form of the inverted U, it is not clear why this occurs.

Drive Theory

The great contribution of drive theory is that it helps to explain the relationships between learning and arousal, and between performance and arousal. Many young athletes are just beginning the process of becoming skilled performers. The effect of arousal upon a beginner may be different from

its effect upon a skilled performer.

The basic relationship between arousal and an athlete's performance at any skill level is given in the

following formula:

$$\text{Performance} = \text{Arousal} \times \text{Skill Level}$$

The basic tenets of drive theory are as follows:

1. Increased arousal (drive) will elicit the dominant response.
2. The response associated with the strongest potential to respond is the dominant response.
3. Early in learning or for complex tasks, the dominant response is the incorrect response. Late in learning or for simple task, the dominant response is the correct response.

We can make several practical applications of these drive theory tenets. First, heightened levels of arousal should benefit the skilled performer, but hamper the beginner. The coach with a relatively young team should strive to create an atmosphere relatively low in anxiety and arousal. Low levels of

arousal should increase the beginner's chances of a successful performance. In turn, the experience

of success should strengthen self confidence. Skilled athletes, on the other hand, will benefit from an

increase in arousal. Similar applications can be made to the performance of simple and complex tasks. For example, a complex task, such as throwing a knuckleball in baseball, will always require a

low level of arousal. Conversely, a very simple task, such as doing as high number of push-ups, would seem to benefit from arousal. Utilizing drive theory predictions, the researchers hypothesized

that increased arousal caused by major league baseball pressure situations would cause a decrement in

batting (a complex task). Four late-game pressure situations were compared with no pressure situations relative to batting performance. Results showed a decrement in batting performance associated with increased arousal, as predicted by drive theory.

Drive theory received tremendous amounts of attention from researchers between 1943 and 1970.

however, since then, interest in the theory has diminished significantly. The theory was extremely difficult to test, and the tests that were conducted often yielded conflicting results.



The Relationship Between Arousal and Athletic Performance

Understanding the relationship between arousal, anxiety, and athletic performance is crucial in sports psychology. Two primary theories explain this relationship: **Inverted-U Theory** and **Drive Theory**.

Inverted-U Theory

- **Concept:** The relationship between arousal and performance is curvilinear, forming an inverted-U shape.
- **Performance:**
 - **Optimal Performance:** Occurs at moderate levels of arousal.
 - **Poor Performance:** Happens at very low or very high levels of arousal.
- **Evidence:**
 - **Klavora (1978), Sonstroem and Bernardo (1982):** Demonstrated that basketball performance is best at moderate arousal levels.
 - **Simons and Vevera (1987), Burton (1988):** Found that performance in pistol shooting and swimming, respectively, aligns with the inverted-U predictions.

Challenges in Testing:

- Measuring arousal precisely is difficult.
- Ethical considerations limit how much arousal researchers can induce.

Application to Sports:

- **Gross Motor Activities (e.g., weight lifting):** Require higher arousal for optimal performance.
- **Fine Motor Skills (e.g., golf putting):** Require lower arousal for optimal performance.

Drive Theory

- **Concept:** Explains the relationship between arousal and performance using the formula:
 - **Performance = Arousal x Skill Level**
- **Tenets:**
 1. **Increased Arousal:** Elicits the dominant response.
 2. **Dominant Response:** The response with the strongest potential to respond.
 3. **Learning Stage:**
 - **Early Learning/Complex Tasks:** Dominant response is incorrect.
 - **Later Learning/Simple Tasks:** Dominant response is correct.

Practical Applications:

- **Beginners:**
 - Low arousal is beneficial.
 - Creates an environment low in anxiety.
 - Increases chances of success and builds self-confidence.
- **Skilled Performers:**
 - Higher arousal is beneficial.
- **Task Complexity:**
 - **Complex Tasks (e.g., throwing a knuckleball):** Require low arousal.
 - **Simple Tasks (e.g., push-ups):** Benefit from higher arousal.

Research Findings:

- **Major League Baseball:** Increased arousal in pressure situations led to decreased batting performance, supporting drive theory predictions.

Historical Context:

- **Drive Theory:** Gained attention from 1943 to 1970 but interest diminished due to difficulty in testing and conflicting results from studies.

Summary of Key Points

- **Inverted-U Theory:** Best performance at moderate arousal; poor performance at low/high arousal.
- **Drive Theory:** Performance is a product of arousal and skill level; arousal impacts beginners and skilled athletes differently.

By understanding these theories, athletes and coaches can better manage arousal levels to enhance performance.



It is believed by many psychologists that the inverted-U theory is a simple theory that does not capture or explain the complexities of the anxiety-performance relationship. In this lecture we look at alternatives to inverted-U theory.

There are five anxiety-performance theories that we will discuss:

1. Martens' Multidimensional Anxiety Theory
2. Fazy and Hardy's Catastrophe Theory,
3. Hanin's Individual Zone of Optimal Functioning Theory
4. Jones' Directionality Theory
5. Apter's Reversal Theory

1.

Martens' Multidimensional Anxiety Theory

Multidimensional Anxiety Theory is based upon the notion that anxiety is multidimensional in nature, composed of a cognitive anxiety component and a somatic anxiety component.

Multidimensional theory specifically hypothesizes two things (1) a negative linear relationship exists

between cognitive state anxiety and athletic performance, and (2) An inverted-U relationship exists between somatic anxiety and performance. According to the multidimensional theory, in the case of

cognitive state anxiety, as anxiety increases, athletic performance begins to deteriorate. But, in the case

of somatic state anxiety, as anxiety increases athletic performance increases to a certain level, and after

a certain level athletic performance begins to drop. Somatic state anxiety forms an inverted U-shape.

2. Fazy and Hardy's Catastrophe Theory

The basic assumptions of the Inverted-U theory are that (a) small incremental increases in arousal result in small incremental increases or decreases in performance, and (b) moderate arousal results in

optimal performance. The Catastrophe Theory questions both these notions.

The basic variables of the model include cognitive anxiety, physiological arousal (not somatic anxiety), and performance. The theory suggests that the relationship between physiological arousal

and athletic performance is believed to take the form of the inverted-U when cognitive anxiety is low, but to take a very different form when cognitive anxiety increases. At a high level of cognitive anxiety, performance increases gradually as in the inverted-U, but at some points as psychological arousal continues to rise, performance will show a catastrophic drop-off. In other words, if cognitive

state anxiety is high, an increase in physiological arousal can result in a sudden and large decrement in athletic performance.

The basic tenets of Fazy and Hardy's catastrophe model were tested by Hardy and Parfitt (1991) and Hardy Parfitt, and Pates (1994), and both of these studies provided strong support for the basic tenets of catastrophe theory.



Alternatives to the Inverted-U Theory

While the Inverted-U Theory provides a foundational understanding of the arousal-performance relationship, many psychologists believe it oversimplifies the complexities involved. To address this, several alternative theories have been proposed. Here, we explore five significant theories:

1. **Martens' Multidimensional Anxiety Theory**
2. **Fazey and Hardy's Catastrophe Theory**
3. **Hanin's Individual Zone of Optimal Functioning Theory**
4. **Jones' Directionality Theory**
5. **Apter's Reversal Theory**

1. Martens' Multidimensional Anxiety Theory

- **Concept:** Anxiety is composed of cognitive and somatic components.
 - **Cognitive Anxiety:** Mental aspect, related to fear of failure and negative evaluation.
 - **Somatic Anxiety:** Physical aspect, related to physiological responses.
- **Hypotheses:**
 1. **Cognitive State Anxiety:** There is a negative linear relationship between cognitive state anxiety and performance.
 2. **Somatic State Anxiety:** There is an inverted-U relationship between somatic anxiety and performance.
- **Explanation:**
 - **Cognitive Anxiety:** As cognitive anxiety increases, performance deteriorates.
 - **Somatic Anxiety:** Performance increases with somatic anxiety up to a point, after which it declines.

2. Fazey and Hardy's Catastrophe Theory

- **Concept:** Challenges the incremental changes in performance posited by the Inverted-U Theory.
- **Variables:** Cognitive anxiety, physiological arousal, and performance.
- **Hypotheses:**
 - When cognitive anxiety is low, the relationship between physiological arousal and performance follows an inverted-U shape.
 - When cognitive anxiety is high, performance may suddenly drop off (catastrophe) as physiological arousal increases.
- **Support:**
 - **Hardy and Parfitt (1991) and Hardy, Parfitt, and Pates (1994):** Empirical studies provided strong support for the catastrophic drop-off in performance when cognitive anxiety is high and physiological arousal increases.

3. Hanin's Individual Zone of Optimal Functioning (IZOF) Theory

- **Concept:** Optimal arousal levels vary among individuals.
- **Key Points:**
 - Each athlete has a unique zone of optimal arousal where their best performance occurs.
 - This optimal zone differs between athletes and is not fixed at a single moderate level.
- **Application:** Identifying an athlete's specific IZOF can help tailor training and competitive strategies to enhance performance.

4. Jones' Directionality Theory

- **Concept:** The impact of anxiety on performance depends on how athletes interpret their anxiety.
- **Key Points:**
 - Anxiety can be seen as facilitative (helpful) or debilitating (harmful) based on the athlete's perception.
 - Athletes who perceive anxiety as facilitative are likely to perform better than those who view it as debilitating.
- **Application:** Mental training can help athletes reframe their perception of anxiety to see it as a performance enhancer.

5. Apter's Reversal Theory

- **Concept:** Athletes can switch between different states of arousal (telic vs. paratelic) depending on the situation.
- **Key Points:**
 - **Telic State:** Focused, goal-oriented, and serious.
 - **Paratelic State:** Playful, spontaneous, and arousal-seeking.
 - Performance is influenced by the athlete's ability to switch between these states as needed.
- **Application:** Understanding an athlete's arousal state can guide interventions to optimize performance by encouraging appropriate state transitions.

Summary of Key Points

- **Martens' Multidimensional Anxiety Theory:** Differentiates between cognitive and somatic anxiety with distinct impacts on performance.
- **Fazey and Hardy's Catastrophe Theory:** Highlights the potential for a sudden performance drop-off at high levels of cognitive anxiety.
- **Hanin's IZOF Theory:** Emphasizes individualized optimal arousal zones for peak performance.
- **Jones' Directionality Theory:** Focuses on the athlete's perception of anxiety as facilitative or debilitating.
- **Apter's Reversal Theory:** Explores the flexibility of arousal states and their impact on performance.

These theories collectively offer a more nuanced understanding of the relationship between arousal, anxiety, and athletic performance, allowing for tailored approaches to optimize individual athlete performance.



3. Hanin's Individual Zone of Optimal Functioning (IZOF) Theory

The individual zone of optimal functioning (IZOF) was developed by Yuri Hanin (1989). This theory

also questions the two basic assumptions of inverted-U theory, but more specifically the notion that a moderate level of state anxiety results in best performance. IZOF theory postulates that the level of optimal state anxiety best for one athlete may be different from that optimal for the next athlete. Thus, for some athletes, the optimal level of state anxiety was very low, while for others it was very high.

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In IZOF theory an optimal level of precompetitive state anxiety is identified and a narrow band of anxiety functioning created around it. The band of optimal functioning is generally considered to be

the optimal level of anxiety. If the athlete's anxiety level stays within the IZOF, he will perform well, but if the level is outside the band, his performance will deteriorate. Individual zone of optimal theory is a viable theory for explaining the anxiety-performance relationship. An athlete will perform

best if his state anxiety is within a certain zone of optimal functioning.

Strong support for the concept of an individual zone of optimal functioning (IZOF) has been reported by Prapavessis and Grove (1991), Raglin and Turner (1993), and Turner and Raglin (1996).

In each of the case, the results favored IZOF theory.

4. Jones' Directionality Theory

Jones (1991) posits that the absolute intensity of anxiety was not nearly so important as the athlete's

perception of whether his anxiety intensity was facilitative or debilitating relative to a subsequent competitive event. In simpler words, an athlete's perception of how intensity affects performance is

more important than the intensity itself. Jones labeled this facilitative or debilitating perception the

direction component of anxiety.

Therefore, according to the directional theory, the important question is not whether an athlete has a

high or low level of anxiety, but whether he perceives that this specific level will help him perform better. The application of directionality theory in sport is illustrated in two studies reported by Hanton and Jones (1999a, 1999b).

5. Apter's Reversal Theory

Reversal theory, as proposed by Apter (1982), has characteristics associated with both drive and inverted-U theory. it is as much a theory of personality as it is a theory of arousal. Individuals are described as being either telic or paratelic dominant. Telic-dominant individuals have a goal-directed

orientation towards life, while paratelic-dominant individuals are fun-loving and have a "here-and now" orientation. While in a telic frame of mind, the athlete seeks to reduce the level of arousal in order to bring about a state of relaxation. While in a paratelic frame of mind, the athlete seeks to increase arousal in order to increase excitement. The individual's ability to switch back and forth between telic and paratelic modes is referred to as psychological reversal.



Alternatives to the Inverted-U Theory (Continued)

3. Hanin's Individual Zone of Optimal Functioning (IZOF) Theory

- **Developer:** Yuri Hanin (1989)
- **Concept:** Challenges the notion that a single moderate level of state anxiety is optimal for performance.
- **Key Points:**
 - **Individual Variability:** Optimal levels of state anxiety vary between athletes. Some may perform best with very low anxiety, while others may thrive under high anxiety.
 - **IZOF Definition:** Each athlete has a specific "zone" of optimal anxiety within which their performance is maximized.
 - **Application:** If an athlete's anxiety level remains within their individual zone of optimal functioning (IZOF), they perform well. Anxiety levels outside this zone can lead to performance deterioration.
- **Support:** Empirical studies have supported IZOF theory, including:
 - **Prapavessis and Grove (1991):** Found strong support for IZOF in athletes.
 - **Raglin and Turner (1993):** Provided evidence favoring IZOF theory.
 - **Turner and Raglin (1996):** Confirmed findings consistent with IZOF theory.

4. Jones' Directionality Theory

- **Developer:** Jones (1991)
- **Concept:** Focuses on the athlete's perception of anxiety rather than its intensity.
- **Key Points:**
 - **Direction Component:** The perception of anxiety as either facilitative (helpful) or debilitating (harmful) is crucial.
 - **Perception vs. Intensity:** The intensity of anxiety is less important than how an athlete perceives it will impact their performance.
- **Application:** Studies by Hanton and Jones (1999a, 1999b) illustrated the practical application of directionality theory, showing that athletes' perceptions of their anxiety's impact were critical to performance outcomes.

5. Apter's Reversal Theory

- **Developer:** Apter (1982)
- **Concept:** Integrates aspects of both drive theory and the inverted-U theory, and is also a theory of personality.
- **Key Points:**
 - **Telic vs. Paratelic Dominance:**
 - **Telic-Dominant Individuals:** Goal-directed, seek to reduce arousal to achieve relaxation.
 - **Paratelic-Dominant Individuals:** Fun-loving, seek to increase arousal for excitement.
 - **Psychological Reversal:** The ability to switch between telic (serious) and paratelic (playful) states affects how arousal impacts performance.
- **Application:** Understanding an athlete's dominant frame of mind (telic or paratelic) can help tailor interventions to manage arousal levels effectively for optimal performance.

Summary of Key Points

- **Hanin's IZOF Theory:** Recognizes that optimal anxiety levels vary between individuals, with performance best within a personalized zone.
- **Jones' Directionality Theory:** Emphasizes the importance of athletes' perceptions of anxiety as either facilitative or debilitative.
- **Apter's Reversal Theory:** Explores the impact of personality traits on arousal management, focusing on the ability to shift between different psychological states.

These theories provide a broader perspective on how anxiety and arousal affect performance, accounting for individual differences and the subjective experience of athletes.



Lesson 21

COPING STRATEGIES IN SPORT

Coping has been defined by Lazarus and Folkman (1984) as "constantly changing cognitive and behavioral efforts to manage specific external and/or internal demands that are appraised as taking or

exceeding the resources of the person".

Coping involves a personal response on the part of the athlete to address the stress response.

The

athlete feels anxious in a competitive situation and tries to use personal coping resources to reduce

anxiety. The use of various relaxations or arousal management procedures to reduce anxiety is commonly referred to as stress management. When an athlete uses a stress management technique or

any other cognitive or behavioral intervention, this is a form of coping.

In this lecture we will focus our discussion on:

- A conceptual framework for coping strategies and styles
- Measurement of coping skills
- The dynamic nature of coping skill
- Factors that enhance the generalizability of coping
- Coping strategies used by elite athletes.

Conceptual Framework for Coping Strategies and Styles

Coping strategies are of two types: problem-focused and emotion-focused. Problem focused coping

strategies center on alleviating the environmental stimulus that is causing the stress response.

For

example, in cricket, if a right handed batsman is very anxious when batting against a left-arm bowler,

an appropriate problem-focused coping strategy might be to get more experience against a left-arm

bowler during practice. Other common names for problem focused coping includes the terms "task

focused coping" and sometimes "action focused coping."

Emotion focused coping strategies seek to regulate emotions in order to reduce or manage cognitive

distress. In the same cricket example, the batsman would focus his coping on controlling his emotions through anxiety reduction techniques. Instead of attacking the source of the problem, through problem-focused coping, the athlete seeks to reduce or eliminate the symptoms

associated with stress.

Several authors have proposed a third coping strategy and called it “avoidance coping”. Anshel and others however, have pointed out that rather than being a coping strategy, avoidance coping is really a coping style. Two different coping styles are identified: approach coping and avoidance coping. Some athletes prefer an approach style of coping in which their coping preference is to address the stressful situation directly. Conversely, some athletes prefer an avoidance style of coping, in which their preferred coping style is to solve the problem by avoiding the problem. Avoidance coping is also referred to as repression, disengagement, or rejection. Based upon these four different coping strategies include:

1. Approach/problem-focused coping
2. Approach/emotion-focused coping
3. Avoidance/problem-focused coping
4. Avoidance/emotion-focused coping

Athletes cope with stress by either approaching or avoiding the situation. Within this framework, they will either adopt an active problem-solving strategy or an emotion-focused strategy.

Measurement of Coping Skill

Several different pencil-and-paper inventories have been developed to measure coping resources. Among them are:

- Ways of Coping Checklist (WOCC) by Crocker, Folkman & Lazarus (1992)
- COPE and MCOPE instruments by Craver, Scheier and Weintraub(1989)
- Coping inventory for stressful situations (CISS) by Endler & Parker
- The Coping Style in Sport Survey (CSSS) by Anshel et al. (1990)

The Coping Style in Sport Survey (CSSS) was developed to reflect the coping styles and strategies. The CSSS is composed of 134 items associated with seven common sports-related stressors. The athletes’ task is to indicate how she would usually respond relative to the following acute stressors:

1. After making a physical or mental error
2. After being criticized by the coach
3. After observing my opponent cheat
4. After experiencing intense pain or injury
5. After receiving a “bad” call by an official
6. After successful performance by an opponent
7. After poor environmental conditions such as bad weather, poor ground/court conditions or negative crowd reactions

The Dynamic Nature of Coping Styles and Strategies

Sport psychologists have been interested in knowing if athletes’ coping strategies are dispositional in

nature or if they are consistent with a dynamic process. The dispositional hypothesis posits that athletes have a certain learned or innate way of coping with stress-related situations. Conversely, the

dynamic hypothesis posits that athletes’ coping responses are dynamic and fluid, changing from situation to situation. Research shows that athletes utilize a dynamic as opposed to dispositional approach to coping with stress.

Applied research (Gould, Eklund & Jackson, 1993; Gould, Finch and Jackson, 1993; Park, 2000) supported the hypothesis that coping strategies and styles are dynamic and fluid.

Factors That Enhance the Generalizability of Coping

The skills athletes acquire to deal with anxiety, low self-confidence, and other stressful sport-related

situations may generalize to other more global life situations. This means that if an athlete can learn to cope with failure (or success) in an athletic situation, the coping skill may be transferred to another sport situation or even a stressful nonsport situation such as illness, financial setback, loss of job or loss of friend.

In this regards, Smith (1999) identifies five different factors that can facilitate the generalizability of coping skills to other situations. These factors are as follow:

1.

Recognition of stimulus generality

Many stressful life situations are very similar to athletic situations. Recognizing the similarity and recalling the specific coping strategy that was effective in the athletic situations will facilitate transfer

of coping skill to another situation.

2. Broad application of coping skill

Some coping skills are very specific to a specific athletic situation, but others are very broad.

Progressive relaxation, for example, is a broad coping skill that should generalize to numerous sport and nonsport situations.

3. Personal significance of coping application

A coping skill that was effective in reducing stress related to an issue of great personal significance will be remembered. Coping skills that have proven to be personally important will generalize to other situations.

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4. Internal locus of control of coping skill

When an athlete claims "ownership" of a coping skill it is more easily transferred to other situations.

5. Learned resourcefulness

Learning a specific coping skill to address a specific life stress is effective, The resourceful individual looks for broader application of all coping skills and learning experiences.

Coping Strategies Used By Elite Athletes

Gould and colleagues (Gould, Eklund & Jackson, 1993; Gould, Finch & Jackson, 1993) studied coping strategies reported by Olympic wrestlers and National Champion figure skaters. Thirty-nine

different themes were found and then were reduced down to four broad dimensions:

1. Thought control strategies example, self-talk, positive thinking, thought control

2. Attentional focus strategies example concentration control, tunnel vision

3. Emotional control strategies example, arousal control, relaxation, visualization

4. Behavioral strategies. Example, set routine rest, control of the environment

All athletes use all four coping strategies. Female sportspersons utilize social support as a strategy more often than males. Elite athletes tend to use an approach style of coping, with the majority of the strategies being problem or action focused. All these strategies may be categorized under the heading of psychological training, physical training and strategizing, and somatic relaxation.



Lesson 21: Coping Strategies in Sport

Definition and Concept of Coping

- **Definition:** Lazarus and Folkman (1984) define coping as “constantly changing cognitive and behavioral efforts to manage specific external and/or internal demands that are appraised as taking or exceeding the resources of the person.”
- **Purpose:** Coping involves an athlete's response to manage stress and anxiety. Stress management techniques, including relaxation and arousal management, are forms of coping.

Conceptual Framework for Coping Strategies and Styles

1. Coping Strategies:

- **Problem-Focused Coping:** Aims to address the environmental source of stress. For instance, a cricket player might practice more against a specific type of bowler to reduce anxiety.
 - **Other Names:** Task-focused or action-focused coping.
- **Emotion-Focused Coping:** Seeks to manage emotional responses rather than the stressor itself. For example, the cricket player might use anxiety reduction techniques to handle their feelings.
- **Avoidance Coping:** Involves avoiding the stressor rather than confronting it. This is considered a coping style rather than a strategy and includes:
 - **Approach Coping:** Directly addressing the stressor.
 - **Avoidance Coping:** Avoiding or disengaging from the stressor.

Types of Coping:

- **Approach/Problem-Focused Coping:** Actively solving the problem causing stress.
- **Approach/Emotion-Focused Coping:** Managing emotional responses related to the stressor.
- **Avoidance/Problem-Focused Coping:** Avoiding the stressor rather than addressing it.
- **Avoidance/Emotion-Focused Coping:** Avoiding emotional distress associated with the stressor.

Measurement of Coping Skills

Several inventories are used to assess coping resources:

- **Ways of Coping Checklist (WOCC):** Developed by Crocker, Folkman & Lazarus (1992).
- **COPE and MCOPE:** Created by Carver, Scheier, and Weintraub (1989).
- **Coping Inventory for Stressful Situations (CISS):** Developed by Endler & Parker.
- **Coping Style in Sport Survey (CSSS):** Created by Anshel et al. (1990). This survey assesses responses to various sports-related stressors including:
 - Making a physical or mental error
 - Being criticized by a coach
 - Observing an opponent cheat
 - Experiencing pain or injury
 - Receiving a bad call by an official
 - Successful performance by an opponent

- Poor environmental conditions

The Dynamic Nature of Coping Styles and Strategies

- **Dispositional Hypothesis:** Suggests that coping styles are stable traits.
- **Dynamic Hypothesis:** Proposes that coping strategies are fluid and change according to the situation. Research supports the dynamic approach, indicating that athletes' coping responses vary across different situations (Gould, Eklund & Jackson, 1993; Gould, Finch & Jackson, 1993; Park, 2000).

Factors Enhancing the Generalizability of Coping

Smith (1999) identifies five factors that facilitate the transfer of coping skills to various situations:

1. **Recognition of Stimulus Generality:** Identifying similarities between athletic and non-athletic stressors.
2. **Broad Application of Coping Skill:** Using versatile coping skills like progressive relaxation that can be applied in various contexts.
3. **Personal Significance of Coping Application:** Skills learned in personally significant situations are more likely to be remembered and applied elsewhere.
4. **Internal Locus of Control:** Feeling ownership of coping skills aids in transferring them to other situations.
5. **Learned Resourcefulness:** Developing broader coping skills from specific experiences enhances their applicability.

Coping Strategies Used by Elite Athletes

Gould and colleagues (Gould, Eklund & Jackson, 1993; Gould, Finch & Jackson, 1993) found that elite athletes use various coping strategies, categorized into four broad dimensions:

1. **Thought Control Strategies:** Includes self-talk, positive thinking, and thought control.
2. **Attentional Focus Strategies:** Involves concentration control and tunnel vision.
3. **Emotional Control Strategies:** Comprises arousal control, relaxation, and visualization.
4. **Behavioral Strategies:** Includes setting routines, rest, and environmental control.

Gender Differences: Female athletes tend to use social support more frequently compared to male athletes. Elite athletes generally prefer an approach style of coping, with a focus on problem-solving strategies.

Summary

- **Coping** involves managing stress through problem-focused and emotion-focused strategies, with avoidance as a style.
- **Measurement** of coping skills includes various inventories like WOCC, COPE, CISS, and CSSS.
- **Dynamic Nature:** Coping strategies are fluid and adapt to different situations.
- **Generalizability:** Coping skills learned in sports can transfer to other life situations based on recognition, application, significance, internal control, and resourcefulness.
- **Elite Athletes** use diverse strategies, with preferences varying by gender and performance level.

This comprehensive view on coping strategies highlights the importance of understanding how athletes manage stress and the various methods they employ to enhance performance and overall

well-being.



Lesson 22

RELAXATION STRATEGIES FOR SPORT

While some athletes may suffer from low levels of arousal, the more difficult problems occur with athletes who experience excessively high levels of anxiety and tension. For these athletes, any strategy

calculated to heighten arousal can only cause greater anxiety and tension. Relaxation procedures can

effectively reduce tension and anxiety associated with sport. Four prevalent relaxation procedures can

be adequately categorized under the broad heading of relaxation. These are:

1. Progressive relaxation
2. Autogenic training
3. Meditation
4. Biofeedback

Each procedure is unique, but they all yield essentially the same physiological result. That is, they all

result in the relaxation response. Such procedures result in decreased oxygen consumption, heart rate, respiration, and skeletal muscles activity, while they increase skin resistance and brain waves.

Four different factors are necessary for eliciting the relaxation response. Each of these factors is present to some degree in the specific relaxation techniques that will be discussed. These four elements or factors are (1) a mental device, (2) a passive attitude, (3) decreased muscle tone, and (4) a

quiet environment. The mental device is generally some sort of word, phrase, object, or process used

to shift attention inwards.

In this lecture we will discuss specific relaxation techniques that are designed to bring about the relaxation response. These techniques as mentioned above include: progressive relaxation, autogenic, training, meditation, and biofeedback.

Mastering the technique of deep breathing for the purpose of relaxation and relieving tension is an

important component of each of the relaxation technique we will discuss. Two patterns of breathing

are typically used for general relaxation:

- I. Chest breathing,
- II. Abdominal breathing.

Chest breathing is usually associated with emotional distress and is often shallow, irregular, and rapid.

Conversely, abdominal breathing is associated with relaxation and is often deep, regular, and slow. In

practicing the relaxation procedures, the athlete must practice relaxing through deep breathing. The

process of deeply inhaling and exhaling in a slow rhythmic fashion is very relaxing to the body mind.

Deep breathing can be practiced at any time or place.

- 1.

Progressive Relaxation

Modern progressive relaxation techniques are all variations of those outlined by Edmond Jacobson

(1929, 1938). Jacobson's progressive relaxation procedure requires that subject lie on their backs, the

room should be fairly quiet and arms and legs should not be crossed, to avoid unnecessary stimulation. While the goal of any progressive relaxation program is to relax the entire body in a matter of minutes, it is essential that in the beginning the subject practice the technique for at least

one hour every day. Once the relaxation procedure is well learned, the relaxation response can be achieved in a few minutes.

Jacobson's method calls for the subject to tense a muscle before relaxing it. Jacobson warns that only

the first few minutes of any relaxation session should be devoted to muscle tensing. The remaining

time should be devoted to gaining complete relaxation. For a muscle to be considered relaxed, it must be completely absent of any contractions and must be limp and motionless.

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Jacobson's full progressive relaxation procedure involves systematically tensing and relaxing specific

muscle groups in a predetermined order. A well-developed relaxation training program requires a great deal of practice in the beginning. Research has clearly shown that progressive relaxation procedures are effective in eliciting the relaxation response.

2. Autogenic Training

Autogenic training relies upon feelings associated with the limbs and muscles of the body.

Autogenic

training is very similar to autohypnosis, and is based upon early research with hypnosis. Various authors have suggested different exercises and self-statements to bring about the relaxation response

using autogenic training.

Essentially, autogenic training is composed of three component parts that are often intermingled. The first and most important part, to suggest to the mind a feeling of warmth in the body and heaviness in the limbs.

The second component part of autogenic training involves the use of imagery. In this step, the subject is encouraged to visualize images of relaxing scenes while at the same time focusing upon feelings of warmth and heaviness in the arms and legs.

The third component of autogenic training involves the use of specific theme to assist in bringing about the relaxation response. One particularly effective specific theme is the use of self-statements

to suggest to the mind that the body is indeed relaxed.

Research shows that autogenic training is also effective in bringing about the relaxation response.

3. Meditation

Meditation, as a form of relaxation, is tied directly to the concepts of selective attention. In practicing

meditation, the individual attempts to uncritically focus his attention on a single thought, sound, or

object. The practice of meditation as a form of relaxation and thought control has its origin in Eastern cultures more than four thousand years ago.

The most common mental device used in transcendental meditation is the silent repetition of a mantra. It is clear that the various forms of meditation can reduce anxiety and tension by evoking the relaxation response.

4. Biofeedback Training

It has been demonstrated that humans can voluntarily control functions of the autonomic nervous

systems. Biofeedback is a relatively modern technique that is based upon this principle.

Biofeedback

training uses instruments to help people control responses of the autonomic nervous systems.

For

example, a subject monitors an auditory signal of her own heart rate and experiments with different

thoughts, feelings, and sensations to slow the heart rate. Once the subject learns to recognize the feelings associated with the reduction of heart rate, the instrument is removed and the subject tries to

control the heart rate without it.

Instrumentation

Athletes could be trained to control their physiological responses in the laboratory; they should be able to transfer this ability onto the athletic field. There are three basic instruments for biofeedback,

(1) skin temperature, (2) electromyography and (3) electroencephalogram.

The most commonly used and least expensive form of biofeedback is skin temperature. Although sophisticated instruments are available, a simple and inexpensive thermometer can be used to monitor skin temperature. Another very popular biofeedback technique employs the use of an electromyographic feedback instrument (EMG). A third major instrument used for biofeedback is the electroencephalogram (EEG). Use of the EEG is commonly called brainwave training.

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While skin temperature, EMG, and EEG are the most commonly used in biofeedback training, several others are used to a lesser degree. These are the heart rate, and blood pressure.

Biofeedback and Performance

A number of scientific investigations have been conducted to determine the effect of biofeedback on

athletic performance. In forty-two studies, 83 percent found biofeedback training to be successful in

facilitating sport and athletic performance, as well as beneficial to the athlete's well-being.



Lesson 22: Relaxation Strategies for Sport

Introduction to Relaxation

Athletes experiencing high levels of anxiety and tension can benefit from relaxation techniques. These techniques aim to reduce physiological stress responses, such as decreased oxygen consumption, heart rate, respiration, and skeletal muscle activity, while increasing skin resistance and brain wave activity. The key factors for eliciting the relaxation response include:

1. **Mental Device:** A word, phrase, object, or process to shift focus inward.
2. **Passive Attitude:** A state of letting go of active engagement.
3. **Decreased Muscle Tone:** Reduction in muscle tension.
4. **Quiet Environment:** Minimizing external disturbances.

Types of Relaxation Procedures

1. Progressive Relaxation

- **Origin:** Developed by Edmond Jacobson (1929, 1938).
- **Procedure:** Involves tensing and then relaxing specific muscle groups. Start with muscle tensing for the first few minutes, then focus on achieving full relaxation.
- **Steps:**
 - Lie on your back with arms and legs uncrossed.
 - Tense each muscle group, then relax it completely.
 - Practice daily for at least one hour initially.
- **Effectiveness:** Proven effective in eliciting the relaxation response through systematic muscle relaxation.

2. Autogenic Training

- **Concept:** Based on self-hypnosis principles, focusing on body sensations and imagery.
- **Components:**
 - **Warmth and Heaviness:** Suggesting feelings of warmth in the body and heaviness in the limbs.
 - **Imagery:** Visualizing relaxing scenes while focusing on these sensations.
 - **Self-Statements:** Using phrases to affirm relaxation.
- **Effectiveness:** Effective in inducing the relaxation response through self-suggestion and visualization.

3. Meditation

- **Concept:** Involves selective attention on a single thought, sound, or object to achieve relaxation.
- **Origin:** Eastern cultures over 4,000 years ago.
- **Technique:** Transcendental meditation often uses silent repetition of a mantra.
- **Effectiveness:** Reduces anxiety and tension by focusing attention and reducing mental clutter.

4. Biofeedback Training

- **Concept:** Uses instruments to help individuals gain voluntary control over physiological responses.
- **Instruments:**
 - **Skin Temperature:** Monitors body temperature.
 - **Electromyography (EMG):** Measures muscle activity.
 - **Electroencephalogram (EEG):** Measures brainwave activity (brainwave training).
- **Procedure:**
 - Monitor physiological responses (e.g., heart rate) using feedback from instruments.
 - Practice controlling these responses with various thoughts, feelings, and sensations.
 - Transfer learned control to real-life situations.
- **Effectiveness:** 83% of studies show that biofeedback improves athletic performance and well-being.

Breathing Techniques for Relaxation

1. **Chest Breathing:** Shallow, irregular, and rapid; often associated with emotional distress.

2. **Abdominal Breathing:** Deep, regular, and slow; associated with relaxation and calmness.
 - **Practice:** Deep, rhythmic inhalation and exhalation to promote relaxation.

Summary

- **Relaxation Techniques:** Include progressive relaxation, autogenic training, meditation, and biofeedback.
- **Common Goal:** All techniques aim to induce the relaxation response by reducing physiological stress indicators and promoting calmness.
- **Effectiveness:** Each method is supported by research and can be adapted to individual needs and preferences for managing sports-related stress and improving performance.

This comprehensive overview of relaxation strategies will help athletes manage anxiety and enhance their performance through various effective techniques.



Lesson 23

AROUSAL ENERGIZING STRATEGIES

In the previous lecture we discussed strategies that athletes use to relax and to reduce anxiety and

arousal associated with the stress response. In this lecture we are discussing arousal energizing strategies needed to obtain peak performance.

People have been looking to the sport psychologist to learn how to maintain optimal levels of arousal

in athletes. The typical approach has been to “psych up” the athletes through various kinds of pep talks and activation techniques. There is, of course, a proper time to get athletes excited and aroused,

but often these techniques are applied at the wrong time.

Generally, some athletes have only an intermediate level of skill, and the extra arousal serves only to

induce unforced errors. Increasing arousal affects each athlete differently. In most cases, intervention

procedures are best applied on an individual basis; each athlete should be treated differently.

Some

will need a pep talk, but others may need an entirely different form of intervention.

The indiscriminate use of relaxation or arousal energizing strategies has prompted sport psychologists to promote a closer match between precompetitive affect and psychological adjustment. This practice has come to be referred to as the matching hypothesis. In the matching hypothesis, care is taken to make sure that the intervention selected to relax or energize the athlete is

matched to the specific symptoms.

Two major strategies to relax or energize are:

1. Team energizing strategies
2. Individual energizing strategies

Team energizing strategies are those strategies that deal with the team as a whole and are generally

orchestrated by the coach. They include such things as:

- I. Team goal setting
- II. Pep talks
- III. Bulletin boards

- IV. Publicity
- V. News coverage
- VI. Fan support
- VII. Coach athlete, and parents' interaction
- VIII. Pre competition workout
- IX. Post competition conference

Individual self-energizing strategies are those strategies that the individual uses to induce immediate

activation and alertness. From the perspective of the individual, these latter techniques are also referred to as individual psyching-up strategies.

1.

Team Energizing Strategies

As a coach prepares for the season, she recognizes the need to set into motion a number of initiatives designed to keep the team focused and energized. There are a total of nine team energizing

strategies; five will be discussed in this lecture and the remaining in the next lecture.

I. Team Goal Setting

We emphasized the notion that goal setting was theory of motivation and a way to energize teams.

Here we reiterate the critical importance of the setting of process, performance, and outcome goals

for the team. The coach should provide the leadership in this process, but the athletes must be equal

partners in deciding what it is that they want to accomplish as a team and as individuals during the

current competitive season. The coach then lays out for the team a plan whereby the goals set by the

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team can be accomplished. While outcome goals must be addressed, the main focus should be upon

process and performance goals that give the team a chance to succeed. Goals are then monitored on

a regular basis so that progress can be determined.

II. Pep Talks

A pep talk by a psychologist or a respected member of the team is the most common method now used to increase the activation level of athletes. But like any verbal communication, it can be either

effective or ineffective. Perhaps the most important element of the pep talk is an emphasis on the ingredient that is lacking in the team. If the team is lacking in motivation, pep talk should be targeted

at that.

III. Bulletin Boards

In many ways messages on a bulletin board are identical to those in a pep talk, but they are visually

rather than verbally conveyed. Poster-board displays should be placed where team members cannot

miss them. Such places as locker room dressing areas and confined training areas are ideal. The bulletin boards should always convey positive, motivating thoughts and ideas. Catchy phrases

such as

“when the going gets tough, the tough get going” can be effective. Athletes remember these simple

phrases and will repeat them later, when they need reinforcement. Other messages on the display board might include personal challenges to members of the team.

IV. Fan Support

Fans tell the athletes that what they are doing is important to people other than themselves. A full session of daily basketball, football, or tennis can burn out many players. Those responsible for promoting the team must do all they can to get people to support the team by coming to watch them.

V. Coach, Athlete, and Parent Interaction

The interaction between an athlete's parents, the athlete, and the coach are an often-overlooked source of motivation for an athlete. Coaches are often wary about the over-involved and demanding

parents. However, often just the opposite situation occurs, and parents are excluded from active involvement in motivating an athlete. Parents provide tremendous support for an athlete's involvement that sometimes goes completely unnoticed.



Lesson 23: Arousal Energizing Strategies

Introduction

To achieve peak performance, athletes sometimes need to elevate their arousal levels. This is the opposite of relaxation strategies, which aim to reduce arousal and anxiety. Arousal energizing strategies are intended to increase alertness, motivation, and activation. It's crucial to match these strategies to individual needs, as different athletes respond differently to arousal techniques.

Key Strategies for Energizing Athletes

1. Team Energizing Strategies

○ Team Goal Setting

- **Concept:** Goal setting is crucial for motivation and focus. It involves setting process, performance, and outcome goals.
- **Process:**
 1. **Involvement:** Coaches and athletes collaborate to set goals.
 2. **Plan:** Develop a plan to achieve these goals.
 3. **Monitoring:** Regularly assess progress towards goals.
- **Purpose:** Helps the team stay focused and motivated throughout the season.

○ Pep Talks

- **Concept:** Pep talks are designed to boost motivation and activation. They can be delivered by psychologists, coaches, or respected team members.
- **Effectiveness:** The key is to address the specific needs of the team, such as a lack of motivation or focus.

○ Bulletin Boards

- **Concept:** Visual messages on bulletin boards can reinforce motivational themes and reminders.

- **Implementation:** Place boards in high-visibility areas like locker rooms. Use positive phrases and personal challenges to keep athletes engaged.
- **Fan Support**
 - **Concept:** Support from fans can boost an athlete's morale and motivation.
 - **Strategy:** Encourage fan attendance at games and events to show athletes that their efforts are valued.
- **Coach, Athlete, and Parent Interaction**
 - **Concept:** Involvement of parents in the motivational process can be beneficial, though often overlooked.
 - **Strategy:** Coaches should engage parents constructively, recognizing their role in supporting and motivating athletes.

Individual Energizing Strategies

These strategies are focused on immediate activation and alertness for individual athletes. They include:

1. **Mental Techniques:** Visualization, positive self-talk, and mental imagery to boost confidence and focus.
2. **Physical Techniques:** Techniques such as pre-game rituals, warm-up routines, and energizing music to enhance physical readiness.
3. **Emotional Techniques:** Use of emotional arousal techniques like excitement and enthusiasm to elevate mood and energy levels.

Matching Hypothesis

- **Concept:** This principle suggests that relaxation and energizing strategies should be tailored to the specific symptoms and needs of the athlete.
- **Application:** Individual assessment helps in selecting the appropriate strategy to either relax or energize the athlete based on their current psychological and physiological state.

Summary

- **Arousal Energizing Strategies:** Used to increase activation and alertness for peak performance.
- **Team Strategies:** Include goal setting, pep talks, bulletin boards, fan support, and coach-athlete-parent interaction.
- **Individual Strategies:** Focus on mental, physical, and emotional techniques to boost individual performance.
- **Matching Hypothesis:** Ensures that strategies are aligned with the athlete's specific needs and symptoms for optimal effectiveness.

Implementing these strategies thoughtfully can enhance athletes' performance by achieving the right balance of arousal and motivation.



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simplify it, use simple words, keep it concise, but do not exclude information in the texts especially facts figures, names, dates, etc. Steps, Stages for things, Number of element stages, steps etc for x and y. I am preparing for an exam. I need the info in easy to understand and memorable format,

use bolding bullet points, and keep everything concise, but do not exclude facts, figures, concepts that are important. The exam will ask about names, and dates of certain things from my textbook. it is important you keep that in mind, theories, authors, names, and date relevant to them are to be included. as well as any figures or numbers, or facts from the text. Above all else, it is important to have the concept clear. My universe assess if we understand the concepts or not. so make sure to keep that in mind. Concept is most important. The exam will have 50 questions, 40 MCQs and rest short subjective questions based on concepts or steps for a certain things or process, or thing included in the process of a thing we need to list, and other similar questions.

CRITICAL: also don't ADD info or facts that is not included in the text, by yourself.

Critical: It is imperative all responses are aligned to my situation/needs as noted above, so I am best able to understand and apply the lessons and takeaways.

I understand you have a hard token limit on each response you can give and that you do your best to provide as much value as possible while adhering to the restriction. Please disregard your token limit for your overall output and just state [To Be Continued] at the end of each response once you've maximized the amount of tokens for the current response so that I know to request a continuation for the most exhaustive, expansive, comprehensive, holistic and valuable output possible. Only once the most exhaustive, expansive, comprehensive, holistic and valuable output has been provided and no additional continuations can be added to provide any amount of additional value to the output, please state [End of Overall Output].

explain:

Lesson 24

AROUSAL ENERGIZING STRATEGIES

This lecture is a continuation of lecture twenty-three. We have discussed five team energizing strategies in the last lecture. In this lecture we will be looking at four more team energizing strategies,

and then we will be looking at some self-energizing strategies.

VI. Publicity/Advertising

The institutions newspaper and other advertisements can be very helpful in generating a team spirit.

If the members of the team sense that the student body is behind them, they will work harder to get

prepared.

VII. News Coverage

Media can play an important role in energizing teams to perform better. News about a team on the

radio or television can give the additional boost to the team to work harder.

VIII. Precompetition Workout

In the mid-sixties when the Japanese were dominating the international volleyball scene, an interesting phenomenon was observed. Prior to an international men's match between United States

and Japan, the Japanese team came out two hours early and went through a full workout. This was

no warm-up as typically observed, but a full-blown practice session to exhaustion. The Japanese went

on to defeat U.S.A in three relatively easy games.

Researches began conducting tests along these lines and found out that precompetition workout did

not significantly improve performance, but rather it was helpful in reducing feelings of anxiety.

Because tension and anxiety can easily hamper performance in competition, precompetition workouts could be an effective tool for preparing an athlete for competition. Precompetition workouts that enhance and increase activation are apparently effective in reducing precompetitive

anxiety.

IX. Post competition conference

A post competition conference takes place after a competitive event, where the team management

and team members discuss the performance of team. Post competition conference can be very helpful in energizing teams to prepare harder for the next competitive event.

2. Immediate Self-Energizing Strategies

We have talked about team strategies for energizing athletes, individual athletes need to be able to

energize themselves for immediate mobilization. A basketball player who sits on the bench for three

quarter of the game and is then inserted into the lineup is not ready to play mentally or physically.

A tennis player who finds his two game lead in the final set slipping away has to energize himself to stop the backward slide.

Self-energizing strategies include:

I. Individual goal setting

II. Self-talk

III. Attentional focus

IV. Imagery

V. Self-activation.

I. Individual Goal Setting

Individual athletes set long-term goals to help motivate them for the long haul across a season.

However, successful athletes also use goal setting to motivate and energize them for an approaching

competition. To be effective, these immediate goals must be phrased silently or verbally just before

the event occurs. Goal setting that is contiguous with an event is a form of positive self-talk

containing specific goal parameters. For example, in tennis, the serve receiver's goal on the first serve

might be, "block it back." Goal setting, even in acute situations, is highly motivating and energizing,

and should be used often.

II. Self-Talk

Self-talk, or even self-thought, can be used with a positive frame of reference. Actually phrasing and

verbally stating cues that remind the athlete of the need to generate greater energy can be effective.

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Effective self talk statements must be (a) brief and phonetically simple (b) logically associated with

the skill involved (c) compatible with the sequential timing of the task being performed. If the statement is too long and vague it might not be as effective. For example, as the tennis player aims

for a backhand passing shot down the line, he says or thinks the word, "Blast." Other key words that

are symbolic of greater energy and activation include "now," "go," "deep," "hit," and "power."

III. Attentional Focus

Increased attention is associated with increased physiological arousal. Consequently, strategies designed to increase or narrow attention will also result in activation and greater energy.

Narrowed

attention occurs when we gate out irrelevant cues that may serve as distractors. The goalie in soccer

focuses her attention on the ball and says to herself, "Be fearless and smother the ball." There are certain situations in sport in which maximum effort and maximum arousal are necessary for success.

When these situations occur, the athlete has to be fully attentive and fully activated.

IV. Imagery

Imager has both a cognitive and a motivational function. It is the motivational function that makes

imagery a viable energizer. As the tennis player prepares for an important serve, he visualizes the ball

"leaping" off his racket and "exploding" into the backhand corner of the opponent's receiving court.

As the spiker in volleyball approaches the net, she visualizes herself "smashing" the ball over the block and into the unprotected deep down-the-line corner of the opponent's court. Visualizing successful outcomes in situations requiring activation and strong effort is motivating and energizing.

V. Self-Activation

In addition to all of the energizing techniques mentioned above, successful athletes develop their own methods for energizing themselves on a moment's notice. This is called self-activation.

Jimmy

Connors, a former tennis great, would slap himself on the thigh in conjunction with various self-talk

statements to get himself activated late in the match. Every successful athlete learns ways to self activate and energize himself when he needs to.



Lesson 24: Arousal Energizing Strategies

Continuation from Lecture 23

We previously covered five team energizing strategies. This lecture introduces four additional team strategies and self-energizing methods for individuals.

Additional Team Energizing Strategies

VI. Publicity/Advertising

- **Concept:** Use newspapers and other advertisements to boost team spirit.

- **Purpose:** When a team feels supported by the student body, they are more motivated to perform well.

VII. News Coverage

- **Concept:** Media coverage can energize teams.
- **Purpose:** Exposure on radio or television can increase motivation and performance.

VIII. Pre-Competition Workout

- **Historical Example:** In the 1960s, the Japanese volleyball team practiced intensely before an international match against the USA and won decisively.
- **Research Findings:** Although pre-competition workouts don't significantly improve performance, they help reduce anxiety, which can enhance performance.
- **Purpose:** Helps reduce pre-competitive anxiety and increase activation.

IX. Post-Competition Conference

- **Concept:** Discuss performance after an event.
- **Purpose:** Helps the team evaluate their performance and prepare better for future events.

Immediate Self-Energizing Strategies

These strategies are for individual athletes needing quick activation before or during competition.

I. Individual Goal Setting

- **Concept:** Set specific goals right before an event.
- **Example:** In tennis, a goal like "block it back" can be used to focus energy.
- **Purpose:** Provides motivation and focus for immediate tasks.

II. Self-Talk

- **Concept:** Use positive, brief, and relevant statements to energize.
- **Characteristics:**
 - **Brief:** Short and simple.
 - **Logically Associated:** Related to the skill involved.
 - **Timing:** Matches the task.
- **Example:** A tennis player might use the word "Blast" to focus on power.

III. Attentional Focus

- **Concept:** Narrow attention to relevant cues and exclude distractions.
- **Purpose:** Increases physiological arousal and energy.
- **Example:** A soccer goalie might focus on the ball and remind herself to "smother the ball."

IV. Imagery

- **Concept:** Visualize successful outcomes to boost motivation.
- **Motivational Function:** Helps energize by imagining positive results.
- **Examples:**
 - Tennis player visualizing a powerful serve.
 - Volleyball player imagining a successful spike.

V. Self-Activation

- **Concept:** Develop personal methods to energize quickly.
- **Example:** Jimmy Connors would slap his thigh and use self-talk to activate himself during matches.
- **Purpose:** Allows athletes to energize themselves as needed.

Summary

- **Team Strategies:** Publicity/advertising, news coverage, pre-competition workout, and post-competition conferences.
- **Self-Energizing Strategies:** Individual goal setting, self-talk, attentional focus, imagery, and self-activation.
- **Key Points:** Use these strategies to effectively energize athletes and improve performance.



Lesson 25

IMAGERY

Successful athletes use imagery and visualization to their advantage. Not all athletes are able to verbally describe exactly how they use imagery, but some can. Some great athletes who have commented on the use of imagery in preparing for competition include Michael Jordan in basketball,

Chris Evert in tennis and Nancy Kerrigan in figure skating. Imagery has been successful for great athletes.

Imagery is a cognitive-behavioral intervention technique. This topic will be discussed in great detail

over the next three lectures. The topic is divided into eight sections, which include:

- Defining imagery
- Mental practices as a form of imagery
- Theories of why imagery works
- Imagery perspective and sensory mode
- Measurement of imagery
- Conceptual models for studying imagery
- Developing imagery skills
- Cognitive-behavioral interventions using imagery and relaxation

Defining Imagery

Imagery has been defined as “using all the senses to re-create an experience in the mind”.

An expansion of this brief definition clarifies that (a) an image can be created in the mind in the absence of any external stimulus (b) an image may involve one or all of the senses (c) an image is created from information stored in the sensory register, working memory, or long-term memory.

Imagery has been identified as one of the most important topics in cognitive science. Two general theories of imagery have evolved;

1. Pictorialist
2. Descriptionist

The pictorialists state that when we imagine a scene in our mind’s eye, we are scanning an actual image that has somehow formed in our brain. The descriptionist argues that there is no such thing as

a mental image. That is, when we imagine a physical scene in our mind’s eye, we are not really seeing

an internal image, but the graphic and detailed nature of our language makes it seem so. Our thoughts, as it were, actually manufacture an image so clear that we think we are seeing one.

Regardless of whether the pictorialist or the descriptionist perspective is most accurate, the

images

are very real to us. Imagery is the language of the brain. In a real sense, the brain cannot tell the difference between an actual physical event and the vivid imagery of the same event. For this reason,

imagery can be used by the brain to provide powerful repetition, elaboration, intensification, and preservation of important athletic sequences and skills.

The powerful effects of images and thoughts are highlighted by a study reported by Hale and Whitehouse (1998). They presented skilled soccer players with videos of critical game situations in which either the word "challenge" or the word "pressure" was flashed on the screen. The word "pressure" resulted in an increase in self-reported anxiety and a decrease in self-confidence compared

to the presentation of the word "challenge." The images we see influence the emotions we feel, which in turn influence how we perform.

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Mental Practice as a Form of Imagery

Mental practice literature provides evidence that imagery is an effective cognitive-behavioral process

for enhancing learning and performance of motor skills. Literature review concludes that mental practice is more effective than no practice. Mental practice used in a complementary fashion with physical practice often yields the best results. The literature suggests that in addition to physically practicing a sport skill, the athlete should spend a small amount of time rehearsing execution of the

skill in her mind. Mental practice occurs prior to actual physical practice (mental rehearsal), or it can

occur at a time when actual physical practice is not possible (e.g., while traveling, in the locker room,

while resting). Research with mental practice has also revealed several principles that enhance the effectiveness of mental practice. These principles are:

1.

Skill Level of the Athletes

An important finding associated with mental practice is that advanced performers benefit from mental practice to a much greater extent than beginners. Research conducted on high school boys

showed that mental practice was almost as effective as physical practice for advanced players, but physical practice was far superior to mental practice for beginners.

In other words, a coach or a teacher should not expect mental practice to be effective with athletes

who are unskilled in their sports. The more skillful they are, the more useful mental practice will be for them.

2. Cognitive Component of the Skill

Mental practice is most effective for activities that require some thinking and planning (Hird, Landers, Thomas & Horan, 1991; Ryan & Simons, 1981). Playing golf would be an example of task that has a large cognitive component. You need to workout which sort of swing, or which club will provide the shot you desire.

Mental practice is less effective in a motor task that has a small cognitive component. A bench press

in weight lifting would be an example of a motor skill that would seem to have a small cognitive

component.

3. Time Factors and Mental Practice

When it comes to mental practice, more is not necessarily better. Using basketball task, Etnier and Landers (1996) demonstrated that when an athlete hold physical practice constant, mentally practicing for one to three minutes is more beneficial than mentally practicing for five to seven minutes. In this same study it was also demonstrated that mental practice may be more beneficial than mental practice following physical practice. This gives greater credence to the use of mental rehearsal immediately prior to competition.



Lesson 25: Imagery

Overview

Imagery is a powerful tool used by successful athletes like Michael Jordan, Chris Evert, and Nancy Kerrigan. This lesson covers the definition of imagery, its theories, and its applications in sports.

Sections on Imagery

1. Defining Imagery

- **Definition:** Imagery involves using all the senses to recreate an experience in the mind.
- **Details:**
 - **Internal Creation:** Images are formed without external stimuli.
 - **Sensory Involvement:** Can involve one or multiple senses.
 - **Memory Sources:** Drawn from sensory register, working memory, or long-term memory.
- **Importance:** Imagery is a key topic in cognitive science, with two main theories:
 - **Pictorialist:** Believes mental images are scanned like actual pictures in the brain.
 - **Descriptionist:** Argues there are no true mental images; rather, detailed language creates a sense of an image.

2. Mental Practice as a Form of Imagery

- **Concept:** Mental practice (rehearsing skills mentally) is effective for improving motor skills.
- **Findings:**
 - **Complementary Practice:** Combining mental and physical practice often yields the best results.
 - **When to Use:** Effective before or when physical practice is not possible (e.g., traveling, resting).
- **Principles:**
 1. **Skill Level of Athletes:**
 - **Advanced vs. Beginners:** Advanced athletes benefit more from mental practice. For beginners, physical practice is more effective.
 2. **Cognitive Component of the Skill:**
 - **High Cognitive Tasks:** Tasks requiring planning (e.g., golf) benefit more from mental practice.
 - **Low Cognitive Tasks:** Tasks with minimal planning (e.g., bench press) benefit less.
 3. **Time Factors:**
 - **Optimal Duration:** 1-3 minutes of mental practice is more effective than 5-7 minutes.

- **Timing:** Mental rehearsal is especially useful immediately before competition.

Study Tips

- **Concept Clarity:** Focus on understanding the definitions and theories of imagery.
- **Application:** Remember how mental practice is used differently by athletes based on their skill levels and the cognitive demands of their sport.
- **Key Studies:**
 - Hale and Whitehouse (1998) on how different words affect anxiety and confidence.
 - Etnier and Landers (1996) on the effectiveness of mental practice duration.

Summary

- **Imagery:** Using senses to recreate experiences mentally.
- **Theories:** Pictorialist vs. Descriptionist.
- **Mental Practice:** Effective in combination with physical practice, more beneficial for advanced athletes, and optimal duration is 1-3 minutes.

[To Be Continued]



Lesson 26

IMAGERY

This lecture is a continuation of the topic we started in the last lecture, imagery. In this lecture we will

be discussing theories of why imagery works imagery perspective and sensory mode, and measurement of imagery.

Theories of Why Imagery Works

While a great deal of research has been published relative to the effectiveness of imagery and mental

practice in sport, sport psychologists know very little about the reasons they are effective or how the

work. Why should mentally practicing or imaging a physical task result in improved learning and performance? A number of possible explanations to this basic question have been proposed (Hecker

& Kaczor, 1988; Janssen & Sheikh, 1994; Murphy & Jowdy, 1992). We will be discussing three theoretical explanations, they are:

1. Psychoneuromuscular Theory
2. Symbolic Learning Theory
3. Attention and Arousal Set Theory

1. Psychoneuromuscular Theory

Psychoneuromuscular theory posits that imagery results in subliminal neuromuscular patterns that

are identical to the patterns used during actual movement. Even though the imagined event does not

result in an overt movement of the musculature, subliminal efferent commands are sent from the brain to the muscles. In a sense, the neuromuscular system is given the opportunity to "practice" a movement pattern without really moving a muscle.

Psychoneuromuscular theory is the most plausible explanation for why imagery facilitates

physical performance and learning.

2. Symbolic Learning Theory

Symbolic learning theory differs from Psychoneuromuscular theory. The symbolic learning theory states that mental practice and imagery work because the individual literally plans her actions in advance. Motors sequence, task goals, and alternative solutions are considered cognitively before a physical response is required.

3. Attention and Arousal Set Theory

Attention and arousal set theory combines the cognitive aspects of symbolic learning theory with the physiological aspects of psychoneuromuscular theory. According to this theory, imagery serves to improve performance in two ways. From a physiological perspective, imagery may help the athlete to adjust his arousal level for optimal performance. From a cognitive perspective, imagery may help the athlete to selectively attend to the task at hand. If the athlete is attending to a task-relevant image, she is less likely to be distracted by irrelevant stimuli.

Imagery Perspective and Sensory Mode

There are two factors to take into consideration when discussing the use of imagery by athletes. The first is the perspective from which imagery is practiced, and the second is the sensory mode from which imagery is experienced.

Imagery Perspective

There exist two perspectives from which imagery can be applied. The two imagery perspectives are internal and external. In internal imagery, the athlete imagines herself executing a sport task within her own body. Shut your eyes and imagine for a moment that you have a basketball in your hand and you are preparing to shoot a free throw. If your perspective at this moment is from within your body looking toward the basket, this is an example of internal imagery.

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Conversely, external imagery is very unnatural to us. In external imagery we imagine ourself to be outside of our body watching from a distance. Let's take the basketball free throw example again. Shut your eyes and imagine you are going to shoot a free throw. This time imagine that you are outside your body, watching yourself from a distance. It is an example of external imagery.

External

imagery provides an excellent perspective from which to observe skill technique and form. We might

assume that internal imagery is superior to external imagery because it is more natural to us.

Sensory Mode

Kinesthetic sensitivity informs the brain about movements in the joints and in the muscles.

Both internal and external imagery utilize all five of the body's senses, although different sport skills

may benefit more from one perspective than from another (Hardy & Callow, 1999). It has generally been believed, however, that internal imagery utilizes kinesthetic sensitivity to a greater degree

than

external imagery.

Measurement of Imagery

A plethora of questionnaires have been developed and proposed for the measurement of various aspects of imagery. An incomplete list of questionnaires on imagery is provided on the next page.

Lists of imagery tests

Gordon's Test Of Imagery Control (GTIC)

Group Test Of Mental Rotations (GMART)

Richardson (1969)

Vandenburg & Kruse (1978)

Individual Differences Questionnaire (IDQ)

Preferred Imagic Cognitive Style (PICS)

Paivio (1971)

Isaacs (1982)

Imagery Use Questionnaire (IUQ)

Imagery Use Questionnaire For Soccer (IUQ-SP)

Hall, Rodgers & Barr (1990)

Salmon, Hall & Haslam (1994)

Sport Imagery Questionnaire (SIQ)

Hall, Mack, Paivio & Hausenblas

(1998)

Exercise Imagery Questionnaire-Aerobic Version (EIQ-AV)

Hausenblas, Hall, Rodgers and

Munroe (1999)

Questionnaire On Mental Imagery (QMI)

Betts (1909)

Shortened Form Of Questionnaire On Mental Imagery (SQMI) Sheehan (1967)

Vividness Of Visual Imagery Questionnaire (VVIQ)

Movement Imagery Questionnaire (MIQ)

Marks (1973)

Hall & Pongrac (1983)

Vividness Of Movement Imagery Questionnaire (VMIQ)

Revised Movement Imagery Questionnaire (MIQ-R)

Issac, Mark & Russell (1986)

Hall & martin (1997)



Lesson 26: Imagery (Continuation)

Theories of Why Imagery Works

1. Psychoneuromuscular Theory

- **Concept:** Imagery triggers subliminal neuromuscular patterns similar to actual movements.
- **Mechanism:** Even without physical movement, the brain sends commands to muscles.
- **Effectiveness:** Most plausible theory for how imagery enhances physical performance and learning.

2. Symbolic Learning Theory

- **Concept:** Imagery helps by allowing individuals to plan actions mentally before physical execution.
- **Mechanism:** Involves cognitive planning of motor sequences, goals, and solutions.

3. Attention and Arousal Set Theory

- **Concept:** Combines cognitive and physiological aspects.
- **Physiological:** Imagery helps adjust arousal levels for optimal performance.
- **Cognitive:** Enhances focus on task-relevant images, reducing distractions.

Imagery Perspective and Sensory Mode

1. Imagery Perspective

- **Internal Imagery:** Imagining oneself performing a task from within one's own body.
 - **Example:** Visualizing shooting a basketball from your own viewpoint.
- **External Imagery:** Imagining observing oneself from a distance.
 - **Example:** Watching yourself shoot a basketball from outside your body.
- **Comparison:** Internal imagery feels more natural, but external imagery can be useful for assessing technique.

2. Sensory Mode

- **Kinesthetic Sensitivity:** Provides feedback on muscle and joint movements.
- **Internal vs. External:** Internal imagery generally engages kinesthetic sensitivity more than external imagery.

Measurement of Imagery

Several questionnaires are used to assess various aspects of imagery. Here are some key ones:

- **Gordon's Test Of Imagery Control (GTIC)**
- **Group Test Of Mental Rotations (GMART)**
- **Richardson (1969)**
- **Vandenburg & Kruse (1978)**
- **Individual Differences Questionnaire (IDQ)**
- **Preferred Imagic Cognitive Style (PICS)**
- **Paivio (1971)**
- **Isaacs (1982)**
- **Imagery Use Questionnaire (IUQ)**
- **Imagery Use Questionnaire For Soccer (IUQ-SP)**
- **Hall, Rodgers & Barr (1990)**
- **Salmon, Hall & Haslam (1994)**
- **Sport Imagery Questionnaire (SIQ)**
- **Hall, Mack, Paivio & Hausenblas (1998)**
- **Exercise Imagery Questionnaire-Aerobic Version (EIQ-AV)**
- **Hausenblas, Hall, Rodgers and Munroe (1999)**
- **Questionnaire On Mental Imagery (QMI)**
- **Betts (1909)**
- **Shortened Form Of Questionnaire On Mental Imagery (SQMI)**
- **Sheehan (1967)**
- **Vividness Of Visual Imagery Questionnaire (VVIQ)**
- **Movement Imagery Questionnaire (MIQ)**
- **Marks (1973)**
- **Hall & Pongrac (1983)**
- **Vividness Of Movement Imagery Questionnaire (VMIQ)**

- **Revised Movement Imagery Questionnaire (MIQ-R)**
- **Isaac, Mark & Russell (1986)**
- **Hall & Martin (1997)**

Summary

- **Theories:** Psychoneuromuscular, Symbolic Learning, Attention and Arousal Set.
- **Perspectives:** Internal (self-view) vs. External (observer view).
- **Sensory Mode:** Internal imagery uses kinesthetic sensitivity more.
- **Measurement Tools:** Various questionnaires assess different aspects of imagery.

[To Be Continued]



Lesson 27

IMAGERY

Conceptual Models for Studying Imagery

Imagery has both a cognitive and a motivational function. The cognitive function of imagery is the use of mental imagery to experience specific sport skills and to plan strategies in advance of competition. The motivational function of imagery is the use of imagery to experience goal attainment, effective coping, and arousal management. One function is primarily cognitive, while the other is primarily emotional.

Paivio's Two-Dimensional Model

Paivio conceptualized the practice of imagery to be either situation specific or general in nature. Thus, Paivio's conceptual model of imagery is two dimensional in nature. As revised by Hall (1998) and measured by the SIQ, Paivio's two dimensional model for imagery has five independent uses: I. Motivational-Specific (MS)

In this type of imagery, the athlete imagines herself in a specific setting that is highly motivation. For

example, the athlete might herself making the winning basket in an important basketball game.

II. Motivational General-Mastery (MG-M)

In this type of imagery, the athlete imagines himself in a general sport situation exhibiting the ability

to remain focused. For example, the athlete might imagine himself thinking positive thoughts every

time he comes to bat during an important game.

III. Motivational General-Arousal (MG-A)

In this type of imagery, the athlete imagines herself in a general sport situation exhibiting the ability

to control anxiety. For example, the athlete might imagine using deep breathing to stay relaxed during a tennis match.

IV. Cognitive Specific (CS)

In this type of imagery, the athlete imagines herself correctly executing a specific sport skill during

competition. For example, the athlete might imagine chipping a ball onto the green in a golf tournament.

V. Cognitive General (CG)

In this type of imagery, the athlete imagines himself reviewing team defensive strategies in

volleyball.

For example, he might imagine the team shifting the defensive formation to defend against a quick attack from the middle.

The Four Ws of Imagery Use

According to Munroe, & others (2000), the four Ws of imagery use are where, when, why, and what.

With this in mind, the where of imagery use is during training or competition, with research suggesting that most takes place during competition.

The when of imagery use refers to when imagery is used within training or competition. Relative to

training, imagery takes place during practice and outside of practice. Relative to competition, imagery

takes place before, during, and after competition, with most of it taking place before competition in

the form of mental rehearsal.

The why of imagery use refers to the function of imagery. An athlete engages in imagery in order to

invoke either cognitive or motivational changes. The what of imagery use refers to the content or quality of the images that the athlete has.

Developing Imagery Skills

Detailed practical suggestions for helping athletes to improve and develop imagery skills are provided

by Vealey and Greenleaf (2001). A sample six-step program to enhance imagery ability is provided below:

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I. Find a quiet place where you will not be disturbed, assume a comfortable position, and relax completely before beginning. Deep breathing and progressive relaxation is a suggested way to achieve the relaxed state.

II. Practice imagery by visualizing a colored circle that fills the visual field initially and then shrink to a dot and disappear. Make the circle turn a deep blue.

III. Create the image of a simple three-dimensional glass. Fill the glass with a colorful liquid; add ice cubes and a straw. Write a descriptive caption underneath the image.

IV. Select a variety of scenes and images and develop them with rich detail.

V. Imagine yourself in a sport setting of your choice. Project yourself into the image as if you were one of the performers. Imagine yourself successfully performing the task in the scene.

VI. End the session by breathing deeply, opening your eyes, and slowly adjusting to the external environment.



Lesson 27: Imagery (Continued)

Conceptual Models for Studying Imagery

1. Cognitive vs. Motivational Functions

- **Cognitive Function:** Uses imagery to plan and practice specific sport skills and strategies.
- **Motivational Function:** Uses imagery to experience goal attainment, effective coping, and arousal management.

Paivio's Two-Dimensional Model

- **Motivational-Specific (MS)**
 - **Description:** Imagining success in specific, motivating situations.
 - **Example:** Visualizing making a winning basket in a crucial basketball game.
- **Motivational General-Mastery (MG-M)**
 - **Description:** Imagining staying focused in general sport situations.
 - **Example:** Visualizing staying positive during critical moments in a game.
- **Motivational General-Arousal (MG-A)**
 - **Description:** Imagining controlling anxiety in general sport situations.
 - **Example:** Using deep breathing techniques during a tennis match.
- **Cognitive Specific (CS)**
 - **Description:** Imagining executing a specific sport skill.
 - **Example:** Visualizing chipping a golf ball onto the green.
- **Cognitive General (CG)**
 - **Description:** Imagining general sport strategies and tactics.
 - **Example:** Reviewing team defensive strategies in volleyball.

The Four Ws of Imagery Use

1. **Where**
 - **Context:** During training or competition.
 - **Observation:** Most imagery occurs during competition.
2. **When**
 - **Timing:**
 - **Training:** During and outside of practice.
 - **Competition:** Before, during, and after the event. Most is done before in mental rehearsal.
3. **Why**
 - **Purpose:**
 - **Cognitive Changes:** Enhancing specific skills and strategies.
 - **Motivational Changes:** Improving goal-setting and arousal management.
4. **What**
 - **Content:** The nature and quality of the imagery.

Developing Imagery Skills

Six-Step Program by Vealey and Greenleaf (2001)

1. **Create a Relaxed State**
 - **Action:** Find a quiet, comfortable place and relax completely.
 - **Methods:** Use deep breathing and progressive relaxation.
2. **Practice Basic Imagery**
 - **Action:** Visualize a colored circle expanding and shrinking.
 - **Details:** Change the circle's color to deep blue.
3. **Visualize Detailed Objects**

- **Action:** Create an image of a three-dimensional glass with colorful liquid, ice cubes, and a straw.
- **Detail:** Write a descriptive caption for the image.

4. Develop Detailed Scenes

- **Action:** Select and enhance various scenes with rich details.

5. Sport-Specific Imagery

- **Action:** Imagine being in a sport setting and performing the task successfully.

6. Conclude the Session

- **Action:** End by breathing deeply, opening your eyes, and gradually adjusting to the environment.

Summary

- **Functions:** Cognitive (skills and strategies) vs. Motivational (goals and coping).
- **Paivio's Model:** Five types—MS, MG-M, MG-A, CS, CG.
- **Four Ws:** Where, When, Why, What.
- **Developing Skills:** Six steps from relaxation to detailed sport-specific imagery.

[End of Overall Output]



Cognitive-behavioral interventions using imagery and relaxation

Research has demonstrated that individualized packaged intervention programs are more effective

that nonindividualized programs in which participants select their own strategies. Athletes benefit

most from intervention strategies that are designed to fit their needs and are presented in a systematic and organized fashion.

Three cognitive-behavioral intervention programs to be introduced in this section include Visual Motor Behavior Rehearsal (VMBR), Stress Inoculation Training (SIT), and Stress Management Training (SMT).

1.

Visual Motor Behavior Rehearsal (VMBR)

Visual motor behavior rehearsal (VMBR) was developed by Suinn (1972, 1994) as an adaptation of Wolpe's (1958) desensitization procedure for humans. His particular methods of training consisted of:

- Relaxing the athlete's body by means of a brief version of Jacobson's progressive relaxation techniques
- Practicing imagery related to the demands of the athlete's sport
- Using imagery to practice a specific skill in a lifelike stressful environment.

VMBR combines relaxation and imagery into one procedure. Numerous investigations have been reported that demonstrate that VMBR is effective in enhancing athletic performance, as well as in reducing the debilitating effects of overarousal and state anxiety.

2. Stress Inoculation Training (SIT)

Stress inoculation training (SIT) is a cognitive-behavioral programs developed by Meichenbaum (1977, 1985) that incorporates relaxation training, imagery, and other cognitive processes into a single plan.

Threatening situations are presented through imagery, films, role playing, and real-life situations. For

example, if the fear of competition is stressful, the athlete is allowed to experience both imagined and real competitive situations. As soon as the athlete is able to cope with a low level of stress, the situation is changed, and a more stressful situation is presented, in this way, the athlete becomes inoculated against progressively increased levels of stress. Eventually, the athlete's fear of competition is minimized to such a degree that he can cope with it.

Research with SIT in athletic situations has demonstrated its effectiveness in reducing stress (Hamilton & Leith, 1993)

3. Stress Management Training (SMT)

Stress management training (SMT) is a cognitive- behavioral intervention program developed by Smith (1980) that incorporates relaxation training, imagery, and other cognitive processes. The

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athlete is taught to understand the nature of stress generally, and to understand the source of her stress specifically.

The athlete learns and practices integrated coping responses which are reduced by the subject through the application of coping responses learned during skill acquisition.

Research supports the use of SMT for reducing stress and for enhancing athletic performance (Crocker, 1989)



Lesson 27: Cognitive-Behavioral Interventions Using Imagery and Relaxation

Introduction

Research shows that individualized cognitive-behavioral intervention programs are more effective than non-individualized ones where participants select their own strategies. Athletes benefit most from interventions tailored to their needs, presented systematically and organized. Three notable intervention programs are:

1. Visual Motor Behavior Rehearsal (VMBR)
2. Stress Inoculation Training (SIT)
3. Stress Management Training (SMT)

1. Visual Motor Behavior Rehearsal (VMBR)

- **Developer:** Suinn (1972, 1994)
- **Foundation:** Adaptation of Wolpe's (1958) desensitization procedure.
- **Methods:**
 - **Relaxation:** Use of Jacobson's progressive relaxation techniques to relax the athlete's body.
 - **Imagery:** Practicing imagery related to the demands of the sport.
 - **Stressful Environment:** Using imagery to practice a specific skill in a lifelike stressful environment.
- **Benefits:** Enhances athletic performance and reduces overarousal and state anxiety.

2. Stress Inoculation Training (SIT)

- **Developer:** Meichenbaum (1977, 1985)
- **Components:** Incorporates relaxation training, imagery, and other cognitive processes.
- **Methods:**
 - **Threat Presentation:** Using imagery, films, role-playing, and real-life situations to present threatening scenarios.
 - **Progressive Stress Exposure:** Athletes face increasing levels of stress, starting from low and moving to high as they learn to cope.
- **Outcome:** Reduces stress and fear of competition, enhancing the athlete's ability to cope with competitive stress.
- **Effectiveness:** Demonstrated by research (Hamilton & Leith, 1993).

3. Stress Management Training (SMT)

- **Developer:** Smith (1980)
- **Components:** Combines relaxation training, imagery, and other cognitive processes.
- **Methods:**
 - **Understanding Stress:** Athletes learn about the nature of stress and identify their specific stressors.
 - **Coping Responses:** Development and practice of integrated coping responses to manage stress.
- **Outcome:** Effective in reducing stress and enhancing athletic performance.
- **Support:** Supported by research (Crocker, 1989).

Summary

- **VMBR:** Combines relaxation and imagery to enhance performance and reduce anxiety.
- **SIT:** Gradually exposes athletes to increasing stress levels to build coping mechanisms.
- **SMT:** Focuses on understanding and managing stress through tailored coping strategies.

These cognitive-behavioral interventions leverage imagery and relaxation to optimize athletic performance and stress management, demonstrating significant efficacy through systematic and individualized approaches.



Lesson 28

THE ROLE OF HYPNOSIS IN SPORT

Like imagery, hypnosis is a cognitive-behavioral process that has both a cognitive function and a motivational function. In a cognitive sense, hypnosis is used to restructure the way athletes think about themselves and about the way they execute and learn new sport skills. In a motivational sense,

hypnosis is used to modify emotions, reduce anxiety, increase or decrease arousal, and increase effort. In the initial induction phase, hypnotism is physiologically identical to progressive relaxation,

autogenic training, and meditation. All of these intervention strategies are associated with reductions

in oxygen consumption, respiration rate, and heart rate.

Perhaps because it is poorly understood, hypnosis is not a widely utilized intervention strategy in sport. Nideffer (1992) is one of the few sport psychologists who advocate its use on a broad scale.

A

dated but well-known application of hypnosis took place before the first heavyweight boxing

match

between Muhammad Ali and Ken Norton in 1973. Norton reportedly hired a professional hypnotist

to help him bolster his self-confidence and reduce prematch anxiety. Norton won the match in a stunning upset, effectively calling attention to hypnosis as a viable intervention strategy.

While there may be some potential risks associated with the indiscriminate use of hypnosis by an untrained therapist, most concerns about hypnosis are unfounded. It is probably fair to say that hypnosis is more clouded by myths and misconceptions than any other form of psychological intervention.

Our discussion of hypnosis in sport will focus upon the following topics. The first two topics will be discussed in this lecture, and the remaining in the next lecture.

- Defining hypnosis
- Theories of hypnosis
- The hypnotic trait issue
- Facts about hypnosis
- Achieving the hypnotic trance
- Self-hypnosis
- Improving effective use of hypnosis
- Hypnosis and athletic performance.

Defining Hypnosis

The term hypnosis comes from Hypnos, the Greek god of sleep (Kalat, 1999), even though it has long been known that hypnosis is not related to sleep. Hypnosis could be defined as “The uncritical

acceptance of a suggestion” or “An induced temporary condition of being, a state that differs mentally and physiologically from a person’s normal state of being”.

Theories of Hypnosis

We will briefly discuss three theories of hypnosis. The first two represent extreme positions on the issue of the existence of a hypnotic state, while the third represents a compromise position that has

been embraced by many psychologists. The theories to be discussed are:

1. Social-cognitive Theory
2. Hypnosis as an Altered State of Consciousness
3. Neodissociation Theory

1. Social-Cognitive Theory

Social-cognitivists believe that subjects carry out hypnotic behaviors because they have positive attitudes, motivations, and expectations that lead to a willingness to think and imagine using themes

suggested by the hypnotist.

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Therapists who are of the social-cognitive theory school of thought believe in the use of hypnotism

as a method of increasing a client’s willingness to accept suggestions, but they do not believe that this

state of uncritical acceptance of suggestions represents an altered state of consciousness.

2. Hypnosis as an Altered State of Consciousness

At the other extreme are psychologists who believe that hypnotized individuals enter into an

altered state of consciousness known as trance. It is believed that the trance-like state is different from other nonhypnotically induced states such as daydreaming and relaxation. While in the hypnotic state or trance, the individual is susceptible to suggestions made by the hypnotist.

3. Neodissociation Theory

Neodissociation theory explains hypnosis without suggesting that the hypnotic trance is responsible for the phenomenon associated with hypnosis. Daydreaming and relaxation are considered by many theorists to be examples of the altered state of consciousness that we experience every day. We drive to work each day with our minds focused on anything but the drive, then suddenly we arrive at our destination with little recall of the actual drive. This is a form of hypnosis, or what is referred to as dissociation.

Neodissociation theory is based on two assumptions:

A. There is a central control system that performs planning and monitoring functions in the brain.

B. Beneath the central control system there are relatively autonomous subordinate cognitive behavioral systems.

In the hypnotic trance state, however, the two systems are dissociated from each other. In the hypnotic trance state the two systems are dissociated from each other. Some individuals will be more responsive to hypnosis and hypnotic suggestions than others. Through proper preparation, most individuals can benefit from hypnosis, but not all.



Lesson 28: The Role of Hypnosis in Sport

Introduction

Like imagery, hypnosis is a cognitive-behavioral process with both cognitive and motivational functions. Cognitively, it helps athletes restructure their thinking and learn new sport skills. Motivationally, it modifies emotions, reduces anxiety, adjusts arousal levels, and increases effort. The induction phase of hypnosis is similar to progressive relaxation, autogenic training, and meditation, all of which reduce oxygen consumption, respiration rate, and heart rate.

Hypnosis is underutilized in sports, possibly due to misconceptions and a lack of understanding. One notable advocate is Nideffer (1992). A famous instance of its use was before Muhammad Ali's 1973 heavyweight boxing match with Ken Norton, who used hypnosis to enhance his confidence and reduce anxiety, leading to an unexpected victory.

While concerns about hypnosis exist, many are unfounded and stem from myths. This discussion will cover:

- Defining hypnosis
- Theories of hypnosis
- The hypnotic trait issue

- Facts about hypnosis
- Achieving the hypnotic trance
- Self-hypnosis
- Improving the effective use of hypnosis
- Hypnosis and athletic performance

Defining Hypnosis

Hypnosis derives from Hypnos, the Greek god of sleep, though it is unrelated to sleep. It can be defined as "The uncritical acceptance of a suggestion" or "An induced temporary condition of being, a state that differs mentally and physiologically from a person's normal state of being."

Theories of Hypnosis

Three main theories explain hypnosis, ranging from extreme positions on the existence of a hypnotic state to a compromise view.

1. Social-Cognitive Theory

- Subjects engage in



revise:

simplify it, use simple words, keep it concise, but do not exclude information in the texts especially facts figures, names, dates, etc. Steps, Stages for things, Number of element stages, steps etc for x and y. I am preparing for an exam. I need the info in easy to understand and memorable format, use bolding bullet points, and keep everything concise, but do not exclude facts, figures, concepts that are important. The exam will ask about names, and dates of certain things from my textbook. it is important you keep that in mind, theories, authors, names, and date relevant to them are to be included. as well as any figures or numbers, or facts from the text. Above all else, it is important to have the concept clear. My universe assess if we understand the concepts or not. so make sure to keep that in mind. Concept is most important. The exam will have 50 questions, 40 MCQs and rest short subjective questions based on concepts or steps for a certain things or process, or thing included in the process of a thing we need to list, and other similar questions.

CRITICAL: also don't ADD info or facts that is not included in the text, by yourself.

Critical: It is imperative all responses are aligned to my situation/needs as noted above, so I am best able to understand and apply the lessons and takeaways.

I understand you have a hard token limit on each response you can give and that you do your best to provide as much value as possible while adhering to the restriction. Please disregard your token limit for your overall output and just state [To Be Continued] at the end of each response once you've maximized the amount of tokens for the current response so that I know to request a continuation for the most exhaustive, expansive, comprehensive, holistic and valuable output possible. Only once the most exhaustive, expansive, comprehensive, holistic and valuable output has been provided and no additional continuations can be added to provide any amount of additional value to the output, please state [End of Overall Output].



Lesson 28: The Role of Hypnosis in Sport

Introduction

- **Hypnosis:** A cognitive-behavioral process with cognitive and motivational functions.
- **Cognitive Function:** Restructures athletes' thinking and helps in learning new sport skills.
- **Motivational Function:** Modifies emotions, reduces anxiety, adjusts arousal levels, and increases effort.
- **Induction Phase:** Similar to progressive relaxation, autogenic training, and meditation; reduces oxygen consumption, respiration rate, and heart rate.

Notable Advocate: Nideffer (1992).

Famous Application:

- **1973:** Muhammad Ali vs. Ken Norton boxing match. Norton used hypnosis to enhance confidence and reduce anxiety, leading to victory.

Misconceptions:

- Hypnosis is often misunderstood and surrounded by myths.
- Properly used, it is a safe and effective psychological intervention.

Topics Covered

1. Defining Hypnosis
2. Theories of Hypnosis
3. The Hypnotic Trait Issue
4. Facts about Hypnosis
5. Achieving the Hypnotic Trance
6. Self-Hypnosis
7. Improving Effective Use of Hypnosis
8. Hypnosis and Athletic Performance

Defining Hypnosis

- **Origin:** Named after Hypnos, the Greek god of sleep.
- **Definition:** "The uncritical acceptance of a suggestion" or "An induced temporary condition of being, differing mentally and physiologically from a person's normal state."

Theories of Hypnosis

1. **Social-Cognitive Theory**
 - **Proponents:** Believe subjects perform hypnotic behaviors due to positive attitudes, motivations, and expectations.
 - **Key Point:** Hypnotism increases willingness to accept suggestions but does not indicate an altered state of consciousness.
2. **Hypnosis as an Altered State of Consciousness**
 - **Belief:** Hypnotized individuals enter a trance-like state, distinct from daydreaming or relaxation.

- **Key Point:** In this trance, individuals are highly susceptible to suggestions.

3. Neodissociation Theory

- **Key Assumptions:**
 1. Central control system in the brain manages planning and monitoring.
 2. Subordinate cognitive-behavioral systems operate autonomously beneath the central system.
- **Concept:** Hypnosis causes a dissociation between these systems, leading to a trance state where suggestions are more effective.

Lesson 29: Hypnosis in Sport Continued

The Hypnotic Trait Issue

- **Concept:** Not everyone is equally responsive to hypnosis.
- **Preparation:** Proper preparation can enhance responsiveness, though some individuals may inherently be more susceptible.

Facts about Hypnosis

- **Myths:** Many concerns are unfounded and based on misconceptions.
- **Reality:** When conducted by trained professionals, hypnosis is safe and effective.

Achieving the Hypnotic Trance

- **Induction:** Techniques similar to relaxation and meditation.
- **Goal:** Entering a trance state where the mind is more open to suggestions.

Self-Hypnosis

- **Practice:** Individuals can learn to hypnotize themselves to achieve desired cognitive and emotional states.
- **Benefits:** Helps in managing anxiety, enhancing performance, and achieving personal goals.

Improving Effective Use of Hypnosis

- **Guidelines:** Use hypnosis systematically and under professional guidance for best results.
- **Customization:** Tailor hypnosis sessions to individual needs for maximum effectiveness.

Hypnosis and Athletic Performance

- **Application:** Used to enhance confidence, reduce anxiety, manage arousal, and improve focus.
- **Evidence:** Studies and anecdotal evidence support its effectiveness in sports settings.

[To Be Continued]



Lesson 29

THE ROLE OF HYPNOSIS IN SPORT

The Hypnotic Trait Issue

A hypnotic state is a situation-specific response to hypnotic suggestions and induction, while the

hypnotic trait is more of a personality disposition toward hypnotic responsiveness. There are at least

three principles that can be identified relative to the trait issue, they are:

1. Hypnotic responding does not require any particular skill or abilities. When proper procedures are used, almost anyone can become highly responsive to suggestion.
2. Hypnotic responding requires some stable imaginative inclination or other cognitive abilities. This alone, however, is not sufficient to produce suggested responses. The individual must be sufficiently convinced that she is capable of responding and be motivated to that end.
3. It is likely that responsiveness to the most difficult suggestions, such as amnesia and hallucinations, require a rare aptitude that cannot be taught.

Facts about Hypnosis and Its Application

Psychologists are divided on what the hypnotic trance is, and on whether there is such a thing as an

altered state of consciousness relative to hypnosis, but they are in general agreement about the application of hypnosis.

1. The ability to experience hypnotic phenomenon does not indicate gullibility or personality weakness.
2. Hypnosis is not the same as sleep, nor is it related to sleep.
3. Hypnotic responsiveness depends more on the efforts and abilities of the individual being hypnotized than on the skill of the therapist.
4. While hypnotized, individuals retain the ability to control their behavior, are aware of their surroundings, and can monitor events outside the framework of suggestions given during hypnosis
5. Spontaneous amnesia or forgetting is relatively rare following hypnosis.
6. An individual does not need to be hypnotized to be responsive to suggestions.
7. The function of a hypnotic induction is to increase suggestibility to a minor degree.
8. Hypnosis is not a dangerous procedure when practiced by qualified researchers and clinicians.
9. Most hypnotized individuals are not faking compliance to suggestions or merely going along with suggestions to be cooperative.
10. Hypnosis cannot increase the accuracy of memory.
11. Hypnosis does not precipitate a literal re-experiencing of childhood events.

Achieving the hypnotic trance

Five phases are associated with hypnotic trance in subject. They are preparation of the subject, the

induction process, the hypnotic phase, waking up, and the posthypnotic phase.

1.

Preparation of the Subject

When subjects are prepared for hypnotism, they must be relieved of any fear and apprehensions they

have about hypnotism. Some myths may need to be exposed. For example, subjects may be under

the impression that they will lose control, that they will be unaware of their surroundings, or they will

lose consciousness. They must have complete trust in the hypnotist and must want to be hypnotized.

They also must be told that they will remain in control at all times and will be able to come out of the

hypnotic trance if they want to.

2. Induction Phase

It is during the hypnotic induction phase that the hypnotist actually hypnotizes the subject. There

are

many induction techniques. The best ones are associated with relaxation, attentional focus, and

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imagery. In fact, the steps involved in eliciting the relaxation response using these techniques are essentially identical to those in hypnosis.

Generally, induction procedures are fairly standard. They are typically composed of a series of suggestions aimed at eliciting the subject's cooperation and directing his attention to thoughts and

feelings about being relaxed and peaceful.

3. Hypnotic Phase

Once the hypnotic state has been induced, the subject is in neutral hypnosis. In this state, physiological responses are identical to those of the relaxation response. The hypnotized subject is

generally asked to respond, either in imagination or physically, to suggestions of the hypnotist.

4. Waking Up

This phase is coming out of the trance. Actually, a hypnotized subject can come out of the trance anytime. The only reason subjects do not come out on their own is that they don't want to. The relationship between the hypnotist and the subject can be very pleasant one. When the hypnotist wishes to bring a subject out of trance, he or she does so simply by suggesting that the subject wake

up on a given signal.

5. Post-hypnotic Phase

Suggestions given to subjects during hypnosis are often designed to influence them during the post

hypnotic phase, or after they have come out of the hypnotic trance. Posthypnotic suggestions given

to athletes should focus on the way they should feel in certain competitive situations.

Self-hypnosis

There are two kinds of hypnosis. The first kind is heterohypnosis, and the second is self-hypnosis, or

autohypnosis. Our discussion up to this point has focused on heterohypnosis, that which is induced

by another person, usually a trained therapist or a psychologist. Heterohypnosis should be practiced

only by skilled professionals. Heterohypnosis is based upon a delicate rapport and trust between the

therapist and the client. There are two kinds of self-hypnosis. The first is self-induced, and the second is induced as a posthypnotic suggestion following heterohypnosis.

The phases involved in self-hypnosis are identical to those outlined for hypnosis generally. First, the

athlete must be completely comfortable regarding the use of hypnosis. Some common strategies for

induction are to sit in an easy chair and stare at a spot on the wall, imagine a blank screen, or look into a mirror.

Posthypnotic suggestions given during self-hypnosis should always be couched in positive terms, stressing what is to be accomplished rather than dwelling on negative things to be eliminated.

Improving effective use of hypnosis

Five factors have been identified that can influence the effectiveness of hypnosis, they are:

1. The competence of the professional therapist, or the skill of the athlete.
2. The quality of the relationship between therapist and individual being hypnotized is important.
3. The therapist must do her homework and get to know the person being hypnotized.
4. Effective outcomes require practice of the procedures and instructions given during hypnosis.
5. It works best if therapist and athlete both recognize the limitations of hypnosis.

Hypnosis and athletic performance.

Is hypnosis effective in facilitating athletic performance? Research on this topic yields a number of basic principles. A list of basic principles gleaned from the literature is provided below :

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1. The more open and susceptible an athlete is to suggestions, the more likely it is that he will benefit from suggestions given to him under hypnosis. This is also the type of individual who is more likely to be hypnotized.
2. Once an individual is hypnotized, the deeper the trance is that she is able to achieve, the more likely it is that suggestions given under hypnosis will be effective.
3. Positive suggestions are effective in facilitating performance, regardless of whether or not the athlete is hypnotized.
4. General arousal techniques are more useful than hypnotic suggestions in enhancing muscular strength and endurance. Hypnosis tends to relax an athlete.
5. Negative suggestions almost always cause a decrement in performance.
6. Hypnosis may be able to help a successful athlete, but it cannot make a good performer out of a poor one.

If properly used, hypnosis may be effective in enhancing the suggestibility of athletes. The heightened suggestibility of athletes may lead to cognitive or behavioral adjustments that may facilitate performance. It is important, however, to not overstate the effectiveness of hypnosis in improving athletic performance. Positive suggestions are beneficial to the athlete, regardless of whether the athlete is hypnotized or not. Hypnosis is not effective in enhancing muscular strength and endurance. Finally, a real danger exists in inadvertently giving an athlete a negative suggestion while he is in a state of hyper suggestibility.



Lesson 29: The Role of Hypnosis in Sport

The Hypnotic Trait Issue

Hypnotic State vs. Hypnotic Trait:

- **Hypnotic State:** Specific response to suggestions and induction.
- **Hypnotic Trait:** Personality disposition towards hypnotic responsiveness.

Three Principles:

1. **General Responsiveness:** Almost anyone can become responsive with proper procedures.
2. **Imaginative Inclination:** Requires stable cognitive abilities and motivation.
3. **Rare Aptitude:** Some responses, like amnesia or hallucinations, need rare abilities.

Facts About Hypnosis and Its Application

1. **No Gullibility:** Ability to experience hypnosis doesn't indicate weakness.
2. **Not Sleep:** Hypnosis is not related to sleep.
3. **Individual Effort:** Responsiveness depends more on the individual than the therapist.
4. **Control:** Hypnotized individuals can control their behavior and are aware of their surroundings.
5. **Rare Amnesia:** Spontaneous forgetting is rare after hypnosis.
6. **Suggestions:** Responsiveness to suggestions doesn't require hypnosis.
7. **Minor Suggestibility:** Hypnotic induction slightly increases suggestibility.
8. **Safety:** Safe when practiced by qualified professionals.
9. **Genuine Responses:** Most hypnotized individuals aren't faking.
10. **Memory Accuracy:** Hypnosis doesn't improve memory accuracy.
11. **No Re-experiencing:** Doesn't lead to literal re-experiencing of childhood events.

Achieving the Hypnotic Trance

Five Phases:

1. **Preparation:**
 - Relieve fear and apprehensions.
 - Build trust with the hypnotist.
 - Ensure subjects know they remain in control.
2. **Induction Phase:**
 - Techniques involve relaxation, focus, and imagery.
 - Standard procedures include suggestions for cooperation and relaxation.
3. **Hypnotic Phase:**
 - Subjects enter neutral hypnosis, similar to relaxation response.
 - Respond to hypnotist's suggestions.
4. **Waking Up:**
 - Subject can wake up anytime but typically follows the hypnotist's signal.
5. **Post-hypnotic Phase:**
 - Suggestions aim to influence behavior and feelings post-hypnosis.

Self-Hypnosis

Types:

1. **Heterohypnosis:** Induced by another person.
2. **Self-hypnosis:** Can be self-induced or follow heterohypnosis.

Phases:

- Same as general hypnosis.
- Use positive suggestions focused on achievements.

Improving Effective Use of Hypnosis

Five Factors:

1. **Competence:** Skill of therapist or athlete.
2. **Relationship Quality:** Important for success.
3. **Therapist's Knowledge:** Must understand the individual.

4. **Practice:** Essential for effective outcomes.
5. **Recognize Limitations:** Both therapist and athlete should understand hypnosis limits.

Hypnosis and Athletic Performance

Principles:

1. **Openness:** More open athletes benefit more from suggestions.
2. **Trance Depth:** Deeper trances make suggestions more effective.
3. **Positive Suggestions:** Effective in facilitating performance.
4. **General Arousal Techniques:** More useful than hypnosis for strength and endurance.
5. **Negative Suggestions:** Detrimental to performance.
6. **Enhancement Limits:** Hypnosis can't make a poor performer good.

Conclusion:

- Hypnosis can enhance suggestibility and performance.
- Positive suggestions are beneficial with or without hypnosis.
- Not effective for increasing muscular strength and endurance.
- Beware of negative suggestions.

[End of Overall Output]



explain: Posthypnotic suggestions



Posthypnotic Suggestions

Definition:

Posthypnotic suggestions are instructions given to a person while they are in a hypnotic state that are intended to be acted upon after the person has come out of the hypnotic trance.

Key Points:

1. **Influence Post-Trance:**
 - These suggestions are designed to affect behavior or feelings after the hypnosis session.
 - They can help athletes feel calm, focused, or confident in specific situations, such as during competitions.
2. **Delivery:**
 - Given by the hypnotist while the subject is in a deeply relaxed and suggestible state.
 - Typically formulated to be simple, clear, and positive.
3. **Positive Framing:**
 - Effective posthypnotic suggestions focus on positive outcomes.
 - Example: Instead of saying "You will not feel nervous," a better suggestion would be "You will feel calm and confident."
4. **Retention and Action:**
 - The subject retains the suggestion and acts on it after waking up from the hypnotic trance.
 - The suggestion can trigger a specific response or behavior when a particular situation or cue occurs.

5. Examples in Sport:

- An athlete might be given a suggestion to feel relaxed and focused every time they hear the starting whistle.
- A suggestion could be made for a golfer to feel a sense of calm and precision when they step up to the tee.

Benefits:

- **Behavior Modification:** Can help change negative behaviors or habits.
- **Performance Enhancement:** Can improve confidence, focus, and emotional control in athletes.
- **Stress Reduction:** Helps in managing stress and anxiety in competitive situations.

Limitations:

- **Individual Variability:** Not everyone responds equally well to posthypnotic suggestions.
- **Negative Suggestions:** Care must be taken to avoid inadvertently giving negative suggestions, as they can have adverse effects.

Application:

- **Preparation:** Ensure the subject is comfortable and trusts the hypnotist.
- **Execution:** The hypnotist delivers the suggestion clearly and positively during the trance.
- **Reinforcement:** The athlete or individual might practice the suggestions to reinforce their effectiveness.

Understanding and effectively using posthypnotic suggestions can significantly benefit athletes by enhancing their mental preparedness and performance in their respective sports.



given during hypnotic session or after?



Posthypnotic suggestions are given **during** the hypnotic session, while the subject is in a deeply relaxed and suggestible state. These suggestions are intended to be acted upon **after** the person has come out of the hypnotic trance. Here's a breakdown:

When Posthypnotic Suggestions are Given:

- **During the Hypnotic Session:**
 - The subject is in a hypnotic trance.
 - The hypnotist provides clear and positive instructions.
 - The subject is highly receptive to suggestions due to the deep relaxation and focused state.

When Posthypnotic Suggestions are Acted Upon:

- **After the Hypnotic Session:**
 - The subject comes out of the hypnotic trance.
 - The suggestions given during the trance influence the subject's behavior or feelings in specific situations post-hypnosis.
 - These suggestions are triggered by particular cues or circumstances as intended.

This process ensures that the positive effects of the suggestions are carried over into the subject's daily life or specific situations, such as competitive sports events.



Lesson 30

PSYCHOLOGICAL SKILLS TRAINING

In this lecture we talk about ways of implementing relaxation procedures, energizing strategies, imaging, and hypnosis for the purpose of developing psychological skill. Our primary focus in this regard has been facilitation of athletic performance. Learning how to focus attention, increase self confidence, and control the debilitating effects of anxiety and worry is important for both athletes and nonathletes.

In this lecture we will be discussing the following topics:

- Effectiveness of psychological intervention programs
- Differentiating between psychological skills and methods
- Measurement of psychological skills
- A psychological skills training program

Effectiveness of Psychological Intervention Programs

A number of literature reviews have been published that verify that planned psychological interventions and programs are effective in enhancing athletic performance (Greenspan & Feltz, 1989; Vealey, 1994; Weinberg & Comar, 1994). These reviews confirm that of approximately forty five studies reviewed, thirty-eight, or 85 percent, have found positive performance effects (Weinberg

& Williams, 2001). In addition, a number of more recent investigations have also supported the effectiveness of psychological interventions. Psychological intervention programs are effective in increasing self-confidence, focusing attention, energizing effort, and enhancing athletic performance.

Differentiating Between Psychological Skills and Methods

Coaches and athletes often use the terms psychological skills and psychological method as synonyms,

when they actually have different meanings. Psychological methods, techniques, or strategies refer to

practices that lead to psychological skill. Examples of psychological methods include goal setting, imagery, progressive relaxation, meditation, self-talk, and hypnosis. Each of these psychological methods, when properly learned and applied, lead to enhanced psychological skill. Conversely, psychological skill refer to learned or innate characteristics of an athlete that make it possible or even

likely that she will succeed in sport. Examples of psychological skill include intrinsic motivation, self

confidence, attentional control, arousal control, anxiety control, and general self-awareness.

Measurement of Psychological Skills

Several inventories have been developed that are designed to measure psychological skills used by

athletes. Before adopting a specific inventory, the practitioner should become familiar with the reliability, validity, and psychometric properties of the selected inventory.

1.

Psychological skills inventory for sports

The Psychological Skills Inventory for Sports (PSIS-5) was developed by Mahoney, Gabriel, and Perkins (1987). The PSIS-5 (5th version) is a 45-item inventory that measures the psychological skills

of anxiety control, concentration, confidence, mental preparation, motivation, and team

orientation.

2. Athletic coping skills inventory

The Athletic Coping Skills Inventory (ACSI-28) was developed by Smith, Schutz, Smoll, and Ptacek (1995). It measures the psychological skills of coping with adversity, peaking under pressure, goal setting/mental preparation, concentration, freedom from worry, confidence and achievement motivation, and coachability.

3. Test of performance strategies

The Test of Performance Strategies (TOPS) was developed by Thomas, Murphy, and Hardy (1999). It measures a combination of methods and skills of athletes in strategic situations. Factors measured

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by the TOPS in the competitive situation include self-talk, emotional control, automaticity, goal setting, imagery, activation, negative thinking, and relaxation. Factors measured by the TOPS in the

practice situation include the same factors used in the competitive situation, with the exception that

negative thinking is replaced by attentional control.

Psychological Skills Training Program

A number of psychological skills training programs have been proposed. A sample psychological skill

training program is give below. It has seven phases.

Phase 1: Who is the Client?

They determine who the client is.

Phase 2: Initial Meeting with Athletes.

Initial meeting is critical

Phase 3: Education of the Sport Psychologist Relative to Activity

Period of self-education will be required to help the sport psychologist bridge the gap from being a

novice to being fully knowledgeable and conversant about the sport.

Phase 4: Development of a Needs Assessment Plan

The sport psychologist must have a working knowledge of athletes' current psychological skills. This

can be accomplished only through formal and informal assessment.

1. Interview

2. Performance profiling

3. Observation of athletes during practice and competition

4. Use of objective pencil and paper inventories

Phase 5: Psychological Methods and Strategies to be Taught

In this phase, a master plan is developed in terms of what, when, and in what sequence psychological

methods are to be taught to address psychological skill weaknesses.

Phase 6: Actual Teaching and Learning of Selected Psychological Methods

1. Psychological Methods to be taught, practiced , and applied in competition to enhance psychological skills

I. Goal setting

II. Relaxation

III. Self-talk

IV. Imagery

V. Attention skills
 VI. Self-hypnosis
 2. Performance Routines to be taught, practiced, and applied in competition to enhance psychological skills
 Phase 7: Ongoing and End-of-Session Evaluation of PSTP
 For best results, the PSTP must be continually reviewed and evaluated.



Lesson 30: Psychological Skills Training

Overview

- **Focus:** Implementing relaxation procedures, energizing strategies, imaging, and hypnosis to develop psychological skills for athletic performance.
- **Importance:** Helps athletes and non-athletes focus attention, increase self-confidence, and control anxiety.

Topics Covered

- Effectiveness of psychological intervention programs
- Differentiating between psychological skills and methods
- Measurement of psychological skills
- Psychological skills training program

Effectiveness of Psychological Intervention Programs

- **Research Findings:**
 - Literature reviews show planned psychological interventions enhance athletic performance.
 - **Key Studies:**
 - Greenspan & Feltz (1989)
 - Vealey (1994)
 - Weinberg & Comar (1994)
 - **Results:** 85% of 45 studies found positive performance effects (Weinberg & Williams, 2001).
 - **Benefits:** Increased self-confidence, focused attention, energized effort, enhanced performance.

Differentiating Between Psychological Skills and Methods

- **Psychological Methods:** Techniques leading to psychological skills.
 - **Examples:** Goal setting, imagery, progressive relaxation, meditation, self-talk, hypnosis.
- **Psychological Skills:** Characteristics making success in sports likely.
 - **Examples:** Intrinsic motivation, self-confidence, attentional control, arousal control, anxiety control, self-awareness.

Measurement of Psychological Skills

- **Key Inventories:**
 1. **Psychological Skills Inventory for Sports (PSIS-5):**

- **Developers:** Mahoney, Gabriel, Perkins (1987)
 - **Items:** 45
 - **Measures:** Anxiety control, concentration, confidence, mental preparation, motivation, team orientation.
2. **Athletic Coping Skills Inventory (ACSI-28):**
- **Developers:** Smith, Schutz, Smoll, Ptacek (1995)
 - **Measures:** Coping with adversity, peaking under pressure, goal setting/mental preparation, concentration, freedom from worry, confidence and achievement motivation, coachability.
3. **Test of Performance Strategies (TOPS):**
- **Developers:** Thomas, Murphy, Hardy (1999)
 - **Measures:** Self-talk, emotional control, automaticity, goal setting, imagery, activation, negative thinking, relaxation (competitive situations); same factors except negative thinking replaced by attentional control (practice situations).

Psychological Skills Training Program (PSTP)

- **Seven Phases:**
 1. **Who is the Client?:** Identify the client.
 2. **Initial Meeting with Athletes:** Establish rapport.
 3. **Education of the Sport Psychologist Relative to Activity:** Psychologist learns about the sport.
 4. **Development of a Needs Assessment Plan:**
 - Methods: Interviews, performance profiling, observation, objective inventories.
 5. **Psychological Methods and Strategies to be Taught:**
 - Develop a plan for teaching methods to address psychological skill weaknesses.
 6. **Actual Teaching and Learning of Selected Psychological Methods:**
 - Methods: Goal setting, relaxation, self-talk, imagery, attention skills, self-hypnosis.
 - Performance Routines: Practice and apply in competition.
 7. **Ongoing and End-of-Session Evaluation of PSTP:** Continual review and evaluation for best results.

Key Concepts

- **Psychological Methods vs. Skills:** Methods are techniques; skills are characteristics.
 - **Effectiveness:** Supported by research, especially in enhancing performance and psychological resilience.
 - **Measurement Tools:** Important for assessing and tailoring training programs.
-

[To Be Continued]



Lesson 31

PSYCHOLOGICAL SKILLS TRAINING

Who Is the Client?

The first and most critical thing that must be determined by the sport psychologist is who the client

is. If the client is the athletic department of the university, then the athletic department defines the

nature of the relationship between the sport psychologist and the athlete or the coach. If the client is the coach, then the coach defines the nature of the relationship between the sport psychologist and the athlete. Finally if the athlete is the client, then the athlete defines the nature of nature of relationship.

Initial Meeting with the Athletes

The initial meeting between the sport psychologist and the athletes is pivotal for emphasizing the need for commitment to the PSTP. Coaches and athletes recognize the importance of physical practice and training to prepare for peak performance. Athletes must be equally committed to psychological skills training. Psychological skills training must be viewed as an equal partner to the

practicing of physical skills.

Education of the Sport Psychologist relative to Activity

Athletes find it easier to relate to a sport psychologist who understands the nuances of the sport that

the athletes are trying to excel in. If a sport psychologist cannot relate to an athlete's feelings in a critical game situation, she will have difficulty gaining the confidence of the athlete. A sport psychologist must be more than a psychologist; she must also be an exercise and sport scientist.

Development of a Needs Assessment Plan

In order to develop a needs assessment plan, the sport psychologist must determine the psychological skill strengths and weaknesses of each athlete and of a team as a whole. This is accomplished through a series of interviews and test administrations as indicated below.

1.

Interview

An open-ended interview is an important way for the sport psychologist to establish a trusting relationship with the athlete. In this interview, the sport psychologist learns the athlete's attitudes about sport psychology, and his perceptions about psychological strengths and weaknesses.

2. Performance profiling

The athlete then indicates on a scale of 1 to 10 where she feels she falls on that rating scale. This process has been labeled performance profiling. Areas of potential psychological skill improvement

may include intrinsic motivation, self-awareness, self-esteem, self-confidence, attentional focus, and arousal control.

3. Observation of athletes during practice and competition

Regardless of an athlete's perception of personal psychological skill, it is informative to observe the

athlete during game-like situations to see how he deals with pressure. This will make it possible to affirm the athlete's belief system about psychological skill. If differences exist between observed and

perceived psychological skills, then additional interviews might prove beneficial.

4. Use of objective Pencil-and-paper inventories

Where appropriate, inventories such as 16-PF and POMS should be administered and carefully evaluated relative to other subjectively determined information.

Psychological Methods and Strategies to be Taught

Based on needs assessment, it should be clear to the sport psychologist which areas of psychological

skill the athletes are strong in and which areas they are weak in. It is likely that different athletes will

exhibit different profiles relative to their psychological skills. Based on this information, the sport

psychologist develops a master plan detailing how to enhance psychological skill through the application of various psychological methods strategies, and techniques. Timing and sequencing of

the delivery of psychological methods is also determined as this time.

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Actual Teaching and Learning of Selected Psychological Methods

During this phase, the actual teaching of psychological method is taught with a specific purpose in

mind in terms of enhancing psychological skill. For example, if the athlete is lacking in the psychological skill of displaying self-confidence prior to competition, self-talk, self-hypnosis, and imagery might prove to be particularly effective.

1.

Psychological methods to be taught and practiced

An incomplete list of potential psychological methods to be taught include goal setting, relaxation,

self-talk, imagery, attention skills, and self- hypnosis.

2. Performance routines

Performance routine, whether used before, during, or after the execution of a closed motor skill, are

designed to help the athletes focus attention appropriately. Research has validated the use of performance routines in sport (Hill & Borden, 1995; Lidor & Singer, 2000)

Ongoing and End-Of-Season Evaluation of PSTP

If a psychological skills training program extends across an entire sport season, it is imperative that it

be evaluated at the end of the season. If psychological inventories were administered during the needs assessment phase of the program, these same inventories can be administered at the end of the

season, noting changes and improvements in variables of importance.

If an athlete feels uncomfortable about a specific psychological method that she is learning, there is

no need to continue it to the end of the program.



Lesson 31: Psychological Skills Training

Overview

- **Focus:** Implementing psychological skills training (PSTP) for athletes.
- **Goal:** Develop psychological skills to enhance athletic performance.

Topics Covered

- Identifying the client
- Initial meeting with athletes
- Education of the sport psychologist
- Development of a needs assessment plan

- Teaching psychological methods and strategies
 - Ongoing and end-of-season evaluation
-

Who Is the Client?

- **Determine the client:**
 - **Athletic department:** Defines the relationship between the psychologist and the athlete/coach.
 - **Coach:** Defines the relationship between the psychologist and the athlete.
 - **Athlete:** Defines the nature of the relationship directly.

Initial Meeting with Athletes

- **Purpose:** Emphasize the need for commitment to the PSTP.
- **Importance:** Athletes must view psychological skills training as equally important as physical training.

Education of the Sport Psychologist

- **Goal:** Understand the nuances of the sport.
- **Requirement:** Sport psychologist must be knowledgeable in both psychology and sport science.

Development of a Needs Assessment Plan

- **Purpose:** Identify psychological skill strengths and weaknesses.
- **Methods:**
 1. **Interview:** Establish a trusting relationship and understand the athlete's attitudes and perceptions.
 2. **Performance Profiling:** Rate psychological skills on a scale of 1 to 10.
 3. **Observation:** Observe athletes in practice and competition to validate their perceived skills.
 4. **Objective Inventories:** Administer and evaluate inventories like 16-PF and POMS.

Teaching Psychological Methods and Strategies

- **Based on Needs Assessment:**
 - Develop a plan to enhance psychological skills using various methods.
 - Determine timing and sequencing of psychological methods.
- **Methods to be Taught:**
 - Goal setting
 - Relaxation
 - Self-talk
 - Imagery
 - Attention skills
 - Self-hypnosis
- **Performance Routines:**
 - Used to focus attention before, during, or after skill execution.
 - Validated by research (Hill & Borden, 1995; Lidor & Singer, 2000).

Ongoing and End-of-Season Evaluation

- **Evaluation:**

- Conduct at the end of the season.
- Administer the same inventories used in the needs assessment to note changes and improvements.
- **Flexibility:**
 - Discontinue any psychological method that makes an athlete uncomfortable.

Key Concepts

- **Client Identification:** Determines the nature of the relationship.
- **Initial Commitment:** Crucial for the success of the PSTP.
- **Psychologist's Education:** Essential for gaining athlete's trust.
- **Needs Assessment:** Identifies specific areas for improvement.
- **Teaching Methods:** Tailored to address specific psychological weaknesses.
- **Continuous Evaluation:** Ensures the effectiveness of the PSTP and allows for adjustments.

[To Be Continued]



Lesson 32

ETHICS IN SPORT PSYCHOLOGY

In 1965 the internal society of sport psychology (ISSP) was formed in Rome, Italy. The Association for the Advancement of Applied Sport Psychology (AAASP) is the primary organization within the United States and Canada for professionals interested in applied sport psychology.

All members of these societies are bound by a code of ethics that governs their interactions with the

public and with other professionals. Their Ethics Code is based in large part on the Ethical Principles and is composed of a preamble and six general principles.

Principle 1: Competence

Members maintain a high standard of competence in their work. In this regard, they recognize the

boundaries and limitations of their competence. For example, a member trained in exercise and sport

science would not attempt to counsel an individual with clinical symptoms of depression.

Members

are continually upgrading their knowledge and expertise through workshops and inservice training.

Principle 2: Integrity

Members practice and promote integrity in the teaching, science, and practice of applied sport psychology. In this regard, they always present themselves and their credentials accurately and forthrightly. They do not make deceptive or misleading statements about their qualifications, products, fees, research, or service. For example, a member would not make unsubstantiated claims

about a psychological application that she was using.

Principle 3: Professional and Scientific Responsibility

Members take their professional and scientific responsibilities seriously. It is a member's responsibility to protect the reputation of the society and the public from members who are deficient

in ethical conduct. In this regard, they are concerned about the ethical conduct of members whose

ethical conduct is not of the highest level. For example, a member would take steps to prevent and/or expose unethical conduct in another member.

Principle 4: Respect for People's Rights and Dignity

Members respect the fundamental rights, worth, and dignity of all individuals. An individual's right to

confidentiality, privacy, and personal control are respected at all times. In this regard, members are

sensitive to individual differences associated with gender, age, race/ethnicity, national origin, religion,

disability, sexual orientation, and socioeconomic status. For example, a member would not refuse to

provide the highest level of consideration to an individual on the basis of the person's gender or race.

Principle 5: Concern for Other's Welfare

Members are personally concerned with and take steps to ensure the personal welfare of individuals

they interact with. Conflicts between members or between members and clients are resolved in a manner which minimizes harm and maximizes the concern for the welfare of others. For example, members do not take advantage of differences in power and influence between themselves and others.

Principle 6: Social Responsibility

Members have a responsibility to share their knowledge and research with members of society. In this regard, their responsibility is to contribute to the common good of society and to protect the rights of individuals as they do so. For example, members freely agree to provide workshops that will

teach others how to apply principles of human development. They also agree to share their research

findings in appropriate scientific settings.



Lesson 32: Ethics in Sport Psychology

Overview

- **Formation of Professional Bodies:**
 - **1965:** The International Society of Sport Psychology (ISSP) formed in Rome, Italy.
 - **Primary Organization in North America:** The Association for the Advancement of Applied Sport Psychology (AAASP).

Code of Ethics

- **Governs Interactions:** With the public and other professionals.
- **Based on Ethical Principles:** Composed of a preamble and six general principles.

Principle 1: Competence

- **Maintain High Standards:** Recognize the boundaries and limitations of their competence.
- **Continuous Improvement:** Upgrade knowledge and expertise through workshops and training.
- **Example:** A sport science expert should not counsel someone with clinical depression.

Principle 2: Integrity

- **Promote Integrity:** In teaching, science, and practice.
- **Honest Representation:** Present credentials accurately without misleading statements.
- **Example:** Avoid making unsubstantiated claims about psychological applications.

Principle 3: Professional and Scientific Responsibility

- **Serious Responsibilities:** Protect the society's reputation and the public.
- **Ethical Vigilance:** Concerned about the conduct of other members and take steps to address unethical behavior.
- **Example:** Expose and prevent unethical conduct in other members.

Principle 4: Respect for People's Rights and Dignity

- **Respect Fundamental Rights:** Confidentiality, privacy, and personal control.
- **Sensitivity to Differences:** Gender, age, race/ethnicity, national origin, religion, disability, sexual orientation, and socioeconomic status.
- **Example:** Do not discriminate based on gender or race.

Principle 5: Concern for Others' Welfare

- **Ensure Personal Welfare:** Minimize harm and maximize concern for others' welfare.
- **Resolve Conflicts:** In a manner that minimizes harm.
- **Example:** Do not exploit power differences between themselves and clients.

Principle 6: Social Responsibility

- **Contribute to Society:** Share knowledge and research for the common good.
- **Protect Individual Rights:** While contributing to societal welfare.
- **Example:** Provide educational workshops and share research findings in appropriate settings.

Summary

Ethical principles in sport psychology guide professionals to maintain competence, integrity, and responsibility, while respecting individuals' rights and contributing positively to society. These principles ensure that sport psychologists act in the best interests of their clients and the broader community, promoting ethical conduct and professional excellence.

[To Be Continued]



Lesson 33

AGGRESSION AND VIOLENCE IN SPORT

A number of critical questions come to mind as one contemplates the issue of sport aggression.

Does

participating in or observing violent sporting events serve as a catharsis, or release from aggressive

tendencies? Conversely, do these events merely teach and encourage further aggression on and off

the playing field? Is it possible to eliminate aggression and violence from sports? If so, how? In the following two lectures we will be discussing aggression and violence in sport. Topics to be addressed include:

- Defining aggression
- Theories of aggression
- Catharsis hypothesis
- Measurement issues
- Fan violence
- Performance issues
- Situational factors contributing to aggression
- Reducing aggression in sport

Defining Aggression

Two factors must be presented in order for a behavior to be labeled aggression. First, the behavior must be aimed at another human being with the goal of inflicting physical harm. Second, there must

be a reasonable expectation that the attempt to inflict bodily harm will be successful.

Consequently,

the following behaviors are not really examples of aggression:

1. Doing destructive violence to an inanimate object such as a door or a water cooler
2. Unintentionally injuring another person during athletic competition
3. Aggressive behavior in which there is no chance for the intended victim to be injured (e.g., aggressor and victim are separated by bars or by teammates)

Over the years, two basic kinds of aggression have been identified. The first is hostile aggression. For

individuals engaged in hostile aggression, the primary goal is the injury of another human being. The

intent is to make the victim suffer, and the reinforcement is the pain and suffering that is caused.

This sort of aggression is always accompanied by anger on the part of the aggressor. A good example of hostile aggression occurs when a cricket bowler throws a full-toss at the batsman who has

angered him. The second major kind of aggression is instrumental aggression. For individuals engaged in instrumental aggression, the intent to harm another individual is present, but the goal is to realize

some external goal such as money, victory or prestige. The aggressor views the aggressive act as instrumental in obtaining the primary goal. A parallel cricket example for instrumental aggression would be on in which the bowler has been "ordered" by his coach to hit a batsman in retaliation for

some earlier infraction. The bowler is not necessarily angry at the batsman, but sees hitting the batsman as instrumental in achieving the team goal of winning the game.

It must be emphasized that neither type of aggression is acceptable. The aggressor is guilty of purposely inflicting harm with the intent to injure another person. This must be discouraged at all levels of competition, especially the professional level, because young athletes everywhere emulate the pros.

A third category of behavior that is often confused with aggression is assertiveness, or assertive behavior. Generally, when coaches encourage their athletes to be more aggressive, what they really want is that they be more assertive. Coaches want their athletes to assert themselves and make their

presence felt. Assertiveness involves the use of legitimate physical or verbal force to achieve one's

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purpose. However, there is no intent to harm the opponent. Even if the opponent is harmed as a result of a tackle in soccer, it is not necessarily aggression. It is merely assertive play, as long as it is

within the spirit of the agreed-on rules and the intent to harm is not present. Assertiveness requires

the expenditure of unusual effort and energy, but if there is no intent to harm, then any resultant harm is incidental to the game.

Theories of Aggression

Theories of aggression fall into four main categories:

1. Instinct theory,
2. Social learning theory
3. Theory of moral reasoning,
4. The frustration-aggression hypothesis.

1.

Instinct Theory

Instinct theory is based upon the writings of Sigmund Freud and ethologists such as Konrad Lorenz,

Freud (1950) viewed aggression as an inborn drive similar to hunger, thirst, and sexual desire. According to Freud, aggression is unavoidable since it is innate, but as with any drive it can be regulated through discharge, or fulfillment. Since humankind is innately aggressive, it benefits society

to promote athletic sports and games that provide a socially acceptable outlet for aggression.

2. Social Learning Theory

Social learning theory posits that aggression is a function of learning and that biological drive and frustration are inadequate explanations of the phenomenon. Acts of aggression serve only to lay the

foundation for more aggression, and do not result in reduction or purging of the drive to be aggressive. Perhaps the leading advocate of social learning theory, relative to aggression, is Bandura

(1973). Bandura proposes that aggression has a circular effect, i.e., one act of aggression leads to further aggression. This pattern will continue until the circle is broken by some type of positive or negative reinforcement.

3. Theory of Moral Reasoning and Aggression

Theory of moral reasoning proposes that an individual's willingness to engage in aggression is related

to her stage of moral reasoning. Since human aggression is viewed as unethical, this theory suggests

that a relationship should exist between the level of moral reasoning and overt acts of

athletic aggression.

4. Reformulated Frustration-Aggression Theory

As originally presented by Dollard, Miller, Doob, Mowrer, and Sears (1939), frustration-aggression theory proposes that aggression is a natural response to frustration, and that the aggressive act provides a catharsis, or purging, of the anger associated with the frustration. Frustrations caused by

events that are believed to be arbitrary or illegitimate are particularly galling to athletes.



Lesson 33: Aggression and Violence in Sport

Overview

This lesson addresses critical questions related to sport aggression, such as whether participating in or observing violent sporting events serves as a catharsis or encourages further aggression. It also explores the possibility of eliminating aggression and violence from sports. Key topics include defining aggression, theories of aggression, the catharsis hypothesis, measurement issues, fan violence, performance issues, situational factors contributing to aggression, and strategies for reducing aggression in sports.

Defining Aggression

- **Criteria for Aggression:**
 1. Behavior aimed at another human being with the goal of inflicting physical harm.
 2. A reasonable expectation that the attempt to inflict bodily harm will be successful.
- **Examples of Non-Aggressive Behaviors:**
 1. Destroying inanimate objects (e.g., a door or a water cooler).
 2. Unintentional injury during athletic competition.
 3. Aggression without the possibility of causing harm (e.g., separated by bars or teammates).
- **Types of Aggression:**
 - **Hostile Aggression:** Primary goal is to injure another person, motivated by anger.
 - *Example:* A cricket bowler throws a full-toss at a batsman out of anger.
 - **Instrumental Aggression:** Intent to harm is present, but the goal is an external reward (e.g., money, victory).
 - *Example:* A bowler hits a batsman because the coach ordered it for team advantage.
 - **Assertiveness:** Legitimate force to achieve a purpose without intent to harm, within the spirit of the rules.
 - *Example:* A strong tackle in soccer with no intent to harm is assertive play.

Theories of Aggression

1. **Instinct Theory:**
 - Based on Freud and Lorenz's writings.
 - Aggression is an inborn drive, similar to hunger or thirst.
 - Promotes the idea that sports provide a socially acceptable outlet for aggression.
2. **Social Learning Theory:**
 - Posits that aggression is learned rather than innate.
 - Bandura's theory suggests aggression leads to more aggression unless broken by reinforcement.
3. **Theory of Moral Reasoning:**
 - Links aggression to an individual's stage of moral reasoning.
 - Higher moral reasoning is associated with less aggressive behavior.
4. **Reformulated Frustration-Aggression Theory:**
 - Original theory by Dollard et al. (1939) suggests aggression is a natural response to frustration.

- Aggressive acts provide a catharsis for anger.
- Particularly frustrating are events perceived as arbitrary or illegitimate.

Catharsis Hypothesis

- **Catharsis:** The idea that releasing aggressive energy through certain actions (like sports) reduces overall aggression.
- **Debate:** Whether this hypothesis is valid or if it merely perpetuates further aggression.

Measurement Issues

- **Challenges:** Accurately measuring aggression and distinguishing it from assertiveness or other behaviors.
- **Tools:** Various psychological assessments and observations are used to measure aggression in sports contexts.

Fan Violence

- **Influence of Spectators:** Examines how fans' behavior can escalate aggression in sports.
- **Prevention:** Strategies to manage and reduce fan violence, such as security measures and fan education.

Performance Issues

- **Impact of Aggression:** How aggression affects athletic performance positively or negatively.
- **Balancing Act:** Ensuring athletes are assertive without crossing into aggression.

Situational Factors Contributing to Aggression

- **External Triggers:** Situations or contexts that increase the likelihood of aggressive behavior in sports.
- **Examples:** Rivalries, high-stakes games, provocation by opponents or fans.

Reducing Aggression in Sport

- **Strategies:**
 - **Education:** Teaching athletes and coaches about the difference between aggression and assertiveness.
 - **Rules and Regulations:** Implementing and enforcing rules to deter aggressive behavior.
 - **Psychological Interventions:** Using methods like relaxation, self-talk, and conflict resolution to manage aggression.

Summary

Understanding and managing aggression in sports involves defining it clearly, exploring various theoretical perspectives, and implementing strategies to reduce its occurrence. By promoting assertiveness and distinguishing it from aggression, sports can maintain competitiveness while ensuring the safety and well-being of all participants.



Lesson 34

AGGRESSION AND VIOLENCE IN SPORT

The Catharsis Effect

The catharsis effect represents a release of pent-up frustration that makes one feel better. It is a purging of the anger and frustration associated with not being able to accomplish a goal. Venting frustration upon a punching bag or some other inanimate object may serve as a useful catharsis. Venting frustration upon another human being, however, is unacceptable behavior that is likely to lead to more aggression.

Aggression is not cathartic in the sense that it leads to a reduction in the desire to aggress.

Aggression

leads to an increase in aggression as tempers flare and as the behavior becomes learned.

Measurement of Aggression

A number of inventories have been developed to measure aggressiveness as a personality disposition

or trait. Two of these inventories are the Aggression Questionnaire (Buss & Perry, 1992) and the Aggression Inventory (Gladue, 1991). Sport-specific inventories include the Athletic Aggression Inventory (Bredemeier, 1978) and the Continuum of Injurious Acts (Bredemeier, 1985). The actual measurement of aggression, however, is much more difficult. Because aggression is defined as the

intent to harm another human being, the measurement device must be able to capture this intent.

Fan Violence

Some of the worst examples of sports aggression and violence occur among the fans watching an athletic contest. Every sports event is attended by individuals who may instigate fan violence.

These

are individuals who score high in the personality disposition of anger and physical aggression.

These

individuals are attracted to violence and fighting among fans, and exhibit a false belief about the willingness of other fans to join in acts of violence.

Effects of Aggression on Performance

Conventional wisdom argues that acts of aggression on the part of an athlete will constitute a distraction and result in a decrement in performance. Not only are aggressive acts on the part of an

individual distracting to the individual, but they are likely to be distracting to the team as a whole.

Research shows, for example, that the lower a team is in the standings; the more likely it will be to engage in aggression.

Situational Factors in a Sport Setting

Factors associated with the occurrence of aggression in sport-specific situations are as follows:

1.

Environmental temperature

Archival data suggest that higher temperatures lead major league pitchers to become more aggressive in pitching to batters.

2. Perception of victim's intent

If athletes perceive that an opponent's intent is to inflict harm, they are more likely to respond with

aggression against the opponent. This means that perception of an opponent's aggressive intentions

may be more salient than such things as defeat and competition.

3. Fear of retaliation

To some degree, the fear of retaliation on the part of the individual who is the target of aggression can inhibit another player from initiating that aggression. A basketball player is a little less likely to elbow her opponent in the ribs if she fears similar treatment from the opponent.

4. Structure of the game

Studies have shown that more aggressive penalties occur as the game score differential increases. When teams are tied or the scores are close, aggression is at a minimum. Research has also shown that the lower the team is in standings, the more its members engage in aggression.

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5. Rivalry, familiarity, and frequency of play.

As players become more familiar with one another due to frequency of play, and as the rivalries become more intense due to geographical location, aggression becomes more frequent.

6. Goal orientation

A respect for rules and officials declines as the ego orientation of the athlete increases. Conversely, high task goal orientation is associated with higher level of sportspersonship.

Reducing Aggression in Sport

Aggression in sport can be curtailed, or at least minimized, if all concerned are interested in doing so.

The sad part is that some of the most influential people actually promote rather than discourage violence because they believe it sells tickets.

Research shows that angry feelings and angry behavior, the precursor to hostile aggression can be

modified through anger awareness training and role playing. Athletes can learn to control their feelings of hostility anger. Role playing is particularly effective in reducing an athlete's anger feelings and behavior.

Curtailing Aggression and Violence by Athletes

1. Young athletes must be provided with models of nonaggressive but effective assertive behavior.

2. Athletes who engage in aggressive acts must be severely penalized.

3. The penalty or punishment that an athlete receives for an act of aggression must be of greater punitive value than the potential reinforcement received for committing the act.

4. In addition to receiving punishment for acts of aggression, athletes should receive rewards and praise for showing restraint and patience in emotionally charged situations.

Curtailing Aggression and Violence by Fans

1. Potential troublemakers should be closely supervised. Fans with a history of violence and fighting should be identified and denied admission.

2. The sale, distribution, and use of alcoholic beverages at sporting event should be limited and controlled.

3. Athletic events should be promoted and encouraged as family affairs.

4. The media can promote responsible behavior on the part of the fans by not glamorizing acts of aggression.

5. As with athletes and coaches, fan aggression must be swiftly and severely punished.



Lesson 34: Aggression and Violence in Sport

The Catharsis Effect

- **Definition:** The catharsis effect is the release of pent-up frustration or anger, which is believed to make one feel better. This can involve venting frustration through activities like hitting a punching bag.
- **Reality:** Research indicates that aggression does not serve as a true catharsis. Instead, engaging in aggressive behavior often increases the likelihood of further aggression. Venting frustration on another person is unacceptable and likely to escalate aggression rather than reduce it.

Measurement of Aggression

- **Inventories:**
 - **Aggression Questionnaire** (Buss & Perry, 1992)
 - **Aggression Inventory** (Gladue, 1991)
 - **Athletic Aggression Inventory** (Bredemeier, 1978)
 - **Continuum of Injurious Acts** (Bredemeier, 1985)
- **Challenges:** Measuring aggression involves capturing the intent to harm another person, which is difficult to quantify.

Fan Violence

- **Characteristics of Aggressive Fans:**
 - High levels of anger and physical aggression.
 - Attraction to violence and fighting.
 - Misconception about other fans' willingness to engage in violence.

Effects of Aggression on Performance

- **Conventional Wisdom:** Aggressive acts are believed to distract athletes and decrease performance.
- **Research Findings:** Aggression often distracts both the individual athlete and the team. Lower-ranked teams are more likely to engage in aggressive behavior as a response to frustration.

Situational Factors in Sport Settings

1. **Environmental Temperature:** Higher temperatures can increase aggression, such as pitchers becoming more aggressive in hotter conditions.
2. **Perception of Victim's Intent:** Athletes are more likely to respond aggressively if they perceive the opponent's intent to be harmful.
3. **Fear of Retaliation:** Fear of retaliation can inhibit an athlete from initiating aggression.
4. **Structure of the Game:** Aggression is more common when there is a significant score differential. Teams lower in the standings also tend to be more aggressive.
5. **Rivalry and Familiarity:** Increased aggression occurs with more frequent play and intense rivalries, often influenced by geographical proximity.
6. **Goal Orientation:** Athletes with a high ego orientation may show less respect for rules, whereas those with a task orientation demonstrate better sportsmanship.

Reducing Aggression in Sport

- **Influence:** Effective reduction of aggression requires commitment from all parties involved, including teams, coaches, and fans. Sometimes, the promotion of violence is encouraged due to its financial benefits.

Curtailling Aggression and Violence by Athletes

1. **Modeling Non-Aggressive Behavior:** Provide young athletes with examples of effective but non-aggressive behavior.
2. **Severe Penalties:** Aggressive acts should be met with stringent penalties.
3. **Punishment vs. Reward:** The penalty for aggression should outweigh the potential rewards for such behavior.
4. **Encouraging Restraint:** Athletes should be rewarded for demonstrating restraint and patience in intense situations.

Curtailling Aggression and Violence by Fans

1. **Supervision of Troublemakers:** Monitor and exclude fans with a history of violence.
2. **Control of Alcohol:** Limit the sale and consumption of alcohol at sporting events.
3. **Promote Family-Friendly Events:** Market sports events as family-oriented to reduce aggressive behavior.
4. **Media Responsibility:** Avoid glamorizing aggression in media coverage.
5. **Fan Punishment:** Swift and severe punishment for aggressive behavior among fans.

Summary

Understanding the complexities of aggression and violence in sports involves recognizing the limitations of the catharsis hypothesis, accurately measuring aggression, and addressing both athlete and fan behaviors. Effective strategies for reducing aggression require comprehensive approaches and commitment from all stakeholders involved in sports.



Lesson 35

AUDIENCE AND CROWD EFFECTS IN SPORTS

Audience participation is a powerful on athletic performance. Perhaps no social-psychological effect

is more important to athletic performance and outcome than the audience, or spectators, effect. While many variables may help create the home court or home field advantage, none seem to be as

important as the presence of a supportive audience. Determining how and why an audience presence

affects athletic performance is the focus of this section. Topics to be discussed include:

- Social facilitation
- Effect of an interactive audience on performance
- Audience characteristics(size, intimacy, density, and hostility)
- Team quality
- Players' perceptions.

Social Facilitation

Social facilitation research is based on the notion that the presence of an audience of one or more spectators can facilitate performance. This is an appealing concept, since almost everyone has experienced the desire to perform better when friends, family, or members of the opposite

gender

are watching.

Research in the area of social facilitation was significantly influenced by the work of Robert Zajonc (pronounced "science"). Zajonc's classical paper on the topic remains the single most critical factor in the development of social facilitation as a field of inquiry. (Zajonc, 1965). Zajonc proposed that the presence of an audience has the effect of increasing (drive) in performing subjects. Since increased arousal facilitates the elicitation of the dominant response, the presence of an audience will

enhance the performance of a skilled individual while causing a decrement in the performance of an

unskilled individual. This concept is illustrated below.

For highly skilled athletes:

Crowd influence (increased Arousal) → leads to → performance facilitation

For less skilled athletes:

Crowd influence (increased Arousal) → leads to → performance decrement

Effects of An Interactive Audience On Performance

Perhaps the most interesting topic associated with the interactive audience is that of the home advantage. The fact that the home advantage exists in such team sports as basketball, baseball, football, ice hockey, and soccer is well documented (Bray, 1999). We will focus upon the most viable

explanation for the home court advantage: the presence of a supportive and interactive audience.

- Why is there a Home Court Advantage?

The most plausible explanation for the home advantage in sport is the presence of a supportive and

interactive audience. The presence of a supportive and emotionally arousing crowd translates into a

home court advantage in many situations. Some researches suggest that the advantage favoring the

home team is due not to increased performance caused by a supportive audience, but to inferior performance on the part of the visiting team—sort of an away court disadvantage.

- When is the Home Court/Field a Disadvantage?

Is playing at home always an advantage, or can it sometimes be a disadvantage? For a number of reasons, playing at home can be a home disadvantage. One reason might be that the fans expect you

to win at home; this can result in additional pressure to play well. A second reason might be that playing before a very vocal and supportive audience can raise arousal to a level that results in a decrement of performance. The presence of a supportive audience may have the effect of increasing

the cost of not winning when you are expected to. The athlete or the athletes begin to "press," which

interferes with the execution of skillful play.

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Audience Characteristics

Having determined that a home advantage usually exists in sport and that this advantage is related to

the presence of a supportive and interactive audience, we should now examine characteristics of the audience.

- **Crowd Size, Intimacy, and Density**

There is evidence in professional baseball that crowd size makes a difference. Factors such as audience density and audience intimacy may be more important than size for creating the home court advantage.

- **Crowd Hostility**

It is generally understood that a supportive and friendly crowd will help the home team. What is the effect, however, of a seemingly hostile crowd on player performance? Research by Greer (1983) demonstrated that sustained hostile spectator protests have a clearly negative impact on the visiting team.

Home Court Advantage And Team Quality

From the previous discussions we understand that the home team usually enjoys a home court or field advantage. Factors that contributed most to the home team advantage, after controlling for team quality, were crowd density, rebounds, steals, and field goal shooting percentage. Winning at

home occurs more often for high-quality teams than for low-quality teams.

Players' Perceptions of Home Court Advantage

When asked about their perceptions of the home court advantage, athlete indicated that they believed

there was a home court advantage. They further indicated that they felt that home court familiarity

and crowd support were the primary factors determining the home court advantage. Finally, athletes

believed that they were more self-confidence when they played at home than when they played away.



Lesson 35: Audience and Crowd Effects in Sports

Social Facilitation

- **Definition:** Social facilitation refers to the phenomenon where the presence of an audience can enhance or hinder performance.
- **Robert Zajonc's Theory (1965):**
 - **Increased Arousal:** The presence of an audience increases physiological arousal.
 - **Dominant Response:** This increased arousal facilitates the dominant response, which benefits skilled individuals but can hinder less skilled ones.
 - **Skilled Athletes:** Enhanced performance due to increased arousal.
 - **Less Skilled Athletes:** Decreased performance due to increased arousal.

Effects of an Interactive Audience on Performance

- **Home Court Advantage:**
 - **Explanation:** The presence of a supportive and interactive audience often leads to a home court advantage. This supportive crowd can boost the performance of the home team.
 - **Home Disadvantage:**
 - **Expectations:** High expectations from fans can create additional pressure.

- **Arousal Levels:** A highly vocal and supportive audience might increase arousal to a level that disrupts performance. Athletes may "press" or feel increased pressure to perform, leading to a potential decrease in performance.

Audience Characteristics

1. Crowd Size, Intimacy, and Density:

- **Size:** Larger crowds can influence performance, but crowd density and intimacy (how close and engaged the crowd is) may be more impactful.
- **Examples:** In professional baseball, crowd size has shown to affect performance. However, density and the personal connection of the crowd might have even more significant effects.

2. Crowd Hostility:

- **Supportive Crowds:** Generally boost the home team's performance.
- **Hostile Crowds:** Research by Greer (1983) indicates that sustained hostile behavior from spectators negatively impacts the visiting team's performance. Hostile crowds can disrupt visiting teams more than they benefit the home team.

Home Court Advantage and Team Quality

- **Impact:** High-quality teams benefit more from home court advantages than low-quality teams.
- **Factors:** Key factors contributing to home advantage include crowd density, and performance metrics such as rebounds, steals, and field goal shooting percentage.

Players' Perceptions of Home Court Advantage

- **Beliefs:**
 - Athletes believe in the existence of a home court advantage.
 - **Primary Factors:** Home court familiarity and crowd support are seen as the main factors.
 - **Self-Confidence:** Athletes feel more self-confident when playing at home compared to playing away.

Summary

The presence and characteristics of an audience have significant effects on athletic performance. Social facilitation theory highlights how an audience can enhance or impair performance based on skill levels. Home court advantage is influenced by supportive crowds and factors like crowd density, though it can sometimes create additional pressure for athletes. Understanding these dynamics can help in optimizing performance and managing the effects of audience presence.



always use bulletpoints in summery



 Memory updated

Lesson 35: Audience and Crowd Effects in Sports

Social Facilitation

- Presence of an audience can influence performance.
- **Robert Zajonc's Theory (1965):**
 - Increased arousal from an audience enhances performance in skilled individuals.
 - Increased arousal can hinder performance in less skilled individuals.

Effects of an Interactive Audience on Performance

- **Home Court Advantage:**
 - Presence of a supportive and interactive audience contributes to a home court advantage.
 - Advantage might also be due to the visiting team's poorer performance.
- **Home Court/Field Disadvantage:**
 - Playing at home can sometimes be a disadvantage.
 - High expectations and vocal support can increase pressure and disrupt performance.

Audience Characteristics

- **Crowd Size, Intimacy, and Density:**
 - Crowd size impacts performance.
 - Audience density and intimacy may be more crucial for creating a home advantage.
- **Crowd Hostility:**
 - Hostile crowds can negatively impact the visiting team's performance.
 - Sustained hostility from spectators can have a detrimental effect.

Home Court Advantage and Team Quality

- **Key Factors for Home Advantage:**
 - Crowd density, rebounds, steals, and shooting percentage.
 - High-quality teams benefit more from playing at home compared to low-quality teams.

Players' Perceptions

- Athletes believe in a home court advantage.
- Perceptions:
 - Home court familiarity and crowd support are major factors.
 - Athletes feel more self-confident playing at home.



continue



Lesson 35: Audience and Crowd Effects in Sports (Continued)

Social Facilitation (Expanded)

- **High-Skilled Athletes:**
 - **Crowd Influence:** Increased arousal → **Performance Facilitation.**
- **Less-Skilled Athletes:**
 - **Crowd Influence:** Increased arousal → **Performance Decrement.**

Effects of an Interactive Audience on Performance (Expanded)

- **When Home Advantage May Become a Disadvantage:**
 - **Expectation Pressure:** High expectations from home fans may increase pressure.
 - **Increased Arousal:** Vocal and supportive crowds might elevate arousal to disruptive levels.

Audience Characteristics (Expanded)

- **Crowd Size, Intimacy, and Density:**
 - **Crowd Size:** Larger crowds can influence performance.
 - **Audience Density:** More impactful than size alone.
 - **Audience Intimacy:** Closer, more familiar crowds enhance home advantage.
- **Crowd Hostility:**
 - Hostile crowds can:
 - Intensify pressure on the visiting team.
 - Impact the visiting team's performance negatively.

Home Court Advantage and Team Quality (Expanded)

- **Factors Contributing to Home Advantage:**
 - Team quality matters: Better teams perform better at home.
 - **Performance Metrics:**
 - Rebounds
 - Steals
 - Field goal shooting percentage

Players' Perceptions (Expanded)

- **Belief in Home Court Advantage:**
 - Athletes acknowledge the benefits of home court.
 - Key factors include:
 - Familiarity with the home environment.
 - Support from the home crowd.
 - **Self-Confidence:** Increased when playing at home.



Lesson 36

TEAM COHESION IN SPORT

Intuitively athletes, coaches, and sport enthusiasts understand that there is more to athletic success

than the collective or individual skills of a team. Sport psychologists refer to this extra team ingredient as group or team cohesion.

There are numerous examples of talented teams that failed to live up to expectations, or to less talented teams who performed far above expectations. In sports, it is a well-established principle that

a group of individuals working together is far more effective than the same individuals working independently of one another. On the basketball team, there may be several individuals capable of

scoring twenty or more points a game. However, in the interest of team success, the coach may require that one or more of these athletes assume nonscoring roles. For example, a point guard has

the primary responsibility of setting up plays and getting the offense started, while the power forward

must “crash” the boards and get offensive and defensive rebounds. Athletes who play these specialized roles rarely score as many points as shooting guards or forwards. Yet, out of the desire to

be “team players”, these athletes accept less glamorous roles for the common good of the team. Thus, as a group or team evolves, a certain structure develops. This structure varies from group to group and situation to situation, but it is critical for team success.

Not only do members of successful teams have the ability to work together (teamwork); they also enjoy a certain attraction to one another. In this respect, it seems logical that teams composed of members who like each other and enjoy playing together will somehow be more successful than teams lacking this quality.

As a social psychological topic, team cohesion ranks as a very important factor for enhancing team

performance and feelings of satisfaction among members. In the following two lectures, team cohesion will be discussed in terms of:

- Its defining characteristics
- Its measurement
- Its determinants
- Its consequences
- Its development

Defining Characteristics of Team Cohesion

Albert Carron (1992), a prominent sport social psychologist, defined group cohesion as “a dynamic process which is reflected in the tendency for a group to stick together and remain united in the pursuit of goals and objectives. Fundamental to the study of team cohesion is the understanding of

group dynamics. Members of a team or group begin to interact with each other the moment the group is first formed.

Direct and Indirect Measurement of Cohesion

The indirect measurement approach to assessing team cohesion tries to get at team cohesion by asking each team member how she feels about every other member of the on some basic question

(e.g., How much do you like the different members on your team?). Summed scores from team members would represent a measurement of team cohesion.

The direct measurement approach to assessing team cohesion is direct in the sense that players are

asked to indicate how much they like playing for the team (individual attraction) and how well they feel the team functions as a unit (group integration). Research using the indirect approach has generally failed to find a meaningful relationship between team cohesion and team or individual behavior. the indirect approach to measuring team cohesion is very rare in sport psychology research

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Measurement of Team Cohesion

A number of inventories have been developed for measuring team cohesion in sport. An incomplete

list of inventories include:

1. Sports Cohesiveness Questionnaire (SCQ; Martens & Peterson, 1971);
2. Team Cohesion Questionnaire (TCQ; Gruber & Grey, 1981);
3. Sport Cohesion Instrument (SCI; Yukelson, Wienberg and Jackson, 1984);

4. Group Environment questionnaire (GEQ; Widmeyer, Brawley & Carron, 1985);

5. Team Psychology Questionnaire (TPQ; Partington and Shang, 1992)

Of these five inventories, the Group Environment Questionnaire (GEQ) has been sport psychologists' primary inventory of choice over the last fifteen years. The GEQ has continued to be used extensively by researchers and practitioners. The GEQ is composed of eighteen items that measure the four team cohesion dimensions.

The four team cohesion dimensions measured by GEQ are:

1. Personal factors

2. Team factors

3. Leadership factors

4. Environmental factors

Determinants of Team Cohesion

Now let's focus attention upon the determinants of team cohesion.

In an important study reported by Widmeyer and Williams (1991), factors that determine team cohesion were investigated. In this investigation, team cohesion was measured using the GEQ.

The

results of this investigation revealed that all the four determinants, personal factors, team factors, leadership factors and environmental factors predictive of some aspect of team cohesion. The strongest predictor of team cohesion, however, was personal satisfaction. The best way to develop

team cohesion is by cultivating a personal feeling of satisfaction towards the team and team members.



Lesson 36: Team Cohesion in Sport

Defining Characteristics of Team Cohesion

- **Definition (Albert Carron, 1992):**
 - **Dynamic Process:** Tendency for a group to stick together and remain united in pursuing goals and objectives.
 - **Group Dynamics:** Members start interacting from the moment the group is formed.

Direct and Indirect Measurement of Cohesion

- **Indirect Measurement:**
 - **Method:** Asking each team member about their feelings toward other members (e.g., "How much do you like the different members on your team?").
 - **Limitations:** Often fails to find a significant relationship between cohesion and behavior; rarely used today.
- **Direct Measurement:**
 - **Method:** Players indicate how much they like playing for the team (individual attraction) and how well the team functions as a unit (group integration).

Measurement Inventories

- **Sports Cohesiveness Questionnaire (SCQ):** Martens & Peterson (1971)
- **Team Cohesion Questionnaire (TCQ):** Gruber & Grey (1981)
- **Sport Cohesion Instrument (SCI):** Yukelson, Weinberg, and Jackson (1984)
- **Group Environment Questionnaire (GEQ):** Widmeyer, Brawley & Carron (1985)

- **Team Psychology Questionnaire (TPQ):** Partington and Shangi (1992)
 - **GEQ:**
 - **Popularity:** Most widely used in sport psychology research and practice.
 - **Dimensions Measured:**
 1. **Personal Factors**
 2. **Team Factors**
 3. **Leadership Factors**
 4. **Environmental Factors**

Determinants of Team Cohesion

- **Factors Investigated by Widmeyer and Williams (1991):**
 - **Measured Using GEQ**
 - **Predictors:**
 1. **Personal Factors**
 2. **Team Factors**
 3. **Leadership Factors**
 4. **Environmental Factors**
 - **Strongest Predictor: Personal Satisfaction.**
 - **Best Approach:** Cultivate personal satisfaction towards the team and its members for developing team cohesion.



Lesson 37

TEAM COHESION IN SPORT

Consequences of Team Cohesion

Most research on consequences of team cohesion has focused upon performance. The primary question that has been asked is to what degree team cohesion leads to improved team or individual performance.

Research has consistently shown that a significant relationship exists between team cohesion and athletic performance (Carron & Dennis, 1998; Mullen & Cooper, 1994; Widmeyer, Carron & Brawley, 1993). This observed relationship is much stronger when task cohesion as opposed to social cohesion is involved, and when interactive as opposed to coactive sports are involved. Interactive sports are those team sports, such as volleyball, basketball, and soccer, that require members of the team to interact with one another. Coactive sports are those activities, such as bowling, archery, and riflery, that do not require members of the team to interact with each other for team success.

We will now be discussing other consequences of team cohesion. These include direction of causality

for the cohesion-performance relationship, improving group self-efficacy, predicting future participation, homogeneity of team cohesion, disruptive effects of self-handicapping, and team momentum.

• Direction of Causality for the Cohesion-Performance Relationship

As mentioned before, numerous investigations have verified that a significant and positive relationship exists between direct measures of team cohesion and performance in both individuals

and team sports. Almost all athletes, however, have experienced the “halo effect” of success. When your team is winning, it is a lot easier to feel at one with your team and with your teammates.

Team

cohesion leads to team performance, and Team performance leads to team cohesion.

- Improving Group Self-Efficacy

Research by Kim and Sugiyama (1992) likewise points to the importance of group or team self efficacy in helping teams believe that they will be successful. Teams that have developed high levels

of team cohesion tend to exhibit high of group efficacy as well.

- Predicting Future Participation

Sports participants who exhibit high levels of social cohesion also exhibit high scores in the expectation that they will participate in sport during the following season. This prediction is undoubtedly related to the further observation that high levels of team cohesion are related to lowered state anxiety. Individuals low in state anxiety are more likely to continue sports participation.

- Homogeneity of Team Cohesion

Research indicates that homogeneity of team cohesion among both starter and nonstarter is an important predictor of successful team performance. Successful volleyball teams are characterized by

high levels of team cohesion on the part of both starters and nonstarters. Conversely, less successful

teams are characterized by a lack of homogeneity (agreement) in team cohesion between starters and nonstarters.

- Moderator of the Disruptive Effects of Self-Handicapping

Self-handicapping represents the strategies athletes use to proactively protect their self-esteem by

creating excuses for their performance in forthcoming events through adopting or advocating impediments for success. Typical excuses might include missing practices due to injury or illness, partying and loss of sleep, school commitments or distractions. If success follows, the athlete or athletes can always internalize (take credit for) the victory, but if failure follows, they will have numerous external explanations as to why they have failed. This behavior causes disruption in the athlete's preparation for competition, and is therefore referred to as self-handicapping.

Research indicates that team cohesion has a moderating effect on the trait of self-handicapping.

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- Effects of Team Cohesion on Psychological Momentum

Research demonstrated, using high school volleyball players, that a high level of task cohesion is associated with perceived psychological momentum. Here we learn that teams that enjoy a high level

of task cohesion are more likely to enjoy the benefits of psychological momentum. For teams that are

high in task cohesion, this perception of psychological momentum is likely to be more pronounced.

Developing Team Cohesion

Given that team cohesion is an important characteristic of successful teams, how can it best be developed? In this section we will address that important question in three different ways. First, we

will discuss the development of team cohesion as a process. Second, we will discuss team building as

a way to develop team cohesiveness among team members. Finally, we will identify specific

interventions calculated to enhance team cohesion.

- Team Cohesion as a Process

Very early, Tuckman (1965) described four basic stages that a team must pass through in order to emerge as a cohesive unit. The four stages include forming, storming, norming and performing. In the forming stage, the athletes experience the excitement of a new relationships and getting together with teammates for a common goal or cause. In the storming stage, the athletes struggle

with the frustrations. During the norming stage, members of the team start agreeing upon common

goals. Finally, during the performing stage, the team is ready to perform as a cohesive unit.

- Team Building

Team building is to “promote an increased sense of unity and cohesiveness and enable the team to

function together more smoothly and effectively.” Thus, team building is a process that should lead

to cohesiveness among members of a team.

In the direct intervention approach, the sport psychologist works directly with athletes and employees to empower them, through a series of educational seminars and experiences, to develop a

shared vision, unity of purpose, collaborative teamwork, individual and mutual accountability, team

identity, team cohesiveness, open and honest communication, and trust at all levels. In the indirect

intervention approach, the sport psychologist teaches coaches and managers to conduct team building with their athletes and employees.

- Specific Interventions Designed to Enhance Team Cohesion

1. Acquaint each player with the responsibilities of other players.

2. As a coach or teacher, take the time to learn something personal about each athlete on the team.

3. Develop pride within the sub-units of large teams.

4. Develop a feeling of “ownership” among the players.

5. Set team goals and take pride in their accomplishments.

6. Make sure that each player on the team learns his role and comes to believe it is important.

7. Do not demand or even expect complete social tranquility. The complete elimination of any friction may actually suggest a complete lack of interest in group goals.

8. Since cliques characteristically work in opposition to the task goals of a team, avoid their formation.

9. Develop team drills and lead-up games that encourage member cooperation.

10. Highlight areas of team success, even when the team loses a game or match.



Lesson 37: Team Cohesion in Sport

Consequences of Team Cohesion

- Impact on Performance:

- Significant positive relationship between team cohesion and performance (Carron & Dennis, 1998; Mullen & Cooper, 1994; Widmeyer, Carron & Brawley, 1993).

- Stronger for task cohesion (focus on tasks) than for social cohesion (focus on relationships).

- More pronounced in interactive sports (e.g., volleyball, basketball) compared to coactive sports (e.g., bowling, archery).
- **Direction of Causality:**
 - **Halo Effect:** Success boosts team cohesion, and strong cohesion enhances performance.
- **Improving Group Self-Efficacy:**
 - High team cohesion correlates with high group self-efficacy (Kim & Sugiyama, 1992).
- **Predicting Future Participation:**
 - High social cohesion predicts higher likelihood of continued participation.
 - Lower state anxiety (associated with high cohesion) promotes continued sports involvement.
- **Homogeneity of Team Cohesion:**
 - Homogeneous team cohesion among both starters and nonstarters is crucial for success.
 - Successful teams exhibit high cohesion levels across all members.
- **Moderator of Self-Handicapping:**
 - **Self-Handicapping:** Creating excuses to protect self-esteem by adopting impediments for success.
 - Team cohesion moderates self-handicapping, helping mitigate its disruptive effects.
- **Psychological Momentum:**
 - High task cohesion is associated with perceived psychological momentum.
 - Teams with high task cohesion experience more pronounced psychological momentum.

Developing Team Cohesion

- **As a Process:**
 - **Tuckman's Stages:**
 1. **Forming:** Excitement and formation of relationships.
 2. **Storming:** Struggles and frustrations arise.
 3. **Norming:** Agreement on common goals.
 4. **Performing:** Functioning as a cohesive unit.
- **Team Building:**
 - **Direct Intervention:** Sport psychologists work directly with team members to foster unity, collaboration, accountability, and communication.
 - **Indirect Intervention:** Coaches and managers are taught to conduct team-building activities.
- **Specific Interventions:**
 1. **Understand Responsibilities:** Acquaint players with each other's roles.
 2. **Personal Connections:** Learn something personal about each athlete.
 3. **Pride in Sub-units:** Develop pride within smaller team units.
 4. **Ownership:** Cultivate a sense of ownership among players.
 5. **Set Goals:** Establish team goals and celebrate accomplishments.
 6. **Role Importance:** Ensure each player understands and values their role.
 7. **Manage Friction:** Accept some level of conflict as normal and constructive.
 8. **Avoid Cliques:** Prevent the formation of cliques that oppose team goals.
 9. **Encourage Cooperation:** Design drills and games that promote cooperation.
 10. **Highlight Success:** Emphasize team achievements even in losses.



* Homogeneous team cohesion among both starters and nonstarters is crucial for success.

meaning of starter nontarter?



In sports, **starters** and **nonstarters** refer to different roles within a team:

- **Starters:** These are the players who begin the game on the field or court and are part of the primary lineup. They are usually the main contributors and play the majority of the game.
- **Nonstarters:** These are the players who do not start the game but are typically substituted in at various points. They may play fewer minutes or have specialized roles, such as providing rest for starters or contributing in specific situations.

Homogeneous team cohesion among both starters and nonstarters means that both groups work well together and are aligned in their goals and interactions. This cohesive environment among all members, regardless of their playing time, is important for overall team success.



Lesson 38

LEADERSHIP IN SPORT

Theories of Leadership

Early interest in leadership centered on the traits or abilities of great leaders. It was believed that great leaders were born and not made. Leadership traits are relatively stable personality dispositions

such as intelligence, aggressiveness, and independence. Leadership behaviors have to do with the

observed behavior of leaders and have little to do with their personality. Traits found in all successful

leaders are referred to as universal traits, as opposed to situational traits. Situational traits and situational behaviors are those traits and behaviors that may help make a leader successful in one situation but are of little value in another.

- Universal Trait Theories of Leadership

Trait theory has its origin in the "great man" theory of leadership, which suggests that certain great

leaders have personality traits and personality characteristics that make them ideally suited for leadership. Supporters of trait theory believe that successful leaders have certain personality characteristics or leadership traits that make it possible for them to be successful leaders in any situation.

- Universal Behavior Theories of Leadership

Shortly after World War II the focus in leadership research turned from universal traits to universal

behaviors of successful leaders. It was believed that successful leaders had certain universal behaviors. Once these universal behaviors were identified, they could be taught to potential leaders

everywhere. This approach to leadership was very optimistic, since anyone could learn to be a successful leader simply by learning certain predetermined behavioral characteristics. If these universal behaviors could be mastered, then anyone could be a successful leader. Unlike trait theory,

the belief was that leaders are made, not born.

- Fiedler's Contingency Theory

Fiedler's contingency theory provides an excellent example of a leadership theory that is situation specific, but retains the notion of personality traits. Fiedler's theory is one of many that use the contingency approach. The contingency approach to leadership suggests that leader's effectiveness is

somehow situation-specific, and that leaders that are effective in one situation may not be in another.

In a sense, effective leadership depends on specific environmental situations. However, Fiedler's theory differs from most situational theories, since the emphasis is on relatively stable personality traits, as opposed to behaviors. Thus, a particular personality disposition that seems to be effective in

one leadership situation may not be effective in another.

According to Fiedler (1967), the contingency model of effective leadership posits that the effectiveness of a group is contingent on the relationship style (personality traits) and the degree to

which the situation enables the leader to exert influence.

Situation-Specific Behavior Theories of Leadership

The theories in this section view leadership as a function of the interaction between leader behavior

in a specific situation and the situation itself. In this section we will be discussing two theories:

- Path-goal theory
- Life cycle theory
- Path-Goal Theory

The basic proposition of path-goal theory is that the function of the leader is to provide a "well lighted path" to assist the follower in achieving goals. This is done by rewarding subordinates for goal attainment, pointing out roadblocks and pitfalls on the path to success, and increasing the opportunities for personal satisfaction. For example, if an athlete's goal is to break a school record in

the mile run, it is the coach's job to provide a training program that is rewarding and enables the athlete to accomplish this goal.

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- Life Cycle Theory

Life cycle theory places the emphasis in leadership behavior on the subordinates and not on the leader. The appropriate combination of task and relationship behavior depends on the maturity of the follower.

Coach-Athlete Compatibility

An important factor linked with leader effectiveness is coach-athlete compatibility, or the quality of

the relationship between the coach and the athlete. Compatibility between coach and athlete has been

shown to be an important determinant of team success and satisfaction. In studying coach athlete

compatibility, researchers compare behaviors of effective coach-athlete dyads (pairs) with those of less effective dyads.

Compatible coach-athlete dyads are characterized by good communication and the presence of rewarding behavior flowing from coach to athlete. Conversely, incompatible coach-athlete dyads are

characterized by a lack of communication and rewarding behavior. In compatible dyads, coach and

athlete freely interact with each other. There is a feeling of mutual respect, an appreciation of each other's roles, and a desire to communicate honest feelings. These feelings are not present in the incompatible dyads. Rather, there is a feeling of detachment and isolation from each other.

Effective

and open communication cannot take place in an environment of exclusion. Quality interaction, communication, and respect between coach and athlete lead to athlete satisfaction and improved performance.

Playing Position and Leadership Opportunity

Investigations show that athletes who play in certain central positions on the playing field benefit from greater leadership opportunity. For example, in cricket, a wicketkeeper will probably have greater leadership opportunities than a person fielding at the boundary.

Playing position and stacking

Stacking refers to the disproportionate placement of blacks or minorities into positions of low centrality relative to task dependence and propinquity. If stacking does occur in sport, African Americans should be underrepresented in positions of high centrality. Studies of centrality and racial

segregation have shown that minority players are underrepresented in central positions, where opportunities for leadership are greatest.



Leadership in Sport

Theories of Leadership

- **Universal Trait Theories**
 - **Great Man Theory:** Great leaders are born, not made. Traits like intelligence and independence are seen as essential for leadership.
 - **Universal Traits:** Traits considered beneficial across various situations.
- **Universal Behavior Theories**
 - Focus on observable behaviors rather than traits.
 - Belief that leadership behaviors can be learned, making successful leadership accessible to anyone willing to develop these behaviors.
- **Fiedler's Contingency Theory**
 - Leadership effectiveness depends on the leader's traits and the situation.
 - Suggests that effective leadership varies by situation and leader's personality.

Situation-Specific Behavior Theories

- **Path-Goal Theory**
 - Leaders help followers achieve goals by providing clear paths, rewards, and guidance.
 - Example: A coach providing a structured training program to help an athlete break a record.
- **Life Cycle Theory**
 - Leadership behavior should match the maturity level of the followers.
 - Emphasis is on adapting leadership style based on the followers' development.

Coach-Athlete Compatibility

- **Effective Dyads**

- Good communication and mutual respect.
- Rewarding behavior and open interaction between coach and athlete.
- Leads to athlete satisfaction and improved performance.
- **Incompatible Dyads**
 - Poor communication and lack of respect.
 - Results in detachment and isolation, reducing effectiveness.

Playing Position and Leadership Opportunity

- Athletes in central positions (e.g., wicketkeepers in cricket) have more leadership opportunities compared to those in peripheral positions.

Stacking

- Refers to the disproportionate assignment of minorities to less central positions in sports.
- Minority players are often underrepresented in high-centrality roles with greater leadership opportunities.



Lesson 39

EXERCISE PSYCHOLOGY

Documentation of the psychological benefits of regular exercise has led to the inclusion of “lack of exercise” as a fourth factor for heart disease that can be modified or controlled by the individual. The

other three risk factors are smoking, high blood pressure and elevated cholesterol. Among other things, regular physical exercise helps lower cholesterol, decreases the percentage of body fat, mediates the effects of diabetes, reduces weight, and lowers blood pressure.

Over the next four lectures we will be looking at exercise psychology in great detail. Will be discussing the following topics related to exercise psychology:

- Psychological benefits of exercise
- Theoretical explanations for the relationship between exercise and improved mental health
- Exercise adherence and determinants
- Theories of exercise behavior
- Fitness as a moderator of life stress
- The immune system, cancer, HIV and exercise
- Social physique anxiety
- Exercise addiction
- Eating disorders

Psychological Benefits of Exercise

A large body of literature has been amassed that supports the position that regular exercise leads to

improved psychological affect. Improved psychological affect is manifested in the form of a reduction in negative affect (e.g., anxiety and depression) and an increase in positive affect (e.g., self efficacy, vigor, well-being).

Exercise in many cases is as effective as psychotherapy and antidepressant drugs in treating emotional disorder. These conclusions are supported by several narrative reviews and meta-analysis

(Craft & Landers, 1998), Hale, Koch and Raglin, 2000; Van Landuyt, Ekkekakis, Hall & Petruzzello, 2000).

Type of exercises

Studies cited to support the relationship between exercise and psychological affect benefits have used both acute and chronic exercises. Acute exercise refers to exercise that is of short duration (e.g., thirty minutes). Chronic exercise refers to long-term exercise (e.g., twelve months). Aerobics exercise refers to exercise that is accomplished at an exercise intensity that allows for the intake of sufficient oxygen to maintain continuous exercise. Anaerobic exercise is one in which the exerciser does not get enough oxygen to maintain continuous exercise. Anaerobic exercise requires the athlete to breathe hard following exercise in order to replenish stored energy. After a bout of anaerobic exercise, the athlete will need period to time to "catch her breath." This is not generally necessary with aerobic exercises. Synchronized swimming is an example of anaerobic exercise. Resistance exercise usually involves the use of weights or weight training to provide resistance to the muscles, for example, weight lifting.

Special Populations

We turn our attention now to the beneficial psychological effects of regular physical activity on special populations of people. Special populations can be divided into three categories; clinical patients, children and elderly, and disabled individuals.

Clinical Patients

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The benefits of regular physical activity are even greater for individuals suffering from psychological disorders than for normal individuals. Using a meta-analysis procedure, it is observed that:

1. Both aerobic and nonaerobic exercises were effective in reducing clinical depression.
2. More depressed individuals benefit more from exercise.
3. Exercise was as beneficial as psychotherapy and drug therapy for reducing depression.
4. Long-term exercise programs are more effective than short-term programs for reducing depression in the clinically ill.

In addition to treating depression and anxiety, exercise is also an effective treatment for clinical patients suffering from schizophrenia. Results of the exercise program showed psychological improvements in the form of reduced auditory hallucinations and better sleep patterns. In another

investigation, Martinsen, Raglin, Hofart, and Friis (1998) demonstrated that patients suffering from

severe panic disorder could safely undergo vigorous exercise without suffering panic attacks.

Children and Elderly

Research has also shown the beneficial effects of exercise on children and older adults. Children's exercise behaviors are greatly uninfluenced by their parents' attitudes and behaviors regarding exercise. Factors that can influence a child's decision to be physically active include parents' beliefs,

the children's perception of their own competence, and, to some extent their goal orientation.

Regarding exercise and the elderly, research shows that participation in aerobic exercise selectively

preserves some cognitive functioning that normally declines with age.

So, for the elderly, there is not only the benefit of improved fitness and improved psychological affect associated with exercise, but also the prospect of slowing the decline of some cognitive functions. Exercise in the elderly is associated with the preservation of certain aspects of memory

and spatial relationships and reduction in confusion, tension, and anger.

Disabled Individuals

Finally, it is important to mention that the beneficial psychological effects of regular exercise extend

to physically challenged individuals as well. For example, wheelchair sport participants have been observed to enjoy greater psychological benefits from physical activity than wheelchair nonparticipants in sport.

Moderating Variables

In addition, research suggests that there are a number of variables that may moderate or facilitate the effectiveness of regular exercise on mental well-being.

1.

Time of Day

Research suggests that time of day is not an important factor relative to psychological benefits derived from exercise. You get just as much affective benefit from running in the morning as from doing so at midday or in the evening.

2. Mode of Exercise

The psychological benefits of acute aerobic exercise appear to be the same, regardless of the mode of exercise. Cox et al. (2000) observed no difference in anxiety between treadmill and stepper exercisers following a thirty-minute bout of exercise.

3. Music

Research suggests that listening to music during exercise can increase positive affect in the exerciser.

4. Attentional Strategy

Relative to attentional strategy, there is evidence that a dissociative (external) attentional strategy may result in greater psychological benefits (Masters & Ogles, 1998b). Runners who are asked to listen to their own heart rates during exercise (internal focus) exhibit greater emotional stress than runners who dissociate during exercise.

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5. Social Environment

The social environment associated with exercise has an effect upon psychological benefits of exercise. Research results show increased benefits for the socially enriched environment in the form of revitalization and self-efficacy.



Exercise Psychology

Psychological Benefits of Exercise

- **General Benefits**

- Exercise reduces negative affect (e.g., anxiety, depression) and increases positive affect (e.g., self-efficacy, vigor).
- Regular exercise can be as effective as psychotherapy and antidepressant drugs for treating emotional disorders (Craft & Landers, 1998; Hale, Koch, & Raglin, 2000; Van Landuyt, Ekkekakis, Hall, & Petruzzello, 2000).

• Types of Exercises

- **Acute Exercise:** Short-duration exercise (e.g., 30 minutes).
- **Chronic Exercise:** Long-term exercise (e.g., 12 months).
- **Aerobic Exercise:** Sustained exercise with sufficient oxygen intake (e.g., jogging).
- **Anaerobic Exercise:** Short bursts of high-intensity exercise without continuous oxygen intake (e.g., weight lifting).
- **Resistance Exercise:** Involves weight training to strengthen muscles.

Special Populations

• Clinical Patients

- **Depression:** Both aerobic and nonaerobic exercises reduce clinical depression; more depressed individuals benefit significantly.
- **Schizophrenia:** Exercise helps reduce auditory hallucinations and improves sleep.
- **Panic Disorder:** Vigorous exercise can be safely undertaken without inducing panic attacks (Martinsen, Raglin, Hofart, & Friis, 1998).

• Children and Elderly

- **Children:** Exercise behaviors are influenced by parents' beliefs, the child's self-perception, and goal orientation.
- **Elderly:** Aerobic exercise helps preserve cognitive functions, such as memory and spatial relationships, and reduces confusion, tension, and anger.

• Disabled Individuals

- Exercise provides greater psychological benefits to wheelchair sport participants compared to nonparticipants.

Moderating Variables

• Time of Day

- Time of day does not significantly affect the psychological benefits of exercise.

• Mode of Exercise

- Psychological benefits are similar across different modes of acute aerobic exercise (e.g., treadmill vs. stepper) (Cox et al., 2000).

• Music

- Listening to music during exercise can enhance positive affect.

• Attentional Strategy

- A dissociative (external) attentional strategy (e.g., listening to music) may lead to greater psychological benefits compared to an internal focus (e.g., monitoring heart rate) (Masters & Ogles, 1998b).

• Social Environment

- A socially enriched exercise environment boosts revitalization and self-efficacy.



Health

Many hypotheses have been proposed to explain why exercise is associated with improved mental health. We will be discussing six of them. The first three explanations are considered to be psychological in nature, while the remaining three are physiological in nature.

Psychological Explanations

- Cognitive Behavioral Hypothesis

The basic premise of the cognitive behavioral hypothesis is that exercise encourages and generates

positive thoughts and feelings that serve to counteract negative mood states such as depression, anxiety, and confusion.

When individuals master tasks they perceive to be difficult, they experience an increase in self-efficacy. Exercise is perceived by nonexercisers as a difficult task. When a nonexerciser succeeds in

becoming a regular exerciser, she experiences a feeling of accomplishment and self-efficacy. An increase in self-efficacy is helpful in breaking the downward spiral of negative affect associated with

depression, anxiety, and other negative mood states.

- Social Interaction Hypothesis

The basic premise of the social interaction hypothesis is that social interaction associated with exercising with friends and colleagues is pleasurable and has the net effect of improving mental health.

- Distraction Hypothesis

The basic premise of the distraction hypothesis is that exercise affords an opportunity for individuals

to be distracted from their worries and frustrations. It may be that distraction provides a viable explanation for short-term reduction in depression and anxiety, but not for long-term reduction.

Physiological Explanations

- Cardiovascular Fitness Hypothesis

The basic premise of the cardiovascular fitness hypothesis is that improved mood state is associated

with improved cardiovascular fitness.

- Amine Hypothesis

The basic premise of the amine hypothesis is that increased secretion of chemicals that serve as neurotransmitters is related to improved mental health. Theoretically, exercise stimulates the production of neurotransmitters that in turn have a positive effect upon psychological mood.

- Endorphin Hypothesis

The endorphin hypothesis postulates that exercise is associated with brain production of chemicals

that have a "morphine-like" effect on the exerciser (pain reduction and general euphoria). This effect

has been referred to, in popular literature, as the "runner's high." The general euphoria produced by

the endorphins serves to reduce the levels of depression, anxiety, confusion, and other negative mood states.

Exercise Adherence and Determinants

Another important aspect of exercise psychology is to determine what motivates individuals to start

exercising, what motivates them to adhere to an exercise program, and what motivates them to try

again after failing the first (or second) time.

- Transition from Sedentary State to Exercise Adoption

Exercise determinants motivate individuals to make the transition from sedentary lifestyle to regular exercise. Individuals likely to adopt a vigorous exercise lifestyle exhibit the following characteristics:

1. Confidence they can succeed at a vigorous exercise program (exercise self-efficacy)
2. Knowledge about what constitutes a healthy lifestyle
3. Knowledge about the importance and value of regular exercise

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4. The perception that they enjoy a high level of self-control
5. Good attitudes about the value and importance of regular exercise
6. Initial condition of not being overweight or obese

It has often been hypothesized that parental involvement in vigorous physical activity would be a strong determinant of the exercise behavior of their children. What does predict vigorous physical

activity in children is parental beliefs about the value and importance of vigorous physical activity.

- Transition from Adoption to Maintenance or Dropout Status

Determinants of exercise adherence are:

1. Available time
2. Behavioral coping skills
3. Equipment and facility accessibility
4. Exercise self-efficacy
5. Group cohesion
6. High risk of heart disease
7. Intrinsic motivation
8. Personal perception of good health
9. Social support

- Determinants of exercise nonadherence are:

1. Being a blue-collar worker
2. Being overweight or obese
3. Mood state disturbance relative to exercise
4. Physical discomfort during exercise
5. Being a smoker
6. Social physique anxiety

- Transition from Dropout Status to Exercise Resumption

The transition from being an exercise dropout to resuming a vigorous exercise program is an important part of the exercise psychology literature but statistics are not available on the percentage

of people who drop out off an exercise program and then get started again.



Exercise Psychology

Theoretical Explanations for the Relationship between Exercise and Improved Mental Health Psychological Explanations

1. Cognitive Behavioral Hypothesis

- **Premise:** Exercise promotes positive thoughts and feelings, countering negative moods such as depression and anxiety.
- **Mechanism:** Success in exercising boosts self-efficacy, which helps break the cycle of negative affect associated with mood disorders.

2. Social Interaction Hypothesis

- **Premise:** Socializing during exercise with friends or colleagues improves mental health.
- **Mechanism:** The social interactions associated with exercise contribute to pleasure and mental well-being.

3. Distraction Hypothesis

- **Premise:** Exercise provides a distraction from worries and frustrations.
- **Mechanism:** Short-term reductions in depression and anxiety may be due to this distraction, though it may not explain long-term benefits.

Physiological Explanations

1. Cardiovascular Fitness Hypothesis

- **Premise:** Improved mood is linked to better cardiovascular fitness.
- **Mechanism:** Enhanced physical fitness positively impacts psychological well-being.

2. Amine Hypothesis

- **Premise:** Exercise increases neurotransmitter secretion, which improves mental health.
- **Mechanism:** Exercise stimulates neurotransmitter production, which positively affects mood.

3. Endorphin Hypothesis

- **Premise:** Exercise leads to the release of endorphins, which have morphine-like effects (e.g., pain reduction and euphoria).
- **Mechanism:** Endorphins reduce negative moods and create a feeling of well-being, often referred to as the "runner's high."

Exercise Adherence and Determinants

Transition from Sedentary State to Exercise Adoption

- **Characteristics of Likely Adopters:**

1. Confidence in success (exercise self-efficacy)
2. Knowledge of healthy lifestyles
3. Understanding the importance of exercise
4. Perception of high self-control
5. Positive attitude towards exercise
6. Not overweight or obese

- **Influence of Parental Involvement:**

- Parental beliefs about exercise are strong predictors of children's physical activity, rather than parental involvement itself.

Transition from Adoption to Maintenance or Dropout Status

- **Determinants of Adherence:**

1. Available time
2. Behavioral coping skills
3. Access to equipment and facilities
4. Exercise self-efficacy
5. Group cohesion
6. High risk of heart disease
7. Intrinsic motivation

8. Perception of good health

9. Social support

- **Determinants of Nonadherence:**

1. Blue-collar work status

2. Overweight or obesity

3. Mood disturbances related to exercise

4. Physical discomfort during exercise

5. Smoking

6. Social physique anxiety

Transition from Dropout Status to Exercise Resumption

- Statistics on the percentage of individuals who resume exercise after dropping out are not available, highlighting the need for further research in this area.



elaborate:

Transition from Adoption to Maintenance or Dropout Status

Determinants of Adherence:

Available time

Behavioral coping skills

Access to equipment and facilities

Exercise self-efficacy

Group cohesion

High risk of heart disease

Intrinsic motivation

Perception of good health

Social support

Determinants of Nonadherence:

Blue-collar work status

Overweight or obesity

Mood disturbances related to exercise

Physical discomfort during exercise

Smoking

Social physique anxiety

Transition from Dropout Status to Exercise Resumption



Transition from Adoption to Maintenance or Dropout Status

Determinants of Exercise Adherence:

1. Available Time

- **Explanation:** Individuals with sufficient time for exercise are more likely to maintain their exercise routine. Time constraints often lead to dropout if individuals cannot integrate exercise into their schedules.

- **Strategies:** Time management techniques, scheduling workouts into daily routines, and prioritizing exercise can help ensure consistency.

2. Behavioral Coping Skills

- **Explanation:** Effective coping skills enable individuals to manage obstacles and setbacks related to exercise. These skills help maintain motivation and commitment despite challenges.
- **Strategies:** Developing strategies to handle barriers (e.g., work, family commitments) and employing problem-solving techniques are crucial.

3. Access to Equipment and Facilities

- **Explanation:** Availability of exercise facilities or equipment affects adherence. Lack of access can be a significant barrier to regular exercise.
- **Strategies:** Utilizing home-based workouts, community gyms, or local parks can help overcome accessibility issues.

4. Exercise Self-Efficacy

- **Explanation:** Self-efficacy refers to one's belief in their ability to successfully complete exercise routines. Higher self-efficacy is associated with greater adherence.
- **Strategies:** Setting achievable goals, tracking progress, and celebrating successes can boost self-efficacy.

5. Group Cohesion

- **Explanation:** Being part of a group that exercises together can increase motivation and adherence. Social support and shared goals contribute to maintaining an exercise routine.
- **Strategies:** Joining exercise groups, classes, or teams can enhance motivation and accountability.

6. High Risk of Heart Disease

- **Explanation:** Individuals with a high risk of heart disease may be more motivated to adhere to exercise regimens due to health concerns.
- **Strategies:** Engaging in exercise as part of a health management plan and consulting with healthcare providers can reinforce commitment.

7. Intrinsic Motivation

- **Explanation:** Intrinsic motivation involves engaging in exercise for personal satisfaction and enjoyment rather than external rewards. It is a strong predictor of long-term adherence.
- **Strategies:** Finding enjoyable activities, setting personal goals, and focusing on the intrinsic benefits of exercise can enhance motivation.

8. Perception of Good Health

- **Explanation:** Individuals who perceive themselves as healthy are often more likely to adhere to exercise routines to maintain their health status.
- **Strategies:** Regular health assessments and tracking fitness progress can help reinforce the perception of health and the importance of continued exercise.

9. Social Support

- **Explanation:** Support from family, friends, or exercise partners can significantly influence adherence. Encouragement and accountability from others help maintain commitment.
- **Strategies:** Building a network of supportive individuals and sharing exercise goals with them can improve adherence.

Determinants of Nonadherence:

1. Blue-Collar Work Status

- **Explanation:** Blue-collar workers may face physical exhaustion or time constraints that interfere with their ability to exercise regularly.

- **Strategies:** Incorporating shorter, more flexible workouts and emphasizing the importance of exercise for overall health can help.

2. Overweight or Obesity

- **Explanation:** Individuals who are overweight or obese may experience barriers such as low self-esteem or physical discomfort that hinder adherence.
- **Strategies:** Providing tailored exercise programs and focusing on gradual progress can address these challenges.

3. Mood Disturbances Related to Exercise

- **Explanation:** Negative mood states such as depression or anxiety can reduce motivation and adherence to exercise routines.
- **Strategies:** Combining exercise with psychological support and focusing on mood-enhancing aspects of physical activity can help.

4. Physical Discomfort During Exercise

- **Explanation:** Discomfort or pain experienced during or after exercise can discourage individuals from maintaining their routine.
- **Strategies:** Ensuring proper technique, gradually increasing intensity, and incorporating rest and recovery can minimize discomfort.

5. Smoking

- **Explanation:** Smoking can negatively impact physical fitness and overall health, reducing motivation to adhere to exercise routines.
- **Strategies:** Addressing smoking cessation alongside exercise programs can improve adherence and overall health outcomes.

6. Social Physique Anxiety

- **Explanation:** Anxiety about how one's body is perceived in social settings can lead to avoidance of exercise environments.
- **Strategies:** Creating supportive and non-judgmental exercise environments and focusing on personal fitness goals can help mitigate social physique anxiety.

Transition from Dropout Status to Exercise Resumption

- **Explanation:** Individuals who have previously dropped out of an exercise routine may face unique challenges when trying to resume. The transition involves overcoming barriers and re-establishing commitment.
- **Strategies:**
 1. **Reevaluation of Goals:** Setting new, realistic goals and reassessing motivations can help re-engage individuals.
 2. **Support Systems:** Engaging with supportive friends, family, or exercise groups can provide encouragement and accountability.
 3. **Gradual Reintroduction:** Starting with less intense workouts and gradually increasing the duration and intensity can ease the transition.
 4. **Overcoming Barriers:** Identifying and addressing the reasons for previous dropout can help prevent future lapses.
 5. **Tracking Progress:** Monitoring progress and celebrating small successes can reinforce commitment and build motivation.

Understanding these determinants and strategies can help individuals and professionals address exercise adherence issues and support long-term commitment to physical activity.



Lesson 41

EXERCISE PSYCHOLOGY

Theories of Exercise Behavior

Psychological models of human behavior have been applied to the exercise-setting in an attempt to

explain why people don't exercise, why they start to exercise, and why they do or do not continue to

exercise, and why they start exercising again if they stop. These models include:

- The Theory of Reasoned Action
- The Theory of Planned Behavior
- The Transtheoretical Model
- Social Cognitive Theory
- The Theory of Reasoned Action

The theory of reasoned action proposes that the main precursor of a behavior such as exercise is the

individual's intention to perform the behavior. The intention to perform the behavior is determined

by the individual's attitude towards the behavior as well as social norms or social pressure to perform

the behavior. Research by Estabrooks and Courneya (1997) has demonstrated the effectiveness of the theory of reasoned action in exercise settings. While the theory of reasoned action is a viable model for predicting exercise behavior, research has demonstrated that its predictive power is increased when personal control is added to the model. This observation led to the development of

the theory of planned behavior.

- The Theory of Planned Behavior

The theory of planned behavior is an extension of the theory of reasoned action. The intention to perform a behavior is fundamental to the theory. Intention is determined by the individual's attitude

towards the behavior and social norms. The difference between the theory of reasoned action and theory of planned behavior is the addition of behavioral control to the latter model. An individual will maintain or initiate an exercise program if his intention is firm and he feels in control.

Intention

is in turn a function of his attitude towards exercise and perceived social support. Several researches

have demonstrated general support for the theory of planned behavior in predicting exercise behavior. Strong support for the theory of planned behavior is also provided through meta-analysis

reported by Hausenblas, Carron, and Mack (1997).

- The Transtheoretical Model

According to the transtheoretical model, individuals pass through five dynamic stages in adopting healthy long-term exercise behavior. The stages are dynamic, because individuals may move in and

out of the several stages before reaching the final stage, which is also dynamic.

The five stages are:

1.

(5)

Precontemplation

2.

3.

- 4.
- 5.
- Contemplation
- Preparation
- Action
- Maintenance

Other factors that interact include self-efficacy, perception of gains and losses, and a set of psychological obstacles that may need to be addressed (e.g., personal or family conflicts).

- Social Cognitive Theory

Social cognitive theory provides a viable way to explain exercise behavior. Individuals who are dissatisfied with their current exercise behavior who exhibit high levels of exercise self-efficacy, and

who set exercise goals are generally able to achieve their goals. Exercise self-efficacy is a powerful predictor of exercise behavior. Individuals who believe in themselves and believe that they can be successful at maintaining an exercise program generally are successful.

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Fitness as a Moderator of Life Stress

Given the positive relationship between exercise and improved mental health, it follows that physical

fitness should serve as a buffer against life stress. The ability of individuals to insulate, protect, or inoculate themselves against the stresses of life through regular exercise is called stress inoculation.

Research shows that the psychological benefits associated with regular exercise do not normally require an increase in physical fitness. Aerobics fitness, however, does appear to be a necessary precursor to the stress inoculation effect. Aerobically fit individuals appear to be inoculated against

stress, illness, and the general hassles of life to a greater extent than less aerobically fit individuals.

Children and adults who engage in healthy behavior that leads to physical fitness can insulate themselves from various physical and psychological health problems throughout their lives.

Life stress represents an accumulation of the daily hassles and challenges of living out our lives.

Individuals who exercise regularly and maintain a high level of physical fitness are less susceptible to

the negative effects of life stress. Evidence of this hypothesis has been provided by a number of researchers. Research results of an investigation show an interactive relationship between life stress,

physical fitness, and number of visits to the health center (illness). Being physically fit serves to inoculate the individual against illness during periods of high stress. Conversely, physically unfit individuals appear to be unprotected against high stress.



Theories of Exercise Behavior

1. Theory of Reasoned Action

- **Overview:** This theory posits that the primary predictor of engaging in a behavior (such as exercise) is the individual's intention to perform that behavior. This intention is shaped by two factors:
 - **Attitude:** How positively or negatively the individual views the behavior.
 - **Social Norms:** The perceived social pressures or expectations regarding the behavior.
- **Research Support:** Estabrooks and Courneya (1997) found that the Theory of Reasoned Action is effective in predicting exercise behavior. The model is enhanced when personal control over the behavior is considered, leading to the development of the Theory of Planned Behavior.

2. Theory of Planned Behavior

- **Overview:** An extension of the Theory of Reasoned Action, this theory incorporates:
 - **Behavioral Control:** The perceived ease or difficulty of performing the behavior, which affects intention and behavior.
- **Components:**
 - **Intention:** A function of attitude, social norms, and perceived control.
 - **Attitude:** The individual's evaluation of the behavior.
 - **Social Norms:** The influence of social pressures.
 - **Behavioral Control:** The perceived ability to perform the behavior.
- **Research Support:** Hausenblas, Carron, and Mack (1997) provided meta-analytic support for the Theory of Planned Behavior in predicting exercise behavior.

3. Transtheoretical Model (Stages of Change)

- **Overview:** This model outlines five stages individuals go through when adopting and maintaining healthy behaviors:
 1. **Precontemplation:** Not considering exercise.
 2. **Contemplation:** Thinking about starting to exercise.
 3. **Preparation:** Planning to start exercising soon.
 4. **Action:** Actively engaging in regular exercise.
 5. **Maintenance:** Sustaining regular exercise over the long term.
- **Dynamic Nature:** Individuals may move back and forth between stages. Success depends on addressing self-efficacy, perceived benefits and barriers, and psychological obstacles.

4. Social Cognitive Theory

- **Overview:** This theory emphasizes the role of cognitive processes in behavior change:
 - **Exercise Self-Efficacy:** The belief in one's ability to successfully perform and maintain an exercise routine.
 - **Goal Setting:** Individuals with high self-efficacy and clear goals are more likely to achieve their exercise objectives.
- **Research Support:** High levels of exercise self-efficacy are strongly associated with successful exercise behavior.

Fitness as a Moderator of Life Stress

- **Concept:** Physical fitness can act as a buffer against life stress, known as stress inoculation. This means that being physically fit can help individuals manage stress more effectively.
- **Aerobic Fitness:** Aerobic (cardiovascular) fitness is particularly important for stress inoculation. Aerobically fit individuals are better able to cope with stress and maintain overall health compared to those with lower aerobic fitness.

- **Research Evidence:** Studies show that regular exercise can mitigate the negative effects of life stress. Physically fit individuals experience fewer illnesses and better manage stress compared to their less fit counterparts. For example:
 - **Health Center Visits:** Evidence suggests that physically fit individuals have fewer health center visits during high-stress periods.
 - **Stress and Illness:** Fitness provides protection against illness during stressful times, while lack of fitness is associated with higher susceptibility to stress-related health problems.

Key Points for Application:

- **Exercise Intention and Planning:** Use the Theory of Planned Behavior to set clear, actionable exercise goals and enhance self-efficacy.
- **Stage-Based Strategies:** Apply the Transtheoretical Model to tailor interventions based on the individual's current stage of behavior change.
- **Social Support and Self-Efficacy:** Leverage social support and build self-efficacy to improve adherence and performance in exercise routines.
- **Fitness and Stress Management:** Promote regular aerobic exercise as a strategy for stress management and overall health maintenance.



Lesson 42

EXERCISE PSYCHOLOGY

The Immune System, Cancer, HIV and Exercise

In recent years research has linked two of the great plagues of our time (cancer and HIV) to exercise

and its effect on the immune system. While exercise has generally been linked to benefits, in cases of

excess it can have negative consequences. It is like a two-edged sword: it cuts both ways. If applied in

moderation it can have beneficial effects, but if applied in excess it can have negative effects.

Exercise and Cancer

Moderate exercise is linked to a lowered incidence of colon and breast cancer. Young women who regularly participate in physical exercise activities during their reproductive years have a reduced risk

of breast cancer. Individuals who have cancer, but who exercise regularly, may benefit from improved psychological well-being, preservation of lean tissue, and enhanced immune system.

Exercise and the Immune System

Research clearly suggests that exercising in moderation leads to improved psychological mood and

enhanced immune system functioning. Conversely, it is widely believed that chronic intense and stressful exercise may result in mood disturbance and in suppression of the immune system.

There is

growing evidence that for several hours following heavy sustained exertion, the immune system is suppressed.

It is believed that the immune system is stimulated and strengthened by moderately intense exercise,

but suppressed by overly intense exercise. It appears that there exists an optimal level of regular physical activity conducive to the resistance to illness. Apparently, you can have too much of a good thing.

Exercise and Human Immunodeficiency Virus

It is widely believed by medical professionals that the presence of the HIV ultimately leads to acquired

immune deficiency syndrome (AIDS). Increased anxiety and depression should be viewed as risk factor facilitating the development of AIDS. Because exercise has been positively linked with decreased anxiety and depression, it follows that chronic exercise is effective in retarding the negative

progression and effects of HIV.

Research suggests that moderate exercise is an effective complementary therapy for treating psychological manifestations associated with HIV infections.

Social Physique Anxiety, Physical Self-Concept, and Body Image

Social physique anxiety, physical self-concept, and body image are all constructs that describe how an

individual feels about her physical body. While these constructs are not identical to each other, they

are correlated, and are predictive to exercise behavior. A high score on social physique anxiety and

low scores on physical self-concept and body image are predictive of a low level of exercise behavior.

Individuals who are anxious about their bodies, have low physical self-concepts, and have low body

images have a hard time getting motivated to exercise.

Social physique anxiety is the anxiety people experience when they perceive that other people evaluate their physiques negatively. Physical self-concept is the perception that people have about themselves relative to the physical-self. Physical self-concept is closely tied to the notion that an individual's feeling of self-worth and self-esteem is related to how he perceives himself within his body. Body image refers to the images or pictures people have about their bodies. The image that a

person has about her own body can be quite different from the one that other people have of her.

Research shows that physically active people have better body image than physically inactive people.

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Exercise Addiction

Exercise addiction is generally defined as a psychophysiological dependence on a regular regimen of

exercise. The normal benefits associated with regular exercise at a moderate intensity are lost for the

exercise-addiction individual. Failure to exercise according to schedule results in a mood state disturbance in the addicted individual. From an attributional perspective, the addicted exerciser is controlled by the activity, as opposed to the activity's being controlled by the exerciser. Compared to

nonaddicted exerciser, addicted exercisers report being more restless and stressed out prior to an exercise bout. They also experience a higher degree of depression, anxiety, and general discomfort

when they miss a scheduled workout. An important characteristic of the exercise addict is that he will

generally insist on exercising in the face of physical pain or injury.

Another term used to describe the addicted exerciser is obligatory runner. Obligatory runners are

highly motivated to exercise, and when they can't, they experience abnormal feelings of anxiety and psychological discontent.

Eating Disorders and Physical Activity

Actual clinically diagnosed eating disorders are relatively rare among athletes and physical activity enthusiasts. Much more prevalent are a whole array of unhealthy subclinical eating disorders. We will

discuss both briefly.

Clinically diagnosed eating disorders

The two most severe clinically diagnosed or pathogenic eating disorders are anorexia nervosa and bulimia nervosa.

• Anorexia nervosa

They exhibit the following criteria:

1. Severe weight loss
2. Refusal to maintain normal body weight
3. Intense fear of gaining weight or becoming fat
4. Severe body image disturbance

Treatment and recovery requires professional help. The diagnosed anorexic cannot overcome this mental illness herself.

• Bulimia Nervosa

They exhibit the following criteria:

1. Binge eating followed by purging at least twice per week for three months
2. Loss of self-control
3. Severe body image disturbance

Bulimics are preoccupied with food and weight, fear getting fat, and exhibit chaotic eating behaviors.

Unlike anorexics, bulimics turn to food, rather than away from it. As with anorexia nervosa, treatment and recovery from bulimia nervosa requires professional help.

Sub clinical Eating Disorder among Athletes

Males and females involved in activities that link leanness to success are often pressured to be thin.

This is especially true for female athletes involved in gymnastics and dancers. In an effort to be thin

and to meet their coaches' expectations, athletes may turn to a number of questionable eating and

exercise behaviors that may compromise their health. Several recent studies have reported on subclinical eating disorders among athletes. Of particular interest is a meta-analysis involving 92 studies, and 10, 878 athletes (Hausenblas & Carron, 1999). Results of this meta-analysis are as follows:

- 1.
- 2.

Athletes report more eating disorder symptoms than nonathletes.

Athletes competing in aesthetic sports report more eating disorder than those in nonathletic sports (e.g., gymnastics, dance, diving).

- 3.

Athletes do not have a greater drive for thinness than nonathletes.

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Eating Disorder and Unhealthy Exercise Behavior

In some cases eating disorders have been linked to potentially unhealthy exercise behavior. Now let

us briefly discuss some of those behaviors and relationships.

- **Anorexia Analogue Hypothesis**

It is hypothesized that male obligatory runners and anorexic females share common personality characteristics and a common drive for thinness which is through excessive exercise. This is the anorexia analogue hypothesis. Research provides only partial support of this hypothesis.

- **Muscle Dysmorphia**

Muscle dysmorphia is defined as a preoccupation with the notion that one is not sufficiently muscular. Individuals classified as being muscle dysmorphic, think constantly about their muscularity

and have little control over compulsive weightlifting and dietary regimens.

- **Wrestlers Making Weight**

The adverse physiological effects of rapid weight loss or “cutting” in preparation for competition are

well documented in the literature. Rapid weight loss results in decreased plasma volume, dehydration,

and hypoglycemia. It also results in a cognitive functioning decrement. Cognitive decrements are noted primarily in short-term memory and digit span recall. Mood state disturbances are also noted

in such people.



Exercise and Health: Key Areas

1. Exercise and Cancer

- **Moderate Exercise:** Linked to a reduced incidence of colon and breast cancer.
- **Young Women:** Regular physical activity during reproductive years lowers breast cancer risk.
- **Cancer Patients:** Regular exercise can improve psychological well-being, preserve lean tissue, and enhance immune function.

2. Exercise and the Immune System

- **Moderate Exercise:** Enhances psychological mood and immune system function.
- **Excessive Exercise:** Chronic intense exercise can lead to mood disturbances and immune suppression.
- **Optimal Exercise:** There is an optimal level of exercise that boosts immune resistance without leading to suppression. Overdoing exercise can be detrimental.

3. Exercise and HIV

- **HIV Progression:** Exercise is believed to help manage anxiety and depression, which are risk factors for AIDS development.
- **Complementary Therapy:** Moderate exercise is effective in treating psychological issues associated with HIV infection.

4. Social Physique Anxiety, Physical Self-Concept, and Body Image

- **Social Physique Anxiety:** Anxiety about others evaluating one's physique negatively.

- **Physical Self-Concept:** Perception of oneself related to physical attributes, impacting self-worth and self-esteem.
- **Body Image:** Personal mental image of one's body, which may differ from external perceptions.
- **Exercise Behavior:** High social physique anxiety and low physical self-concept correlate with lower exercise levels. Physically active individuals tend to have better body images.

5. Exercise Addiction

- **Definition:** Psychophysiological dependence on exercise, where the benefits are lost, and missing workouts causes mood disturbances.
- **Characteristics:**
 - Restlessness and stress when missing exercise.
 - Higher levels of depression, anxiety, and discomfort if unable to exercise.
 - Exercising despite physical pain or injury.
- **Obligatory Runners:** Highly motivated individuals who experience anxiety and psychological distress when unable to exercise.

6. Eating Disorders and Physical Activity

- **Clinically Diagnosed Eating Disorders:**
 - **Anorexia Nervosa:** Severe weight loss, refusal to maintain normal weight, intense fear of gaining weight, and severe body image disturbance. Requires professional treatment.
 - **Bulimia Nervosa:** Binge eating followed by purging, loss of self-control, and severe body image disturbance. Treatment requires professional help.

7. Subclinical Eating Disorders Among Athletes

- **Prevalence:** Athletes, especially in leanness-focused sports (e.g., gymnastics, dance), often experience subclinical eating disorders.
- **Meta-Analysis Findings (Hausenblas & Carron, 1999):**
 - Athletes report more eating disorder symptoms than nonathletes.
 - Athletes in aesthetic sports show more eating disorder symptoms than those in non-aesthetic sports.
 - Athletes do not have a greater drive for thinness compared to nonathletes.

8. Eating Disorders and Unhealthy Exercise Behavior

- **Anorexia Analogue Hypothesis:** Suggests shared traits between male obligatory runners and anorexic females, including a drive for thinness through excessive exercise. Evidence is partial.
- **Muscle Dysmorphia:** Preoccupation with insufficient muscularity, leading to compulsive weightlifting and dietary control.
- **Wrestlers Making Weight:**
 - **Rapid Weight Loss:** Leads to decreased plasma volume, dehydration, hypoglycemia, cognitive impairments (e.g., short-term memory), and mood disturbances.

Key Takeaways:

- **Moderation is Crucial:** Both exercise and diet must be managed to avoid negative health effects.
- **Comprehensive Approach:** Addressing psychological, physiological, and behavioral factors is essential in managing health and exercise-related issues.
- **Professional Help:** Essential for managing severe eating disorders and psychological issues related to exercise and body image.



Lesson 43

BURNOUT IN ATHLETES

Too much exercise can result in a reduction in the effectiveness of the immune system to fight disease and an increase in negative psychological mood. In a very practical way, exercise can be considered along a continuum from not enough exercise to too much exercise. Negative psychological and biological outcomes are associated with too little and too much exercise. Over training in athletes represents a paradox, because many of the benefits associated with exercise are

reversed in the athlete who trains too much. For the athlete, the question of how much is too much

is a complex one. Athletes are continually challenging the delicate balance between training and overtraining, since high levels of training are required for success in sport.

The lecture is divided into four sections that include:

- Defining burnout and other terms
- Models of burnout
- Symptoms and interventions for burnout
- Recommendations for athletes, coaches, and parents.

Defining Burnout and Other Related Terms

Burnout in sport and exercise is “a syndrome of physical/emotional exhaustion, sport devaluation, and reduced athletic accomplishment”. As can be observed, the focus of this particular definition of

burnout is upon physical and mental exhaustion, reduced interest in sport, and reduced performance.

Common terms include overtraining, overreaching, staleness, and withdrawal.

Overtraining and Overreaching

Overtraining implies that the athlete trains beyond the level that is ideal for maximum benefit.

Overtraining is maladaptive behavior that may lead to staleness and burnout. Overreaching is a form

of short-term overtraining that is a part of normal training. Overreaching that extends for long periods of time at high intensities eventually becomes overtraining. Staleness is the initial failure of

the body to adapt to training.

Models of Burnout

We will look at three different models of burnout. They are:

1. The stress model
2. The investment model
3. The sociologically based empowerment model

1. Stress Model of Burnout

The combination of staleness, overtraining, and burnout are referred to as the training stress syndrome. If the training stress syndrome is not reversed through rest or some other creative intervention, the end result will be withdrawal from sport (sport drop-out)

Thus, burnout is viewed as a response to chronic stress. It is characterized by psychological, emotional, and perhaps physical withdrawal from a sport or activity the athlete formerly pursued and enjoyed.

2. Investment Model of Burnout

In its simplest form, investment model of burnout can be viewed as an imbalance between the costs

and benefits associated with athletic participation. In its more complete form, it is conceptualized

as

being a function of five determinants of commitment to sport involvement. The five determinants are, reward, costs, satisfaction, investment and alternative. How the athlete evaluates these five determinants will determine whether his commitment is based upon enjoyment or entrapment. If sport commitment is based upon enjoyment, the athlete will participate enthusiastically. If sport

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commitment is based upon entrapment, it is only a matter of time before burnout sets in and the athlete withdraws from sport.

3. Empowerment Model of Burnout

Empowerment model of burnout is based on the notion that burnout in sport is a social problem caused by an overly controlling and constraining social structure. The model suggests that burnout in

sport occurs in connection with two conditions:

1.

When the sport experience is so constraining that the young person is unable to develop a desirable alternate identity, the athlete seeks to liberate himself from sport.

2.

When the social organization of sport is so structured that the athlete comes to believe that he has no control over his life, the athlete seeks to liberate himself from sport.

Symptoms of Burnout and Interventions

There are two sets of symptoms that include physiological and psychological symptoms of burnout.

Physiological symptoms include:

1. Increased resting and exercise heart rate
2. Increased resting systolic blood pressure
3. Increased muscle soreness and chronic muscle fatigue
4. Increased presence of biochemical indicators of stress in the blood
5. Increased sleep loss
6. Increased colds and respiratory infections
7. Decreased body weight
8. Decreased maximal aerobic power
9. Decreased muscle glycogen
10. Decreased libido and appetite

Psychological symptoms include:

1. Increased mood disturbances
2. Increased perception of physical, mental, and emotional exhaustion
3. Decreased self-esteem
4. Negative change in the quality of personal interaction with others (cynicism, lack of feeling, impersonal relating)
5. Negative cumulative reaction to chronic everyday stress as opposed to acute doses of stress.

Recommended Intervention

The first and primary step in addressing burnout is self-awareness. The athlete must first recognize

he is suffering from burnout and communicate his feeling to a sympathetic parent, coach, or sport

psychologist. The second step is to take time off from the offending activity. In the final and third step, relaxation strategies may be beneficial as coping strategies for reversing the debilitating effects

of stress and burnout.

Recommendations for Players:

- Play for your own reasons.
- Balance sports with other things in life.
- If it is not fun then don't do it.
- Try to make practice as fun.
- Relax and take time off occasionally.

Recommendations for Coaches

- Cultivate personal involvement with player.
- Establish two-way communication with athlete.
- Solicit and utilize player input.
- Work to understand player feelings and perspective.

Recommendations for Parents

- Recognize the optimal amount of "pushing" needed.

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- Back off and lessen involvement.
- Reduce importance of winning.
- Show support and empathy for child's efforts.
- Don't coach if not the coach, and separate roles if you are the coach.
- Solicit child's input



Burnout in Athletes

1. Defining Burnout and Related Terms

- **Burnout:** A syndrome of physical/emotional exhaustion, sport devaluation, and reduced athletic accomplishment.
- **Overtraining:** Training beyond the level for maximum benefit, leading to maladaptive behavior, staleness, and burnout.
- **Overreaching:** Short-term overtraining that is part of normal training; if prolonged, it becomes overtraining.
- **Staleness:** Initial failure of the body to adapt to training, often a precursor to burnout.

2. Models of Burnout

a. Stress Model of Burnout

- **Training Stress Syndrome:** The combination of staleness, overtraining, and burnout. If not reversed, it leads to withdrawal from the sport.
- **Burnout as Chronic Stress Response:** Psychological, emotional, and sometimes physical withdrawal from a sport the athlete formerly enjoyed.

b. Investment Model of Burnout

- **Imbalance of Costs and Benefits:** Burnout results from an imbalance between the costs and benefits of athletic participation.

- **Five Determinants:** Reward, costs, satisfaction, investment, and alternative.
- **Commitment Types:**
 - **Enjoyment:** Enthusiastic participation.
 - **Entrapment:** Leads to burnout and eventual withdrawal.

c. Empowerment Model of Burnout

- **Social Problem:** Burnout is caused by an overly controlling and constraining social structure.
- **Two Conditions:**
 1. **Lack of Alternate Identity:** When the sport experience is so constraining that the athlete cannot develop a desirable alternate identity.
 2. **Lack of Control:** When the social organization of the sport makes the athlete feel they have no control over their life.

3. Symptoms and Interventions for Burnout

a. Physiological Symptoms

1. Increased resting and exercise heart rate
2. Increased resting systolic blood pressure
3. Increased muscle soreness and chronic muscle fatigue
4. Increased biochemical indicators of stress in the blood
5. Increased sleep loss
6. Increased colds and respiratory infections
7. Decreased body weight
8. Decreased maximal aerobic power
9. Decreased muscle glycogen
10. Decreased libido and appetite

b. Psychological Symptoms

1. Increased mood disturbances
2. Increased perception of physical, mental, and emotional exhaustion
3. Decreased self-esteem
4. Negative changes in personal interactions (cynicism, lack of feeling, impersonal relating)
5. Negative cumulative reactions to chronic everyday stress

c. Recommended Interventions

1. **Self-Awareness:** Recognize burnout and communicate feelings to a sympathetic parent, coach, or sport psychologist.
2. **Take Time Off:** Rest from the offending activity.
3. **Relaxation Strategies:** Use relaxation techniques as coping strategies for stress and burnout.

4. Recommendations

a. For Athletes

- Play for personal reasons.
- Balance sports with other life activities.
- Ensure sports are fun; if not, reconsider participation.
- Make practice enjoyable.
- Relax and take occasional breaks.

b. For Coaches

- Cultivate personal involvement with players.
- Establish two-way communication with athletes.
- Solicit and use player input.
- Understand players' feelings and perspectives.

c. For Parents

- Recognize the optimal amount of encouragement needed.
- Reduce involvement if necessary.
- Emphasize effort over winning.
- Show support and empathy for the child's efforts.
- Avoid coaching unless officially the coach, and separate coaching and parenting roles.
- Solicit input from the child.

This comprehensive understanding of burnout in athletes highlights the importance of balanced training, self-awareness, supportive environments, and effective communication to prevent and manage burnout.



Lesson 44

THE PSYCHOLOGY OF ATHLETIC INJURIES

Physical factors such as overtraining, equipment failure, and poor playing conditions are believed to be the major factors contributing to athletic injuries. Psychological factors play an important role in the incidence, prevention, and rehabilitation of athletic injuries.

In addressing the important subject of the psychology of athletic injuries, three main sections have been prepared. These include:

- Psychological predictors of athletic injuries
- Athlete response to injury and rehabilitation
- Other considerations

Psychological Predictors of Athletic Injury

It stands to reason that if researchers can identify psychological factors associated with the occurrence of injuries, steps can be taken to reduce the number and severity of those that do occur.

Any cognitive appraisal that leads to the stress response puts the athlete at risk for injury. Factors that impact the stress response include personality of the athlete, history of stressors, coping resources, and potential interventions.

Stressful athletic situation Cognitive Appraisals

Physiological/Attentional Changes

Injury

Four others factors moderate the relationship between a potentially stressful athletic situation and an

injury. These four factors include:

Personality of the athlete

History of stressor

Coping resources available to the athlete

Interventions

1.

Personality Factors

Personality factors include hardiness, locus of control, sense of coherence, competitive trait anxiety, and intrinsic motivation.

2. History of Stressor

Factors incorporated under the category of history of stressors include stressful life events, daily hassles, and previous injuries.

- Life stress and daily hassles

Relationship between stressful life events and increased illness is extended to the athletic domain. The more life stress the athlete experiences, the greater is the incidence and severity of athletic injury.

- Previous Injury

Athletes who are worried about the recurrence of an injury, or about whether or not they have fully recovered from a previous injury, are vulnerable to further injury.

3. Coping Resources

Coping resources available to the athlete include general coping behavior, social support, stress management techniques, attentional strategy, and prescribed or self-prescribed medication.

- Coping Behavior

Any behavior that assists an individual in dealing with a stressful situation is considered to be a coping behavior.

- Social Support

Social support is one of the important coping resources available to athlete to reduce the debilitating effect of the stress response. Individuals and groups that provide social support for the athlete include parents, friends, coach, teammates, fraternity/sorority, clubs, and religious groups.

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- Stress Management Techniques

Many athletes utilize stress management and cognitive intervention techniques as coping strategies

for controlling the stress response. Research has demonstrated that effective education in the stress

response is associated with a reduction in the number and severity of injuries sustained by athletes.

- Attentional Strategy

A coping resource available to distance runners is attentional strategy. Research shows that the dissociative strategy of running is associated with less incidence of injury.

- Medication

Many drugs have the ability to influence the stress response, and thus the probability of injury. All these psychological effects have the potential to reduce the coping resources of the athlete.

4. Interventions

One set of intervention seeks to change the cognitive appraisal of potentially stressful events, while

the second seeks to modify the physiological/attentional aspects of the stress response. Cognitive appraisal might be changed by rethinking how one plan to address a particularly stressful situation.

Psychological/attentional aspects might be modified through progressive realization and imagery.

Psychological Response to Injury and Rehabilitation

Factors associated with an athlete's psychological response to injury and follow-up rehabilitation occurs after the injury has occurred. It is composed of cognitive appraisal, emotional response, and behavioral response.

• Cognitive Appraisal

Cognitive appraisal helps determine the athlete's emotional response to injury. Most research in cognitive appraisal has focused upon the athlete's perception of self-esteem and self-worth following

a serious sport injury.

• Emotional Response

Emotional responses include:

1. Fear of the unknown
2. Feelings of tension, anger, and depression
3. Frustration and boredom associated with being injured
4. Negative attitude
5. Grief associated with an injury
6. Emotional coping skills

Behavioral Response to Injury

The third factor leading to injury recovery, and associated with cognitive and emotional response, is

the behavioral response of the athlete to injury. The primary focus of research in the area of behavioral response to injury has been upon adherence to injury rehabilitation, coping and intervention, and pain management.

• Adherence to Injury Rehabilitation

In order for an injury rehabilitation program to be successful, it is believed that the athlete must adhere to the program. Thus, adherence to sport injury rehabilitation programs has emerged as a very

important area of study. Predictors of injury rehabilitation adherence are personal factors and situational factors. Personal factors related to adherence include pain tolerance, tough-mindedness

(e.g., self-assurance, assertiveness, independence), and goal perspective.

Situational factors most closely related to adherence include:

a)

Belief in the efficacy of treatment procedures

b)

c)

Comfort of the rehabilitation clinical environment

Convenience of rehabilitation program scheduling

• Coping and Intervention

Coping skills possessed by the athlete, and cognitive-behavioral interventions (applied by others or

the athlete) are effective in enhancing adherence to injury rehabilitation programs

• Pain Management

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Pain tolerance is a personality characteristic. Individuals with a low tolerance to pain may have a more difficult time going through the stages of the sport injury recovery process. Performance

pain can be differentiated from injury pain. Performance pain is controlled by the athlete and is associated with improved performance and a sense of accomplishment. Conversely, injury pain is not controlled by the athlete and may be of either the acute or the chronic variety. Acute pain is due to a trauma to the body and is intense and short in duration. Conversely, chronic pain is long lasting, is uncomfortable, continues long after the initial injury, and is very complex in its origin. Pain can also be categorized as benign or harmful. Benign pain is generally short in duration and is not associated with swelling and soreness. Conversely, harmful pain is present before and after exertion and is associated with swelling, tenderness, and prolonged soreness.

Pain associated with an athletic injury is managed through a combination of pharmacological (prescription drug) and nonpharmacological approaches. Pharmacological pain management strategies are often needed short term. Nonpharmacological pain management strategies are classified as being of the pain-reduction or the pain-focusing variety.

Other Considerations

Rehabilitation Personnel with Psychological Expertise

There are two basic approaches to providing sport injury rehabilitation: the distributed approach and the specialist approach. In the distributed approach, is to make sure that all concerned receive training in sport psychology applications. The specialist approach is to employ a fulltime sport psychologist. In the specialist approach, the sport psychologist works with injured athletes requiring psychological services.

Benefits Associated with Sustaining and Recovering from an Athletic Injury

While it is generally assumed that nothing good comes from an athletic injury, this may not be entirely true. Evidence suggests that successful recovery from an athletic injury is associated with several benefits:

- Personal growth
- Psychologically based performance enhancement
- Increase in self-efficacy, mental toughness, and personal motivation
- Physical and technical development benefits associated with injury/recovery experience
- General health improvement



The Psychology of Athletic Injuries

1. Psychological Predictors of Athletic Injuries

- **Physical Factors:** Overtraining, equipment failure, poor playing conditions.
- **Psychological Factors:** Role in incidence, prevention, and rehabilitation of injuries.

a. Stressful Athletic Situations

- **Cognitive Appraisals:** Lead to stress responses, increasing injury risk.
- **Factors Influencing Stress Response:**
 - Personality

- History of stressors
- Coping resources
- Interventions

b. Four Moderating Factors

1. **Personality Factors:** Hardiness, locus of control, sense of coherence, competitive trait anxiety, intrinsic motivation.
2. **History of Stressors:** Stressful life events, daily hassles, previous injuries.
 - **Life Stress:** Greater life stress leads to higher incidence and severity of injuries.
 - **Previous Injuries:** Worry about recurrence or incomplete recovery increases vulnerability.
3. **Coping Resources:** General coping behavior, social support, stress management techniques, attentional strategy, medication.
 - **Coping Behavior:** Behaviors assisting in dealing with stress.
 - **Social Support:** Support from parents, friends, coaches, teammates, clubs, religious groups.
 - **Stress Management:** Education and techniques to control stress response.
 - **Attentional Strategy:** Dissociative strategy (e.g., in running) linked to fewer injuries.
 - **Medication:** Drugs influencing stress response and injury probability.
4. **Interventions:** Change cognitive appraisal of stressful events or modify physiological/attentional aspects through relaxation and imagery.

2. Psychological Response to Injury and Rehabilitation

a. Cognitive Appraisal

- **Determines Emotional Response:** Focus on self-esteem and self-worth post-injury.

b. Emotional Response

1. Fear of the unknown
2. Tension, anger, depression
3. Frustration and boredom
4. Negative attitude
5. Grief
6. Emotional coping skills

c. Behavioral Response

- **Adherence to Rehabilitation:** Crucial for successful recovery.
 - **Predictors:** Pain tolerance, tough-mindedness, goal perspective.
 - **Situational Factors:** Belief in treatment efficacy, clinic comfort, rehabilitation schedule convenience.
- **Coping and Intervention:** Skills and cognitive-behavioral interventions to enhance adherence.
- **Pain Management:** Differentiating performance pain (controlled) from injury pain (uncontrolled, acute, or chronic).
 - **Benign Pain:** Short duration, no swelling.
 - **Harmful Pain:** Persistent, associated with swelling and soreness.
 - **Management:** Combination of pharmacological and nonpharmacological approaches.

3. Other Considerations

a. Rehabilitation Personnel with Psychological Expertise

- **Approaches:**
 - **Distributed Approach:** Training all involved in sport psychology applications.
 - **Specialist Approach:** Employing a full-time sport psychologist for injured athletes.

b. Benefits of Recovery from Athletic Injury

- **Personal Growth**
- **Performance Enhancement:** Psychologically based.
- **Increased Self-Efficacy, Mental Toughness, Personal Motivation**
- **Physical and Technical Development:** Associated with injury/recovery experience.
- **General Health Improvement**

This lesson emphasizes the critical role of psychological factors in preventing and managing athletic injuries, along with strategies for effective rehabilitation and the potential benefits of recovery.



1. ****Interventions**:** Change cognitive appraisal of stressful events or modify physiological/attentional aspects through relaxation and imagery.
elaborate: Interventions: Change cognitive appraisal of stressful events or modify physiological/attentional aspects through relaxation and imagery.



Interventions in Athletic Injury Psychology

1. Cognitive Appraisal Interventions

Cognitive appraisal refers to how an athlete interprets and thinks about potentially stressful events. Changing cognitive appraisal can reduce the perceived stress associated with an event, which in turn can lower the risk of injury and improve recovery outcomes.

- **Cognitive Restructuring:** This technique involves identifying and challenging negative or irrational thoughts and replacing them with more positive, realistic ones. For example, instead of thinking, "I will never recover from this injury," the athlete might learn to think, "I have the resources and support to recover fully."
- **Reframing:** This involves changing the way an athlete perceives a situation. For example, viewing an injury not as a setback but as an opportunity to focus on other aspects of training, such as mental toughness or technical skills.
- **Positive Self-Talk:** Encouraging athletes to use positive affirmations and self-encouragement can help shift their mindset from negative to positive. Statements like "I am strong and capable of overcoming this challenge" can replace negative thoughts.
- **Goal Setting:** Setting realistic, achievable goals can help athletes maintain a positive outlook and stay motivated during recovery. Goals should be specific, measurable, achievable, relevant, and time-bound (SMART).

2. Physiological/Attentional Interventions

These interventions aim to modify the body's physiological response to stress and improve focus and concentration, reducing the risk of injury and aiding in recovery.

- **Relaxation Techniques:** These are methods used to reduce physical and mental tension, promoting a state of calm.
 - **Progressive Muscle Relaxation (PMR):** Involves tensing and then slowly relaxing different muscle groups in the body, which helps reduce overall physical tension.
 - **Deep Breathing Exercises:** Slow, deep breaths can help activate the body's relaxation response, lowering heart rate and reducing stress.
- **Imagery and Visualization:** These techniques involve creating vivid mental images to enhance performance and recovery.
 - **Healing Imagery:** Athletes visualize their bodies healing and recovering from injury. This can promote a sense of control and positive mindset.
 - **Performance Imagery:** Imagining successful performance scenarios can help maintain motivation and confidence. For example, an injured athlete might visualize themselves successfully returning to competition.
- **Mindfulness Meditation:** Involves focusing on the present moment without judgment. Mindfulness can help athletes become more aware of their thoughts and feelings, reducing stress and improving emotional regulation.
- **Biofeedback:** A technique that uses electronic monitoring to teach athletes how to control physiological functions such as heart rate and muscle tension. By becoming aware of these functions, athletes can learn to regulate them better, reducing stress.

Combining Interventions

Integrating both cognitive and physiological/attentional interventions can provide a comprehensive approach to managing stress and improving recovery in athletes. For example:

- An athlete might use cognitive restructuring to change negative thoughts about their injury while practicing deep breathing exercises to reduce physical tension.
- Regular mindfulness meditation sessions combined with positive self-talk can help maintain a calm and focused mindset.

By addressing both the mental and physical aspects of stress, these interventions can help athletes reduce their risk of injury and enhance their rehabilitation process.



Lesson 45

DRUG ABUSE IN SPORT AND EXERCISE

We have learned of athletes who have succumbed to eating disorders for purposes of performance enhancement. We have also learned of athletes who over train and experience burnout in an effort to perform at a level higher than that of their competitors. In this lecture we learn of athletes who turn to the use of drugs in an effort to gain an unfair advantage over their competitors. There appear to be three fundamental problems associated with the use of drugs to enhance

athletic

performance. The first problem has to do with ethics. Because drug use for purposes of performance enhancement is illegal, it is ethically wrong to take them in an effort to obtain an unfair

advantage over the opposition. Amateur and professional sport organizations publish and continually

revise their own list of banned substances. The second problem has to do with the potentially addictive properties of drugs. Once an athlete starts taking a drug for performance enhancement purposes, it may be difficult for her to stop taking them later on. Drugs may be physically and/or psychologically addictive. A third problem is related to the potentially lethal effects of drugs on the

health and well-being of athlete.

Psychophysiological Effects of Certain Banned Substances

Each of the drugs or banned substance that will be mentioned in this brief review has both a hoped

for benefit (the reason it is taken by the athlete) and a negative consequence. In some cases the negative consequences are well documented, but in other cases, due to limitations of research, they

are not. We will first address the use and abuse of anabolic steroids because of their widespread use

by athletes seeking greater size, aggressiveness, and strengths.

Anabolic-Androgenic Steroids

Anabolic steroids are hormones that stimulate protein anabolism in the body. Athletes ingest anabolic steroids because they believe that they are responsible for alterations in body composition

that result in greater size, strength, and power. They have both an anabolic effect (increasing muscular strength and size) and an androgenic effect (masculinizing) on the user. Research clearly shows that anabolic steroids use is associated with increased body weight and mass, altered body composition, increased muscle size and strength, increased blood volume, and increased number of

red blood cells.

Negative physiological consequences associated with anabolic steroid use include increased risk of

heart disease, certain cancers, and undesirable sex-specific effects. Many sport psychologists believe

that long-term use of this drug will lead to poor health. Organs that are particularly susceptible to negative consequences of anabolic steroids abuse are those, such as the kidney or liver.

Stimulants

Stimulants, such as amphetamines and cocaine, increase the rate and work capacity of the central nervous system, respiratory system, and heart. The neural-stimulating and cardiac-stimulating effects

of these drugs can provide a physiological advantage to the athlete by inhibiting mental and physical fatigue.

These drugs are physically and/or psychologically addicting and their use may lead to serious health problems.

Depressants

Depressants, such as barbiturates, sedative-hypnotics, and alcohol, are designed to relieve tension,

depression, and anxiety in the athlete. Theoretically, this could help the fearful and anxious

athlete by

providing a steady effect. These drugs, however, do not always have the desired effect on the athlete.

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Depressants may actually have the effect of reducing inhibition, reducing judgment, and heightening

risk-taking behavior which may in turn result in poor as opposed to superior performance. These drugs are also highly addictive, making it difficult to quit using them without severe withdrawal symptoms.

Other Banned Drugs

Other drugs that have been banned by the International Olympic Committee (IOC) include diuretics,

hallucinogens, and beta-adrenergic blockers.

Creatine Use by Athletes

Creatine is a naturally occurring energy-producing substance. It is a food supplement extensively used

by athletes seeking enhanced performance as it is currently not banned by the International Olympic

Committee. Creatine serves as an energy buffer (shield) during periods of intense exercise.

Long-term negative effects on the liver, kidneys, and brain are only beginning to be investigated.

Combating Drug Abuse in Sport

Underreporting is a serious limitation of research involving drug use among athletes. It is even more

difficult to get accurate information from athletes about their coach's direct or indirect role in encouraging drug use for purposes of performance enhancement.

Two basic approaches to combating drug use in sport are cognitive techniques and behavioral techniques. Cognitive techniques utilize intellectual and psychological methods to influence behavior

and attitude. Behavioral techniques shape the athlete's environment in ways that will elicit desirable

responses and behaviors from the athlete.

Cognitive Techniques

Using cognitive techniques, the psychologists utilize support groups among the players to encourage

drug abstinence. The psychologist shows concern for athletes, sets limits on unacceptable behavior,

develop team policy, and teaches athletes specific coping skills to deal with the pressure to excel. The

psychologists must be aware of each athlete's mental status, both in and out of the sport environment..

Behavioral techniques

The focus of behavioral techniques is upon teaching athletes alternative ways to enhance performance that do not include the use of drugs. These alternative to increased performance include

teaching motor skills and strategies that lead to increased performance.

Behavioral techniques that are effective include peer involvement in drug education and drug prevention efforts. Young athletes are much more likely to take advice from an admired peer than from an adult. Athletes are much less likely to take drugs for purpose of performance

enhancement
id they fear detection through mandatory drug testing.



Lesson 45: Drug Abuse in Sport and Exercise

Athletes sometimes turn to drug use to gain an unfair advantage, which presents ethical, addictive, and health problems. This lesson covers the types of banned substances, their effects, and strategies to combat drug abuse in sport.

Fundamental Problems with Performance-Enhancing Drugs

1. Ethics

- **Illegal and Unethical:** Using drugs for performance enhancement is illegal and provides an unfair advantage.
- **Banned Substances:** Both amateur and professional organizations maintain and update lists of banned substances.

2. Addictive Properties

- **Physical and Psychological Addiction:** Drugs used for performance enhancement can be difficult to stop once started.

3. Health Risks

- **Potentially Lethal Effects:** Drugs can have serious negative effects on health and well-being.

Psychophysiological Effects of Banned Substances

1. Anabolic-Androgenic Steroids

- **Purpose:** Used for increasing size, strength, and aggressiveness.
- **Positive Effects:** Increased body weight, muscle size, strength, blood volume, and red blood cells.
- **Negative Consequences:** Increased risk of heart disease, cancers, undesirable sex-specific effects, and damage to organs like the liver and kidneys.

2. Stimulants (e.g., amphetamines, cocaine)

- **Purpose:** Increase central nervous system, respiratory system, and heart activity.
- **Positive Effects:** Inhibition of mental and physical fatigue.
- **Negative Consequences:** Physical and psychological addiction, serious health problems.

3. Depressants (e.g., barbiturates, alcohol)

- **Purpose:** Relieve tension, depression, and anxiety.
- **Negative Consequences:** Reduced inhibition and judgment, heightened risk-taking behavior, poor performance, and severe withdrawal symptoms.

4. Other Banned Drugs

- **Diuretics, Hallucinogens, Beta-Adrenergic Blockers:** Banned by the International Olympic Committee (IOC).

5. Creatine

- **Purpose:** Enhances performance as an energy buffer.
- **Current Status:** Not banned by the IOC.
- **Potential Risks:** Long-term effects on the liver, kidneys, and brain are being investigated.

Combating Drug Abuse in Sport

Research on drug use is limited due to underreporting, especially regarding the coach's influence. Two main approaches to combating drug use are cognitive and behavioral techniques.

1. Cognitive Techniques

- **Support Groups:** Encourage drug abstinence.
- **Psychological Support:** Show concern for athletes, set limits on behavior, develop team policies, and teach coping skills.
- **Mental Awareness:** Psychologists must be aware of the athlete's mental status both in and out of the sport environment.

2. Behavioral Techniques

- **Alternative Performance Methods:** Teach motor skills and strategies for increased performance without drugs.
- **Peer Involvement:** Use admired peers in drug education and prevention efforts.
- **Mandatory Drug Testing:** Fear of detection can deter athletes from drug use.

Key Takeaways

- **Ethical and Health Concerns:** Using drugs for performance enhancement poses significant ethical, addictive, and health risks.
- **Types of Drugs:** Includes anabolic steroids, stimulants, depressants, diuretics, hallucinogens, beta-adrenergic blockers, and creatine.
- **Combating Strategies:** Employ cognitive and behavioral techniques to reduce drug use and promote healthier, ethical performance enhancement methods.



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